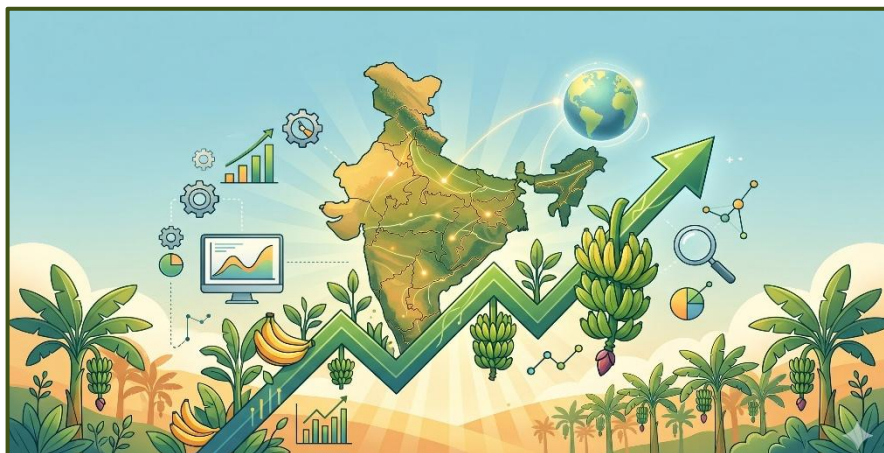


Statistical Analysis and Forecasting of Banana Production: Global to Local Study



A Research Project Report submitted in partial fulfilment
of requirements for the Degree of M.Sc. (Statistics)
Semester – IV



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CERTIFICATE

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Prof. R. L. Shinde

INDEX

Sr. No.	Chapter	Page No.
1.	Introduction	1 – 3
2.	About Banana 2.1 Origin and History 2.2 Botanical Description of Banana 2.3 Nutritional Value of Banana 2.4 Chemistry and Environmental Requirements 2.5 Global Species of Banana 2.6 Indian Banana Species and Cultivars 2.7 Global Banana Production	4 - 10
3.	World Banana Production 3.1 Introduction 3.2 Top 15 Banana Producing Countries 3.3 Global Area Under Cultivation 3.4 Time Series Analysis and Forecasting for World Banana Production	11 – 17
4.	Banana Production and Productivity in India 4.1 Introduction 4.2 Data Sources and Methodology 4.3 Trend in Area, Production and Productivity 4.4 Trend in Area under Cultivation 4.5 Trend in Production 4.6 Trend in Productivity	18 – 23
5.	Time Series Analysis for Major Banana Producing States in India 5.1 Introduction 5.2 Time Series Analysis and Forecasting for India 5.3 Data Description of State wise Banana Production 5.4 Time Series Analysis and Forecasting for Maharashtra 5.5 Time Series Analysis and Forecasting for Andhra Pradesh 5.6 Time Series Analysis and Forecasting for Tamil-Nadu 5.7 Time Series Analysis and Forecasting for Uttar Pradesh 5.8 Time Series Analysis and Forecasting for Gujarat 5.9 Time Series Analysis and Forecasting for Madhya Pradesh	24 – 41
6.	Time Series Analysis for Major Banana Producing Districts in Maharashtra 6.1 Introduction 6.2 Data Description 6.3 Time Series Analysis and Forecasting for Jalgaon 6.4 Time Series Analysis and Forecasting for Solapur 6.5 Time Series Analysis and Forecasting for Nanded	42 – 49
7.	Prices of Banana	50 - 59
8.	Products from Banana	60 – 63
9.	Review of Research Paper (4 Research Paper)	64 – 74
10.	Overall Conclusion	75 – 77
❖	References	78

Chapter 1: Introduction

Banana is one of the most important fruit crops in the world, both in terms of production and consumption. It plays a vital role in ensuring food security, nutrition, and economic stability for millions of farmers globally. In tropical and subtropical regions, banana cultivation is not just an agricultural activity but a major source of livelihood. Due to its year-round availability, high productivity, and adaptability to various climatic conditions, banana has become a dominant fruit crop in many countries, especially in India.

Globally, banana production has shown a remarkable increase over the last six decades. According to available data, global banana production has grown from approximately 22.4 million metric tonnes (MMT) in 1960 to around 142–145 MMT by 2025, reflecting a more than 6-fold increase. This growth can be attributed to advancements in agricultural technology, expansion of cultivation area, improved irrigation methods, and increasing global demand. The steady annual growth rate of approximately 1% to 3% indicates that banana production is a Stable and expanding sector in the global agricultural economy.

India holds a dominant position in global banana production. It contributes nearly 26% – 27% of total world production, making it the largest producer worldwide. In 2024 alone, India produced approximately 36.61 million metric tonnes, which is significantly higher than the second-largest producer, China, which contributed about 12.06 MMT. This clearly highlights the unequal distribution of banana production across countries, where a few major producers dominate the global market. In fact, the top five banana-producing countries contribute more than 55% of global production, demonstrating strong geographical concentration.

Banana is highly nutritious fruit. It provides instant energy due to its high carbohydrate content and contains essential nutrients such as potassium, vitamin B6, and vitamin C. With an average energy value of 89 kcal per 100 grams, banana is considered an ideal food for all age groups. This nutritional significance further increases its demand in both domestic and international markets.

From an economic perspective, banana cultivation plays a crucial role in rural development. In India, approximately 1 million hectares of land is under banana cultivation, making it one of the largest fruit crops in terms of area. States like Maharashtra, Andhra Pradesh, Tamil Nadu, and Gujarat are major contributors. Maharashtra alone accounts for a significant share, with districts like Jalgaon contributing heavily to production. The concentration of production in specific regions indicates the importance of favourable climatic conditions, irrigation facilities, and technological adoption.

Statistical analysis of banana production reveals a consistent upward trend over time. Time series data from 2000 to 2024 shows that global banana production increased from approximately 70 MMT to over 135 MMT, indicating a strong positive trend. Using ARIMA models, it has been estimated that global production will continue to rise, reaching around 155 MMT by 2029. The model also suggests an average annual increase of about

3.09 MMT, which reflects steady growth. However, increasing uncertainty in long-term forecasts indicates that external factors such as climate change, diseases (like TR4), and market fluctuations can influence production.

Similarly, India's banana production has shown significant growth over the past decade. From 28.455 MMT in 2011–12 to 37.807 MMT in 2023–24, the production has increased steadily, with a compound annual growth rate (CAGR) of approximately 2.51%. Productivity has also improved, reaching nearly 37.81 MT per hectare, indicating better farming practices and technological adoption.

Despite these positive trends, banana cultivation faces several risks and challenges. These include climatic uncertainties such as droughts, floods, and temperature fluctuations, as well as biological threats like pests and diseases. For example, extreme temperatures below 10°C or above 38°C can severely affect banana growth, while inadequate water supply can reduce yield significantly. These uncertainties make banana farming a risky venture, especially for small and marginal farmers.

To address these risks, agricultural insurance plays a crucial role in ensuring financial stability for farmers. Banana crop insurance is an important risk management tool that protects farmers against losses due to natural calamities, pests, diseases, and adverse weather conditions. In India, schemes like the Pradhan Mantri Fasal Bima Yojana (PMFBY) provide insurance coverage for banana cultivation. Under such schemes, farmers pay a minimal premium, while the government subsidizes the remaining cost. In case of crop failure, farmers receive compensation based on the extent of damage, helping them recover financially and continue farming.

The inclusion of insurance in banana cultivation has several advantages. Firstly, it reduces the financial risk faced by farmers, encouraging them to invest in better inputs such as fertilizers, irrigation systems, and high-yield varieties. Secondly, it promotes stability in agricultural income, which is essential for rural development. Thirdly, it supports sustainable farming practices by enabling farmers to adopt modern technologies without fear of loss. Statistical evidence suggests that regions with better insurance coverage tend to have more stable production patterns compared to regions without such support.

Moreover, the integration of statistical methods in agricultural studies has become increasingly important. Techniques such as time series analysis, forecasting models, and trend analysis help in understanding production patterns and predicting future outcomes. In this project, statistical tools like the ARIMA model have been used to analyse banana production data. The results indicate that banana production follows a strong upward trend with moderate fluctuations, making it suitable for forecasting and policy planning.

In addition, the study of production distribution reveals the presence of inequality among countries and regions. For instance, while the top 15 countries contribute around 76% of global production. This uneven distribution highlights the importance of resource availability, technological advancement, and government support in determining production levels.

The banana is a globally significant crop with immense economic, nutritional, and agricultural importance. Its production has increased steadily over time, driven by technological advancements and rising demand. India plays a leading role in global banana production, contributing a significant share to the world market. However, the sector faces various challenges, including climatic risks and market uncertainties. The adoption of crop insurance schemes and advanced statistical analysis techniques can help mitigate these risks and ensure sustainable growth. This project focuses on analyzing banana production trends using statistical methods and provides insights into future production patterns, thereby contributing to better planning and decision-making in the agricultural sector.

❖ **Objectives of Study**

- To study the trend of banana production over time.
- To compare banana production across different countries and Indian states.
- To analyze changes in area, production, and productivity of banana.
- To identify major banana-producing regions in the world and India.
- To examine the distribution and concentration of banana production.
- To forecast future banana production using time series models.
- To analyze month-wise banana price trends in India (2020–2024) for Uttar Pradesh, Maharashtra, Tamil Nadu, and Andhra Pradesh using time series analysis.

Chapter 2: About Banana

2.1 Origin and History of Banana

Banana has a long and complex history that dates back thousands of years. It is believed that banana originated in the tropical regions of Southeast Asia and New Guinea, where wild species of *Musa* were first found. Early humans began using bananas as a food source and gradually started cultivating them. Over time, bananas were domesticated through selection and vegetative propagation, which led to the development of many edible varieties. From its place of origin, banana spread to different parts of the world through human migration, trade, and cultural exchange. It was introduced to South Asia, including India, and later spread to Africa and other tropical regions. The movement of banana to Africa and other continents was mainly due to traders and travellers who carried banana plants with them.

Banana cultivation expanded further during historical trade periods, especially through Arab and European traders. Later, banana was introduced to the Americas, where it became an important commercial crop. Today, banana is cultivated in most tropical and subtropical regions of the world and is one of the most important fruit crops for food and economic purposes. The long history of banana domestication shows the important role of humans in selecting, improving, and spreading banana varieties across the world.

2.2 Botanical Description of Banana

Banana is a large perennial herb belonging to the genus *Musa* and the family Musaceae. Although it appears like a tree, the banana plant is actually an herb because it does not have a true woody stem. Instead, it has a pseudo stem, which is formed by tightly packed leaf sheaths. The pseudostem grows from an underground structure called a rhizome, which produces roots and new shoots.

The leaves of the banana plant are large, broad, and arranged spirally around the pseudostem. These leaves play an important role in photosynthesis and plant growth. The banana plant produces a flowering structure called an inflorescence, which emerges from the top of the pseudo stem. The flowers develop into clusters of fruits known as hands, and each individual fruit is called a finger. The banana fruit develops without fertilization in most cultivated varieties, a process known as parthenocarpy. After fruit production, the pseudo stem dies, and new shoots grow from the rhizome to continue the life cycle.

Botanical Characteristics

The banana plant is often mistakenly referred to as a tree, but it is botanically classified as a giant perennial herb.

- **Leaves:** A healthy plant produces large, oblong leaves that can reach up to 2.7 meters in length, providing the massive surface area required for high rates of photosynthesis.

- The Inflorescence: The plant produces a single large flower cluster (heart) that emerges from the centre of the pseudo stem. This develops into a "bunch" made up of "hands" (clusters) and individual "fingers" (the fruit).
- Root System: It possesses a shallow, adventitious root system, making the plant highly sensitive to soil moisture levels and susceptible to being blown over by high winds.

2.3 Nutritional Value of Banana

Banana is a highly nutritious fruit and is an important source of energy. It contains a high amount of carbohydrates, mainly in the form of sugars and starch, which provide instant energy to the body. Banana is also rich in essential vitamins such as vitamin B6 and vitamin C, which are important for maintaining good health. In addition, banana is an excellent source of minerals, especially potassium, which helps in maintaining proper heart function and regulating blood pressure. It also contains dietary Fiber, which aids in digestion and improves digestive health.

Table 2.1: Nutritional Value of Banana

Nutrient	Amount	Importance
Energy	89 kcal	Provides energy to the body.
Carbohydrates	22.8 g	Main source of energy.
Protein	1.1 g	Helps in growth and repair of tissues.
Fat	0.3 g	Very low, good for heart health.
Dietary Fiber	2.6 g	Helps in digestion.
Potassium	358 mg	Maintains heart and muscle function.
Vitamin C	8.7 mg	Improves immunity.
Vitamin B6	0.4 mg	Helps in brain development and metabolism.
Magnesium	27 mg	Supports bone and muscle health.
Calcium	5 mg	Helps in strengthening bones.

Banana is low in fat and easy to digest, making it Suitable for people of all age groups, including children and elderly people. Due to its rich nutritional composition and health benefits, banana is considered one of the most valuable and widely consumed fruits in the world.

2.4 Chemistry and Environmental Requirements

The "chemistry" of banana production involves a precise balance of soil nutrients, climate conditions, and water management. Bananas are "heavy feeders," meaning they require high volumes of nutrients to produce a large fruit bunch.

Soil Chemistry and Quality: Bananas thrive in deep, well-drained loamy soils. The chemical composition of the soil determines the plant's health:

- Soil pH: The ideal range is pH 5.5 to 7.0. If the soil is too acidic (pH below 5.0), the plant becomes susceptible to Panama Disease (Fusarium wilt).

- **Organic Matter:** High organic matter is crucial for moisture retention and slow-release nutrient delivery.
- **Drainage:** Bananas cannot tolerate "wet feet." Waterlogged soil prevents oxygen from reaching the roots, leading to root rot.
- **Phosphorus (P):** Essential for root development and flower formation. It is usually applied at the time of planting.
- **Potassium (K):** The "quality enhancer." Bananas require more Potassium than any other nutrient. It is vital for fruit filling, increasing sugar content, and improving skin strength.
- **Micronutrients:** Small amounts of Boron, Zinc, and Iron are necessary to prevent fruit deformities and maintain photosynthesis.

Table 2.2: Chemical Characteristics of Banana

Parameter	Optimal Condition	Impact of Deviation
Temperature	25°C to 30°C	Growth is retarded below 20°C and stops below 10°C–12°C. Chilling causes "choking" (failed bunch emergence). Heat above 37°C–38°C causes leaf scorch and fruit damage.
Water Requirement / Rainfall	15 to 20 litre/day 1,500–2,000 mm/year	A single banana plant requires 15 to 25 litres of water per day on average when using drip irrigation, though this varies significantly by growth stage. Over its entire 10–12 months life cycle, a plant needs approximately 900 to 1,200 mm of water from rain or irrigation.
Humidity	75% to 85%	Ideal for leaf expansion. Humidity below 50% increases transpiration, leading to leaf curling and browning edges. Very high humidity can encourage fungal diseases like Sigatoka.
Altitude	0 to 1,200 meters	Can grow up to 2,000m (e.g., "Hill Bananas"), but commercial yields are best at lower elevations. Every 100m increase in altitude can add 30–45 days to the production cycle.

2.5 Global Species of Banana

The world's edible bananas are primarily hybrids of two wild species, *Musa acuminata* and *Musa balbisiana*. While there are over 1,000 varieties (cultivars) grown in more than 135 countries, they are generally classified into two main functional groups: dessert bananas (eaten raw) and cooking bananas/plantains (starchy, usually cooked).

➤ Major Banana Species and Cultivars

The following Table outlines the primary biological species and the most common cultivars found worldwide.

➤ Primary Global Sources

- **Food and Agriculture Organization (FAO):** Provides official global production data and general classification between dessert bananas and plantains.

- National Horticulture Board (NHB), India: An authoritative source for the commercial classification and specific regional varieties of bananas, particularly for major producers like India.
- International Tropical Fruits Network (TFNet): Detailed taxonomic data on the *Musa* genus, including the distinctions between the *Eumusa* and *Australimusa* sections.

Table 2.3: Banana Species and Cultivars

Classification	Name/Group	Notable Cultivars	Primary Countries/ Region
Wild Species	<i>Musa acuminata</i>	-	Southeast Asia: Malaysia, Indonesia, Philippines.
	<i>Musa balbisiana</i>	-	South & SE Asia: India, Myanmar, China. Thailand.
Dessert	Cavendish (AAA)	Grand Nain, Robusta, Dwarf, Cavendish.	Global Exports: India, China, Ecuador, Philippines, Brazil, Costa Rica.
	Gros Michel (AAA)	“Big Mike”, Pisang Ambon	Malaysia, Indonesia, Thailand, Uganda, Ecuador
	Lady Finger (AA/AAB)	Sucrier, Pisang Mas, Ney Poovan	Australia, India (Tamil Nadu, Karnataka, Maharashtra), Malaysia, Indonesia.
	Red Banana (AAA)	Red Dacca, Lal Kela	India (Kerala, Tamil Nadu), Brazil, Sri Lanka, Trinidad.
	Apple/Manzano (AAB)	Silk, Rasthali, Latundan	India (West Bengal, Andhra Pradesh), Philippines, Brazil, Colombia.
Cooking	Plantain (AAB/ABB)	Nendran, Orishele, Agbagba	Africa: Nigeria, Uganda, Cameroon, Ghana. India (Kerala) Colombia.
	Blue Java (ABB)	Ice Cream Banana	Southeast Asia, USA (Hawaii, Florida), Fiji, Central America.
	Saba (BBB/ABB)	Cardaba	Philippines (major producer), India, Malaysia, Indonesia.
Speciality	Fe'i Banana	-	Pacific Islands: French Polynesia, Hawaii, Papua New Guinea
	Abaca	<i>Musa Textilis</i>	Philippines

➤ Regional and Agricultural Extension Sources

- Tamil Nadu Agricultural University (TNAU) Agritech Portal: Detailed information on cultivar characteristics, market preferences, and regional varieties in South India.
- Yara India (Crop Nutrition): Information on commercial subgroups like Cavendish and Williams, including their physical height and susceptibility to diseases.

2.6 Indian Banana Species and Cultivars

The Table 2.4 presents important traditional banana cultivars belonging to the species *Musa paradisiaca* along with their regional names and major producing states in India. These cultivars are part of the rich genetic diversity conserved and documented by the ICAR-National Research Centre for Banana. The Table highlights the geographical

distribution of indigenous banana varieties across major banana-growing states such as Kerala, Tamil Nadu, Andhra Pradesh, Telangana, Karnataka, and Maharashtra.

Many cultivars like Nendran, Monthan, and Bontha are primarily used as cooking bananas and are important in regional food systems. Regional names reflect linguistic and cultural diversity, indicating the local importance of these cultivars. The inclusion of Maharashtra shows the wider adaptability of several traditional varieties beyond South India. These cultivars contribute significantly to food security, nutrition, and rural livelihoods.

The Table 2.4 also helps in understanding varietal distribution patterns, which are important for agricultural planning and conservation. Such information is useful for researchers, horticulturists, and policymakers working on banana production and genetic resource management. Furthermore, the Table 2.4 supports statistical and agricultural studies by providing structured data on cultivar distribution. Overall, it reflects the diversity, regional specialization, and economic importance of traditional banana cultivars in India.

Table 2.4: Indian Banana Species and Cultivars

Species	Cultivar Variety /	Regional Name / Synonyms	Major Producing States (ICAR-based)
Musa paradisiaca	Kullan	Rasalu, Kuzhi vazhai, Enugu arati	Kerala, Tamil Nadu, Andhra Pradesh, Maharashtra
Musa paradisiaca	Nendrapadathi	Nendran vannon, Kuthiravalan	Kerala, Tamil Nadu, Maharashtra
Musa paradisiaca	Nana Nendran	Ethakka, Nendra vazha	Kerala, Tamil Nadu, Maharashtra
Musa paradisiaca	Attu Nendran	Yethan, Nedu Nendran	Kerala, Tamil Nadu
Musa paradisiaca	Myndoli	Moirra banana	Goa, Kerala, Maharashtra
Musa paradisiaca	Moongil	Otta Mukil	Kerala, Tamil Nadu
Musa paradisiaca	Pacha Nandan	Kadali, Pachai Vazhai	Kerala, Tamil Nadu, Karnataka, Maharashtra
Musa paradisiaca	Kali	Vannan, Padathi	Kerala, Tamil Nadu
Musa paradisiaca	Kunnan	Kadali, Adukkann	Kerala
Musa paradisiaca	Adakka Kunnan	Veneetu Kunnan	Kerala
Musa paradisiaca	Karim Kadali	Kari Vazha, Anaikomban	Kerala, Tamil Nadu
Musa paradisiaca	Peyan	Peyan Vazhai	Tamil Nadu, Kerala
Musa paradisiaca	Mayil Vazhai	Mayil banana	Tamil Nadu
Musa paradisiaca	Ney Mannan	Mala Vannan, Javari	Kerala, Tamil Nadu

Musa paradisiaca	Venneettu Mannan	Boothi Bale	Karnataka, Kerala
Musa paradisiaca	Monthan	Cooking banana	Tamil Nadu, Karnataka, Andhra Pradesh, Maharashtra
Musa paradisiaca	Pacha Montha Bathees	Batheesa, Bontha Batheesa	Andhra Pradesh, Telangana, Maharashtra
Musa paradisiaca	Sambrani Monthan	Bonya Kilandi	Tamil Nadu, Andhra Pradesh
Musa paradisiaca	Buditha Montha Bathees	Boodithi Batheesa	Andhra Pradesh
Musa paradisiaca	Nalla Bontha	Bontha	Andhra Pradesh, Telangana
Musa paradisiaca	Nalla Bontha Bathees	Isika Bontha	Andhra Pradesh
Musa paradisiaca	Thella Bontha	Boodithi Bontha	Andhra Pradesh
Musa paradisiaca	Kuri Bontha	Bontha type	Andhra Pradesh, Telangana
Musa paradisiaca	Enna Benian	Benian banana	Kerala, Tamil Nadu, Maharashtra

1. Dwarf Cavendish  Short plant with medium-sized fruits. Most common variety in India.	2. Robusta  Large fruits with thick peel. Widely grown in South India.	3. Nendran (Kerala Banana)  Long and thick fruits. Used for cooking, chips and steaming.
4. Red Banana  Red skin with sweet and creamy pulp.	5. Poovan (Mysore Banana)  Small to medium-sized fruits. Aromatic with sweet-tangy taste.	6. Rasthali (Silk Banana)  Very soft, sweet and aromatic. Popular dessert banana.
7. Monthan (Cooking Banana)  Large and starchy fruits. Used for cooking, curries and chips.	8. Karpuravalli  Sweet with high sugar content. Popular in Tamil Nadu.	9. Grand Naine (G-9)  Export quality variety. Uniform size with long shelf life.

2.7 Global Banana Production

Following Table 2.5 shows the Banana Production of the world (in MMT) from year 1960 – 2025.

Table 2.5: Decade wise Global Production of Banana (1960-2025)

Decade / Year	Global Production (MMT)	Top Producer	Key Driver / Event
1960	~22.4	Ecuador / Brazil	Transition from Gros Michel to Cavendish
1970	~34.0	Brazil	Expansion of commercial plantations in Latin America
1980	~42.5	Brazil / India	Rise of Asian production; India begins its climb to #1
1990	~56.7	India	Liberalisation of trade; India hits ~17.8% global share
2000	~101.3	India	Modernization of irrigation (drip) and tissue culture
2010	~116.2	India	India's share rises to 26%; China becomes a major player
2020	135.1	India	Global peak production reached
2025	142.0 – 145.0	India	Projecting 1% annual growth despite TR4 disease risks

Global banana production increased significantly from 22.4 MMT in 1960 to about 142–145 MMT by 2025. Initially, production was dominated by Brazil and Ecuador, but later India emerged as the leading producer. Growth was driven by modern irrigation, tissue culture, and trade expansion. Production peaked around 2020 and continues to grow slowly. However, future growth may be affected by diseases like TR4.

Chapter 3: World Banana Production

3.1 Introduction

In the previous chapter we see the world's banana production through the Table 2.5. In this chapter we are going to study the world's top 15 countries banana production, area under cultivation, their ranks according to their production in last decades and global share of each country. As per 2024 data of Banana production the Table 3.1 shows their ranks.

Table 3.1: World's Banana Production Overview

Category	Production (MMT)	% of Global Total
India (# Rank 1)	36.614	26.23%
Top 5 Countries Combined	72.518	51.95%
Top 15 Countries Combined	106.718	76.45%
Rest of the world (~115 Countries)	32.882	23.55%
World Total (All 130+ Countries)	139.60	100%

❖ Why India Holds a first rank In Banana Production?

- Favourable climate: Tropical and subtropical conditions (optimal temperature, rainfall, humidity) support year-round banana cultivation.
- Large cultivation area: India has the highest area under banana (~1 million hectares), contributing directly to higher total output.
- Advanced farming techniques: Use of drip irrigation, fertigation, and tissue culture increases yield and efficiency.
- Consistent production growth: Stable CAGR (~2–3%) shows continuous increase in production over decades.
- High domestic demand: Large population ensures strong and steady consumption, encouraging production.
- State-level dominance: Major states like Maharashtra, Andhra Pradesh, and Tamil Nadu contribute significantly to total output.

In Table 3.2, the banana production of the top 50 countries in 2024 is presented, with production measured in Million Metric Tons (MMT). These countries represent the major contributors to global banana production out of approximately 135 producing countries worldwide. India ranks first with the highest production, followed by China, Indonesia, Brazil, and Ecuador, which are the other leading producers. The Table shows that banana production is highly concentrated among a relatively small number of countries, with the top producers contributing a substantial portion of total global output. Several countries from Asia and Latin America dominate the list due to favourable climatic conditions and large agricultural areas suitable for banana cultivation. African countries also contribute significantly, although their production levels are generally lower compared to the leading Asian and Latin American producers. The remaining countries in the top 50 produce

comparatively smaller quantities, forming the lower end of the production distribution. This distribution highlights the unequal contribution of countries to global banana production. Overall, Table 3.2 provides a clear overview of the production levels and ranking of the top banana-producing countries in the world in 2024.

Table 3.2: Top 50 Banana Producing Countries out of 135 countries in 2024

Rank	Country	Production (MMT)	Cum Sum	Rank	Country	Production (MMT)	Cum Sum
1	India	36.61	36.61	26	Cameroon	0.87	126.40
2	China	12.06	48.68	27	Bangladesh	0.84	127.24
3	Indonesia	9.34	58.01	28	DR Congo	0.81	128.05
4	Nigeria	7.31	65.32	29	Laos	0.72	128.77
5	Ecuador	7.20	72.52	30	Mali	0.56	129.34
6	Brazil	6.83	79.34	31	Honduras	0.56	129.90
7	Philippines	5.87	85.22	32	Venezuela	0.54	130.44
8	Angola	4.89	90.11	33	Ivory Coast	0.51	130.96
9	Guatemala	4.39	94.50	34	Spain (Canary Islands)	0.47	131.43
10	Tanzania	3.67	98.17	35	South Africa	0.43	131.85
11	Kenya	2.91	101.08	36	Nepal	0.40	132.26
12	Mexico	2.64	103.72	37	Malawi	0.40	132.66
13	Vietnam	2.64	106.36	38	Madagascar	0.40	133.06
14	Colombia	2.55	108.92	39	Cambodia	0.38	133.44
15	Costa Rica	2.53	111.45	40	Australia	0.37	133.82
16	Peru	2.42	113.86	41	Panama	0.37	134.19
17	Rwanda	2.31	116.17	42	Mozambique	0.37	134.56
18	Dominican Republic	1.66	117.83	43	Taiwan	0.36	134.92
19	Thailand	1.30	119.13	44	Zimbabwe	0.31	135.23
20	Papua New Guinea	1.25	120.38	45	Morocco	0.31	135.54
21	Egypt	1.19	121.57	46	Bolivia	0.31	135.85
22	Burundi	1.15	122.72	47	Pakistan	0.29	136.14
23	Sudan	0.95	123.67	48	Malaysia	0.28	136.42
24	Ethiopia	0.94	124.60	49	Cuba	0.25	136.66
25	Turkey	0.93	125.53	50	Guinea	0.24	136.91

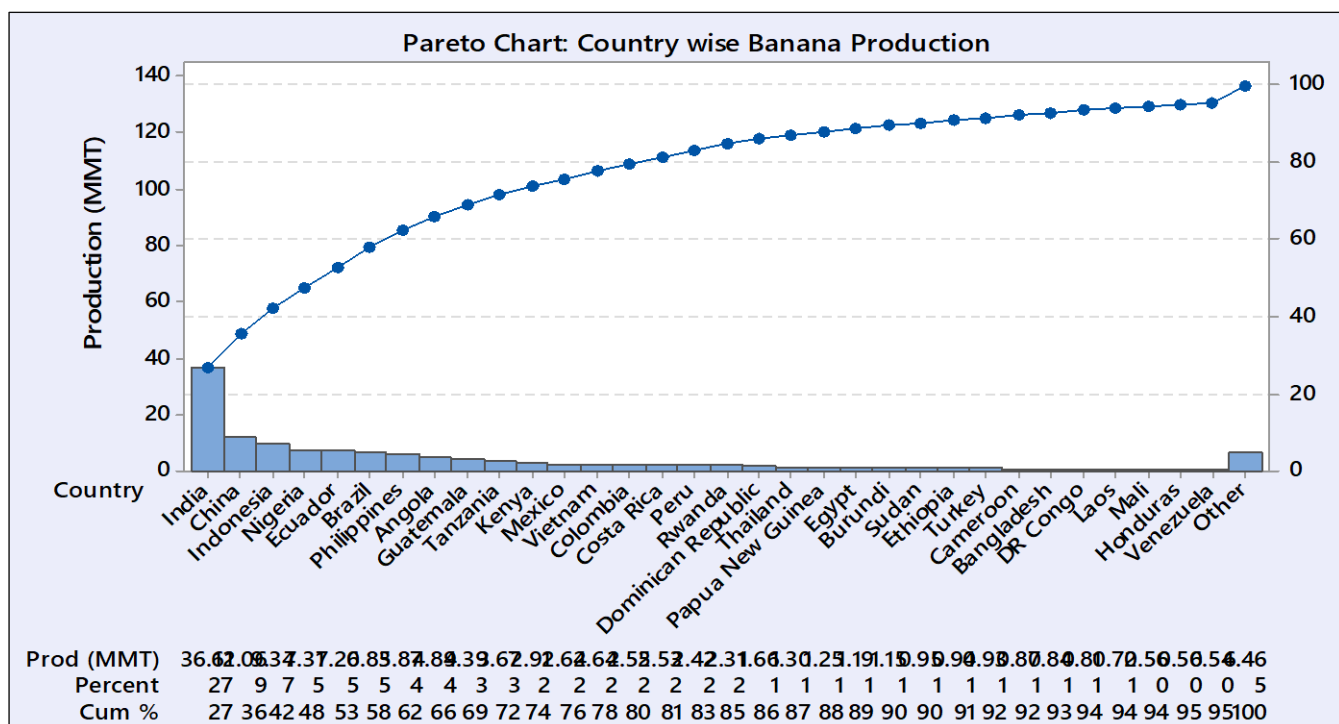


Figure 3.1: Pareto Chart of Top 50 Countries According to their Banana Production

Table 3.3: Global Share Top 5 Banana Producing Countries in 2024

Rank	Country	Production (MMT)	Global Share (%)
1	India	36.61	27.35
2	China	12.06	9.99
3	Indonesia	9.34	6.46
4	Nigeria	7.31	5.82
5	Ecuador	7.20	5.57
Total Global share		72.52	55.19

Through Table 3.3, the global share of the top five banana-producing countries in 2024 is clearly illustrated. These five countries together produce 72.52 million metric tonnes, accounting for 55.19% of total global banana production among approximately 135 producing countries. India is the leading producer with 36.61 million metric tonnes, contributing the highest global share of 27.35%, which confirms its dominant position in the world banana market. China ranks second with 12.06 million metric tonnes, representing 9.99% of the global share. Indonesia holds the third position, producing 9.34 million metric tonnes and contributing 6.46% to global production. Brazil stands in fourth place with 7.31 million metric tonnes and a global share of 5.82%. Similarly, Ecuador ranks fifth, producing 7.20 million metric tonnes and contributing 5.57% to total world production. The Table highlights that more than half of the global banana production is concentrated in these five countries alone. This reflects the strong geographical concentration of banana production in tropical and subtropical regions. It also emphasizes India's significant contribution, producing more than one-quarter of the

world's bananas. Overall, the Table demonstrates the unequal distribution of banana production, with a few countries playing a dominant role in the global supply.

3.2 Top 15 Banana Producing Countries:

In this section we discussed about the top 15 Banana producing countries with their ranks over last decade (from 2014-2024). The Table 3.3 provides a comparative analysis of banana production and global rankings for the world's top 15 producing countries across the years 2014, 2019, and 2024

Table 3.4: Top 15 Banana Producing Countries over a Decade

Country	Year 2014		Year 2019		Year 2024	
	Production	Rank	Production	Rank	Production	Rank
India	30.81	1	30.46	1	36.61	1
China	11.79	2	11.65	2	12.10	2
Indonesia	6.86	4	7.28	4	9.35	3
Nigeria	5.48	7	5.22	3	7.31	4
Ecuador	6.75	5	6.58	6	7.10	5
Brazil	6.95	3	6.83	5	7.04	6
Philippines	5.70	6	6.04	7	5.87	7
Angola	3.48	8	4.03	8	4.80	8
Guatemala	3.42	9	3.91	9	4.39	9
Tanzania	3.19	10	3.40	10	3.67	10
Kenya	1.64	15	1.71	15	2.91	11
Mexico	2.15	12	2.39	13	2.64	12
Vietnam	1.85	13	2.19	14	2.64	13
Colombia	1.77	14	2.91	11	2.55	14
Costa Rica	2.41	11	2.24	12	2.53	15

Throughout this decade, India consistently maintained the top position, significantly increasing its output to 36.61 million tons by 2024.

China held steady as the second-largest producer, while Indonesia showed notable growth, climbing from 7.26 million tons in 2014 to 9.35 million tons in 2024. Nigeria experienced a dramatic rise in the rankings, jumping from 15th place in 2014 to 4th place in 2024 as its production more than doubled. Conversely, the Philippines saw a decline in both production volume and rank over the ten-year period. Other major contributors include Ecuador and Brazil, which remained Stable in the top tier of producers. The data also highlights the rise of African nations like Angola and Kenya, which improved their global standing by 2024. Central and South American countries like Colombia and Costa Rica saw their global ranks fluctuate despite maintaining significant production levels. Overall, the Table illustrates a general upward trend in global banana production and shifting regional dominance.

3.3 Global Area Under Cultivation

The total area dedicated to banana and plantain cultivation is distributed across approximately 135 countries.

- **Total Global Area:** Approximately 5.94 million hectares (as of the most recent FAO records).
- **Acreage Stability:** Global acreage is expected to remain Stable through 2025, with consistent planting in primary regions like India and minor adjustments in countries like the Philippines and Brazil.
- **Regional Concentration:** The top 10 banana-producing countries alone account for nearly 60% to 70% of global banana acreage.

❖ India's Area Under Cultivation

India maintains the largest area under banana cultivation in the world, representing roughly 11% to 15.5% of the global total acreage.

- **Total Area in India:** Approximately 1.0 million hectares (1,000 thousand hectares) as of the 2023-24 period.
- **Growth Trend:** Cultivation area has grown steadily from 796,000 hectares in 2011-12 to the current 1-million-hectare mark in 2024.
- **State-Wise Leading Areas:**
 - **Maharashtra:** Leads with all-time high of 141.53 thousand hectares for the 2024–2025 period. Leading District Jalgaon is known as the "Banana Capital of India," draped in more than 75,000 hectares of plantations. It accounts for approximately 69% of the total banana cultivation area in the state. Jalgaon accounts for approximately 1.26% of global area.
 - **Tamil Nadu:** Leads with approximately 130,000 hectares.
 - **Karnataka:** Roughly 110,000 hectares.
 - **Andhra Pradesh and Maharashtra:** Both utilize approximately 80,000 hectares for banana production.

3.4 Time Series Analysis and Forecasting for Worlds Banana Production

In this section, time series forecasting techniques are applied to historical world banana production data to predict future output levels. The ARIMA (Autoregressive Integrated Moving Average) model is used for forecasting, as it effectively captures patterns such as trend and randomness in time series data. Based on past production data, a Suitable ARIMA model is fitted, and forecasts for the next few years are generated along with confidence intervals to quantify uncertainty.

The time series plot of world's Banana Production (2000-2024) shows there is a clear upward trend in production over time, indicating steady growth in global banana output. Production increases from around 70 million metric tonnes (MMT) in early years to approximately 135+ MMT in recent years. Minor fluctuations are observed around certain periods (e.g., 2011–2013, 2018–2020), but the overall pattern remains increasing. The presence of a strong trend suggests that the series is non-stationary, which justifies differencing in ARIMA modelling.

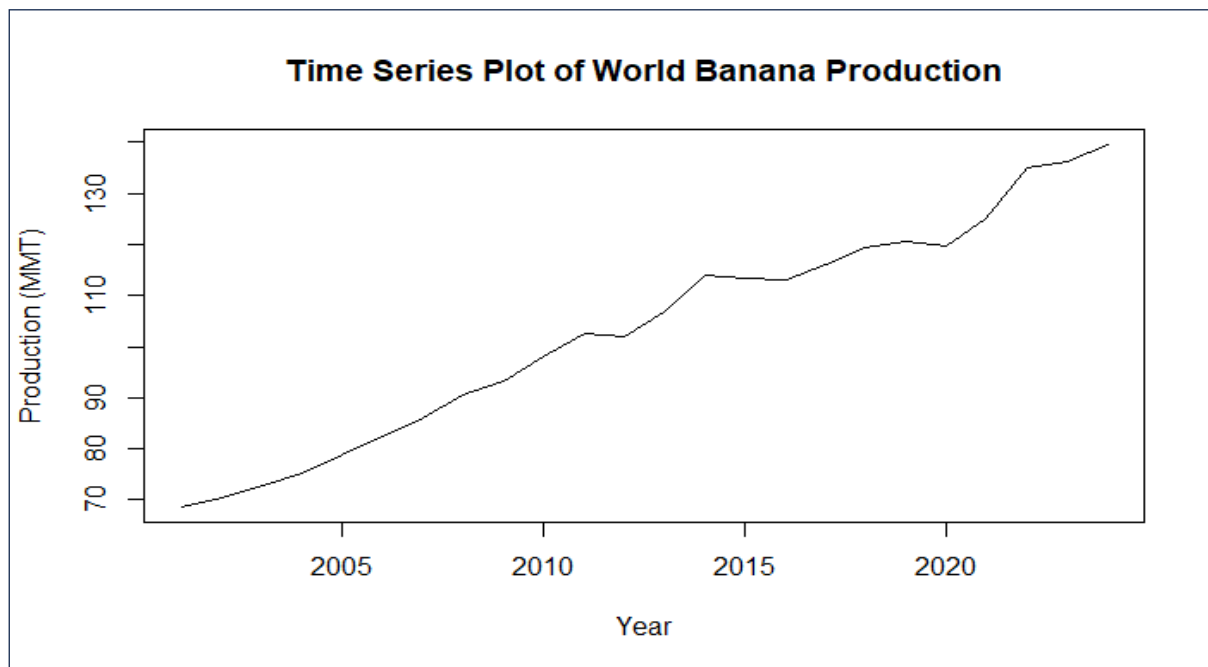


Figure 3.3: Time Series plot of World's Banana Production (2001-2024)

I. Model Fitting:

Series: production_ts

ARIMA(0,1,0) with drift

Coefficients:

drift

3.0870

s.e. 0.5258

sigma² = 6.649: log likelihood = -53.91

AIC=111.82 AICc=112.42 BIC=114.09

Training set error measures:

ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
-3.719145	196.5341	146.9177	-1.162704	10.1418	0.8208851	-0.04006582

The selected model is: ARIMA(0,1,0) with drift

➤ Meaning of the Model

- $d = 1$ → First differencing is applied to remove trend and achieve stationarity.
- $p = 0, q = 0$ → No autoregressive or moving average components.
- Drift term → Represents a constant upward trend in the differenced series.

➤ Estimated Coefficient

- Drift=3.0870 → On average, world banana production increases by about 3.09 MMT per year.

- Standard Error = 0.5258 → The drift estimate is statistically reliable (small error relative to estimate).

➤ Model Adequacy Measures

- $\text{Sigma}^2 = 6.649$ → Indicates variance of residuals (moderate variability).
- Log Likelihood = -53.91 → Used for model comparison.
- AIC = 111.82, BIC = 114.09 → Lower values indicate a better model; suggests this model is reasonably good.

II. Forecast for World's Banana Production in Million Metric Tons

Year	Point Forecast	Lo_80	Hi_80	Lo_95	Hi_95
2025	142.687	139.3825	145.9914	137.6332	147.7407
2026	145.7739	141.1007	150.4472	138.6268	152.921
2027	148.8609	143.1373	154.5844	140.1075	157.6142
2028	151.9478	145.3389	158.5568	141.8403	162.0554
2029	155.0348	147.6457	162.4238	143.7342	166.3353

The forecast results reveal a continued upward trend in world banana production over the next few years. The point forecasts increase steadily from approximately 142.69 million metric tonnes in 2025 to about 155.03 million metric tonnes in 2029, consistent with the historical growth pattern. The forecast plot also shows widening confidence intervals over time, reflecting increasing uncertainty in long-term predictions. The 80% and 95% prediction intervals provide a range within which future production values are likely to fall, ensuring a probabilistic understanding of forecasts. Overall, the model suggests that global banana production will continue to grow steadily, although the degree of uncertainty increases as the forecasting horizon extends.

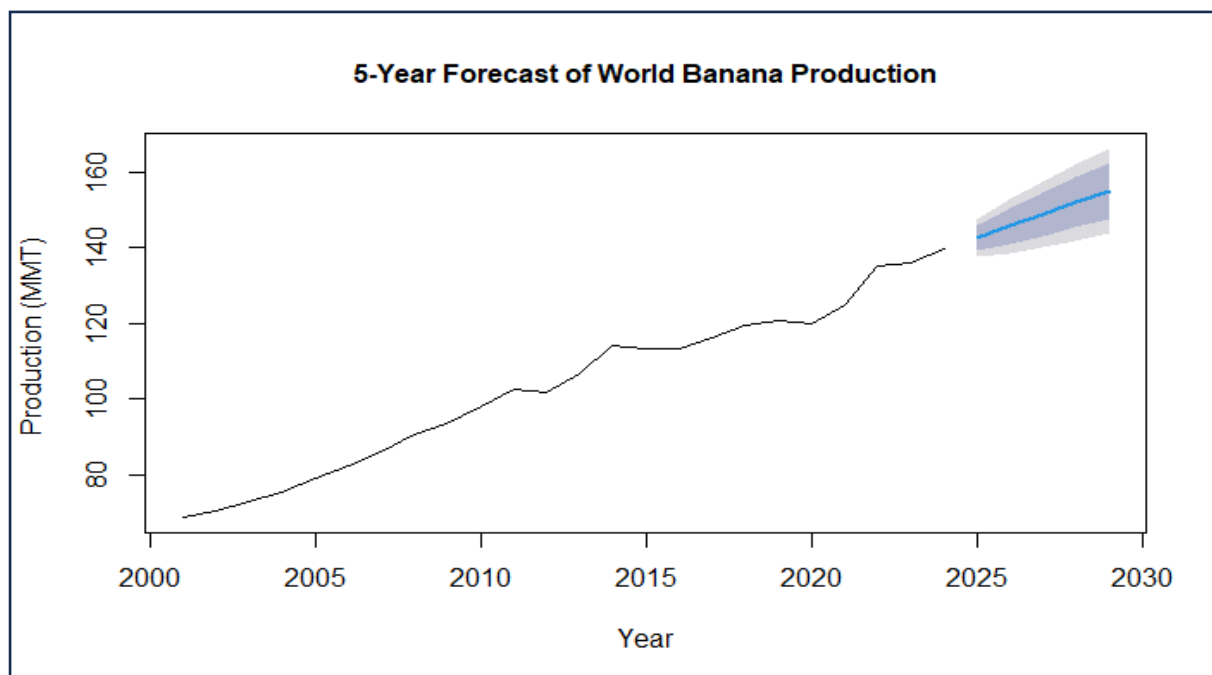


Figure 3.4: Forecast for World's Banana Production

Chapter 4: Banana Production and Productivity in India

4.1 Introduction

While the previous chapter detailed the global landscape of banana production and India's dominant market share, Chapter 4 narrows the focus to the domestic sector. It provides a comprehensive analysis of trends in area, production, and productivity across India, tracking year-wise growth and state-level performance. By examining regional variations and growth rates, this chapter identifies the structural drivers behind the industry's expansion. Additionally, it explores inter-state disparities and the specific factors that have bolstered productivity during the study period. This transition from a global overview to a localized study aims to evaluate the overall efficiency and performance of India's banana sector.

4.2 Data Source and Methodology

The analysis presented in this chapter is based on secondary data on banana cultivation in India. The data related to area under cultivation, total production, and productivity of banana have been collected from published sources such as the Ministry of Agriculture and Farmers Welfare, Government of India, National Horticulture Board (NHB), and Food and Agriculture Organization (FAO). The study covers a period of thirteen years from 2011–12 to 2023–24, which allows for a meaningful assessment of long-term trends.

Table 4.1: Year wise production of Banana in India

Year	Area (^{'000} Hectare)	Production (^{'000} MT)	Productivity (MT/Hectare)
2011-2012	796	28,455	35.80
2012-2013	776	26,509	34.20
2013-2014	803	29,725	37.00
2014-2015	822	29,221	35.50
2015-2016	841	29,135	34.60
2016-2017	860	30,477	35.40
2017-2018	884	30,808	34.90
2018-2019	866	30,460	35.20
2019-2020	897	32,597	36.34
2020-2021	924	33,062	35.78
2021-2022	963	34,528	35.87
2022-2023	994	36,614	36.83
2023-2024	1000	37,807	37.81
CAGR (%)	2.12	2.51	0.37

Table 4.1 presents the year-wise data on area (in thousand hectares), production (in thousand metric tonnes), and productivity (metric tonnes per hectare) of banana in India. Area refers to the total land under banana cultivation during a given year, production

denotes the total output harvested, and productivity is calculated as the ratio of production to area, representing yield per hectare.

4.3 Trend in Area, Production and Productivity

In this section we have studied the trend in Area production and productivity of banana of Indian states. We can see the Table 4.2 for a last decade overview.

Table 4.2: State-wise Area and Production of Banana in India

State	2015			2024		
	Area	Production	Productivity	Area	Production	Productivity
Andhra Pradesh	75.72	3570.62	47.16	110.82	6734.44	60.77
Arunachal Pradesh	5.42	31.64	5.84	3.63	14.49	3.99
Assam	51.10	882.71	17.27	53.86	1022.51	18.98
Bihar	34.80	1535.30	44.12	42.69	1920.96	45.00
Chhattisgarh	25.76	587.42	22.80	23.91	576.47	24.11
Gujarat	64.69	4185.52	64.70	59.9	4025.28	67.20
Himachal Pradesh	0.09	0.42	4.67	0.06	0.21	3.50
Jharkhand	12.53	33.28	2.66	5.77	42.38	7.34
Karnataka	96.63	2370.95	24.54	45.71	1248.89	27.32
Kerala	84.56	1292.41	15.28	47.95	485.2	10.12
Madhya Pradesh	28.35	1758.05	62.01	34.87	2463.36	70.64
Maharashtra	69.55	3025.15	43.50	123.34	7229.86	58.62
Manipur	6.95	93.95	13.52	6.12	81.41	13.30
Meghalaya	7.11	88.71	12.48	7.95	102.88	12.94
Mizoram	10.91	141.03	12.93	11.68	140.5	12.03
Nagaland	7.25	108.51	14.97	7.3	95.27	13.05
Odisha	24.47	462.71	18.91	23.55	508.67	21.60
Punjab	0.11	6.43	58.45	0.24	14.69	61.21
Rajasthan	0.03	0.41	13.67	0.04	0.35	8.75
Sikkim	1.15	3.56	3.10	1.31	5.56	4.24
Tamil Nadu	94.61	4331.65	45.78	114.77	4905.48	42.74
Telangana	4.65	183.70	39.51	1.83	74.68	40.81
Tripura	14.62	153.62	10.51	11.68	114.34	9.79
Uttar Pradesh	67.00	3061.21	45.69	93.29	4757.84	51.00
West Bengal	48.07	1172.34	24.39	65.5	1434.15	21.90
India	836.13	29081.30	34.78	897.77	37999.87	42.32

In figure 4.1 Pareto-style bar chart visualizes India's state-wise banana production in 2024 (in thousand tonnes), with bars sorted descending and a cumulative percentage line showing the 80/20 rule concentration. Maharashtra leads overwhelmingly, followed by Andhra Pradesh and Tamil Nadu, aligning with APEDA data where top states produce ~67-79% of national output (~35-38 million tonnes total). Top 3 states (Maharashtra ~7,230; Andhra Pradesh ~6,734; Tamil Nadu ~4,905) account for ~50% cumulative

share, emphasizing regional dominance in tropical/semi-arid areas. Uttar Pradesh (~4,757), Gujarat (~4,025), and Madhya Pradesh (~2,463) push cumulative to ~80%, with "Other" states at 4-20%.

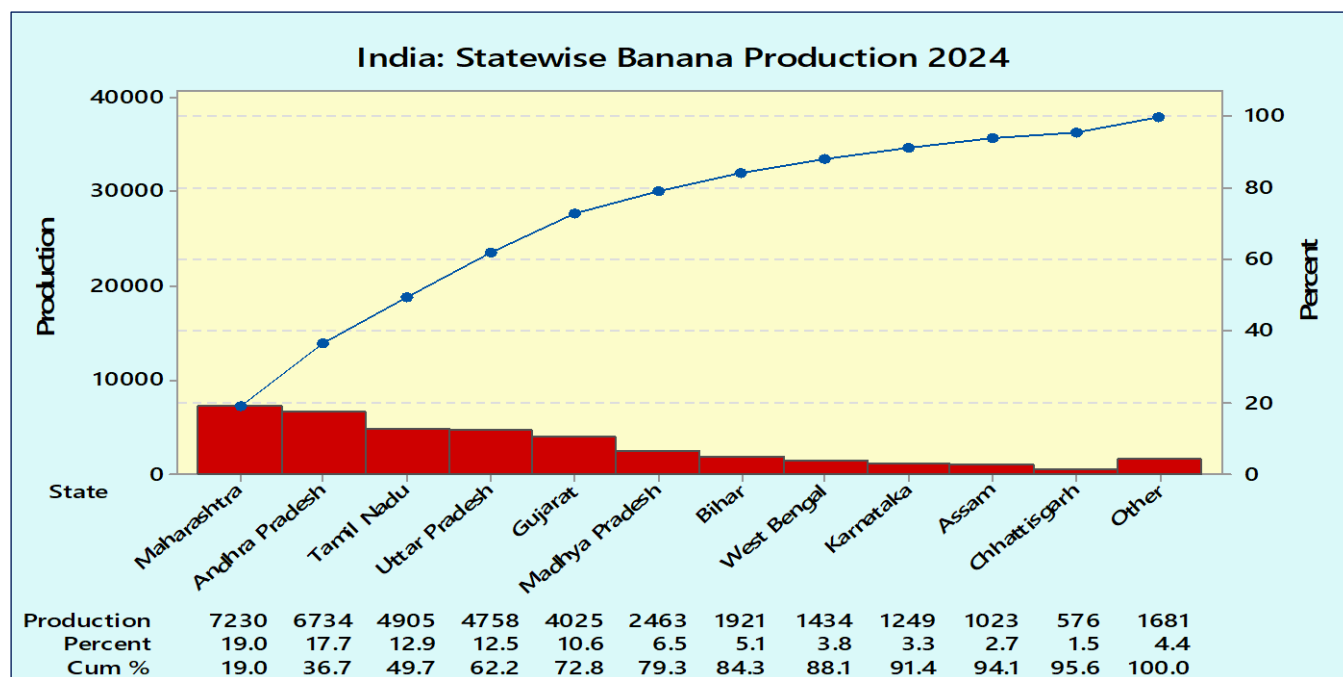


Figure 4.1 Pareto Chart of Banana Production of Indian States.

4.4 Trend in Area under cultivation

Banana cultivation area in India has shown significant expansion over the past two decades, driven by increasing commercial demand and improved farming practices. From 2005 to 2024, major producing states demonstrated varied growth trajectories, with Maharashtra maintaining leadership while northern states like Uttar Pradesh emerged as high-growth regions. Table 4.3 and Figure 4.2 illustrate these dynamics across key states, highlighting both absolute expansions and recent percentage growth rates, reflecting shifting regional priorities in banana production. This trend underscores banana's rising importance in India's horticultural landscape. This Table tracks area expansion in '000 hectares for six major banana-producing states over two decades. However, the growth rates presented are calculated specifically for the period 2020–2024, showing recent trends in expansion.

Uttar Pradesh shows a strong relative increase, while Maharashtra leads in absolute terms. Other states like Madhya Pradesh and Tamil Nadu also exhibit moderate growth during this period. The accompanying bar chart (Fig. 4.2) visualizes these trends with clustered bars for each year (2005, 2010, 2015, 2020, 2024) across the states, plus a separate growth rate bar section. It emphasizes Maharashtra's consistent leadership and Uttar Pradesh's steep rise, using distinct colours for each state to illustrate divergent trajectories in banana cultivation area.

Table 4.3: Year Wise Growth Rate of Area Under Cultivation

State	Area ('000 Hectares)					Growth Rate % (2020-24)
	2005	2010	2015	2020	2024	
Maharashtra	67.40	80.50	69.55	84.26	123.34	46.38
Andhra Pradesh	70.50	77.50	75.72	97.25	110.82	13.95
Tamil Nadu	96.30	123.80	94.61	97.64	114.77	17.54
Uttar Pradesh	24.90	31.20	67.00	73.78	93.29	26.44
Gujarat	47.90	63.50	64.69	59.26	59.9	1.08
Madhya Pradesh	12.60	36.90	28.35	29.48	34.87	18.28
Total	319.60	413.40	399.92	441.67	536.99	21.58

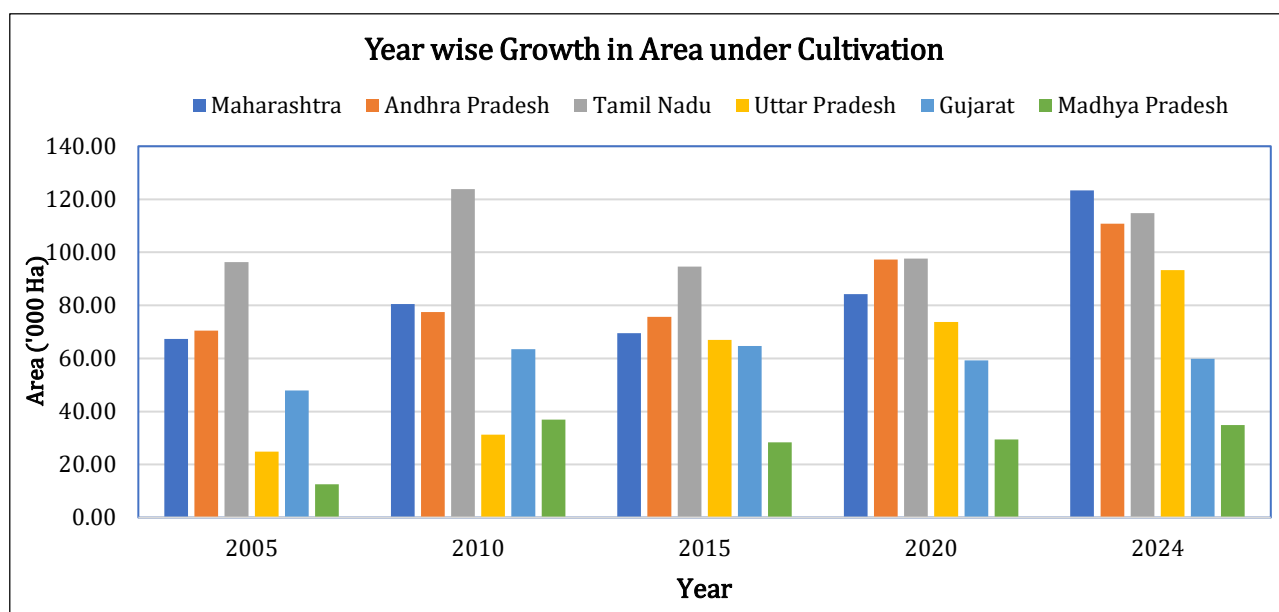


Figure 4.2: Year wise Growth Rate of Area under Cultivation of Banana in India

4.5 Trend in Production

Banana production in India has witnessed robust growth from 2005-2024, fuelled by expanded cultivation areas, improved varieties, and better management practices. This section analyses production trends across major states to identify growth patterns, regional leaders, and factors driving output expansion. Table 4.4 and Figure 4.3 provide a comprehensive view of production changes and growth rates, highlighting banana's pivotal role in India's fruit economy.

This Table presents the banana production (in '000 tonnes) for major states over the period 2005 to 2024. It highlights long-term production trends along with recent growth rates calculated specifically for 2020–2024, reflecting short-term changes in output. Uttar Pradesh shows a strong increase in production with a high recent growth rate, indicating rapid expansion in the last four years. Gujarat exhibits relatively Stable production with minimal growth, while Madhya Pradesh demonstrates moderate improvement during the same period.

Table 4.4: Year wise Growth Rate of Banana Production in India

State	Production ('000 Tons)					Growth Rate % (2020-24)
	2005	2010	2015	2020	2024	
Maharashtra	4535.00	4303.00	3025.15	4223.05	7229.86	71.20
Andhra Pradesh	1438.70	2774.80	3570.62	5834.70	6734.44	15.42
Tamil Nadu	3462.00	3896.83	4331.65	3895.65	4905.48	25.92
Uttar Pradesh	1100.00	1346.10	3061.21	3387.51	4757.84	40.45
Gujarat	1979.00	3978.00	4185.52	3907.21	4025.28	3.02
Madhya Pradesh	750.00	1719.60	1758.05	2025.46	2463.36	21.62
Total	15269.70	20028.33	21947.20	25293.58	32140.26	27.07

Fig. 4.3 shows change in Banana Production - The clustered bar chart tracks production trajectories across years, with Maharashtra maintaining dominance and Andhra Pradesh/Uttar Pradesh showing steep upward trends. Growth rate bars emphasize the exceptional performance of emerging producers.

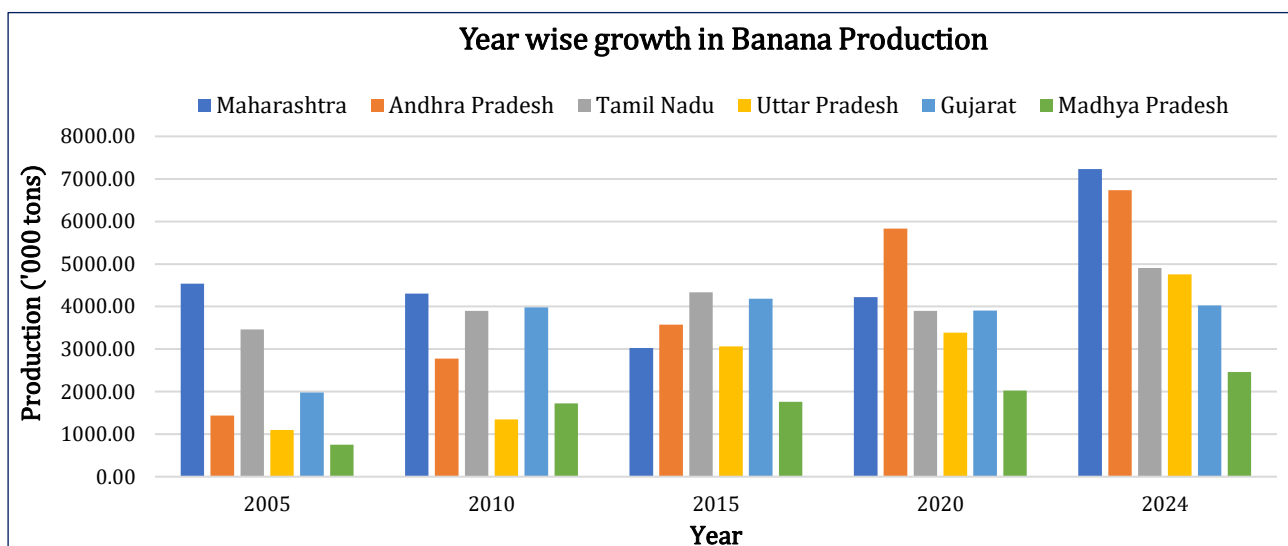


Figure 4.3: Year wise growth in Banana Production

4.6 Trend in Productivity

Productivity trends reveal technological advancements and agronomic improvements in banana cultivation across major states from 2005-2024. This section examines yield per hectare (MT/ha) patterns to assess efficiency gains, with Table 5 and Figure 6 highlighting disparities and progress. The analysis identifies productivity leaders and growth drivers essential for sustainable production enhancement.

Table 4.5 shows productivity (MT/ha) evolution, with Andhra Pradesh surging from 20.4 in 2005 to 60.8 in 2024 (197.8% growth), Madhya Pradesh from 59.5 to 70.6 (18.7%), and Gujarat from 41.3 to 67.2 (62.7%). Uttar Pradesh and Tamil Nadu recorded 15.5%

and 18.9% gains respectively, while Maharashtra saw a slight decline (-12.9%) from 67.3 to 58.6 MT/ha.

Table 4.5: Growth Rate of Banana Productivity

State	Productivity (Ton/Ha)					Growth Rate (%) (2020-24)
	2005	2010	2015	2020	2024	
Maharashtra	67.28	53.45	43.50	50.12	58.62	16.96
Andhra Pradesh	20.41	35.80	47.16	60.00	60.77	1.29
Tamil Nadu	35.95	31.48	45.78	39.90	42.74	7.13
Uttar Pradesh	44.18	43.14	45.69	45.91	51.00	11.08
Gujarat	41.32	62.65	64.70	65.93	67.20	1.92
Madhya Pradesh	59.52	46.60	62.01	68.71	70.64	2.82

Productivity wise change in figure 4.4 the clustered bar graph depicts productivity trends over key years, with Madhya Pradesh and Gujarat consistently high (60-70 MT/ha). Andhra Pradesh's sharp rise stands out, while growth rate bars underscore yield efficiency improvements.

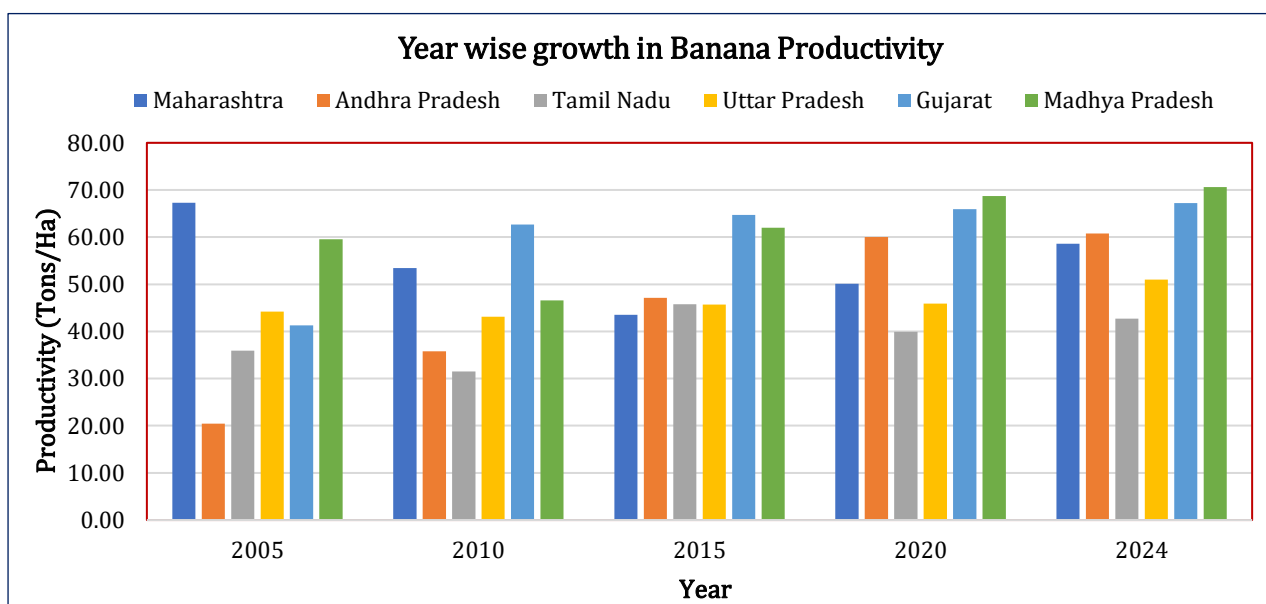


Figure 4.4: Trend of Banana Productivity in Indian States

Key Insights

- **Yield Champion:** Madhya Pradesh maintains top productivity (~70 MT/ha)
- **Biggest Improver:** Andhra Pradesh (nearly tripled from 20.4 to 60.8 MT/ha)
- **Stable Performers:** Gujarat (62.7% growth) and Uttar Pradesh (15.5%)
- **Area for Improvement:** Maharashtra's slight decline suggests need for yield-focused interventions

These patterns indicate technology adoption is accelerating in emerging states.

Chapter 5: Time Series Analysis for Major Banana Producing States in India

5.1 Introduction

Time series analysis is an important statistical tool used to study data collected over time and to identify underlying patterns such as trends, seasonal variations, and random fluctuations. In the context of agricultural production, time series techniques are particularly useful for understanding historical production behavior and for forecasting future output, which supports planning and policy-making.

India is one of the largest producers of bananas in the world, and the crop plays a significant role in the agricultural economy of the country. Banana production varies widely across different states due to differences in climatic conditions, soil fertility, irrigation facilities, and agricultural practices. Understanding these variations over time is essential for effective planning, policy-making, and ensuring sustainable agricultural growth.

In this chapter, we focus on the time series analysis of banana production/productivity for the top six banana-producing states in India. These states are:

- Maharashtra
- Andhra Pradesh
- Tamil Nadu
- Uttar Pradesh
- Gujarat
- Madhya Pradesh

These states are selected based on their high contribution to banana production and their importance in India's agricultural output.

5.2 Time Series Analysis and Forecasting for India

In this section, we present the time series analysis and future projections for the chosen variables within the Indian context. We begin by examining the historical data through a Time Series Plot to identify underlying patterns, followed by the rigorous process of Model Fitting to ensure statistical accuracy. Finally, we provide a five-year forecast, complemented by a visualization of these predicted values to illustrate the expected trend and potential growth.

The time-series plot of India's banana production from 2005 to 2024 reveals a robust and near-linear upward trend, nearly doubling in volume over two decades. This consistent growth justifies the use of an ARIMA(0,1,0) model with a significant drift coefficient of 1,019, indicating a reliable annual increase.

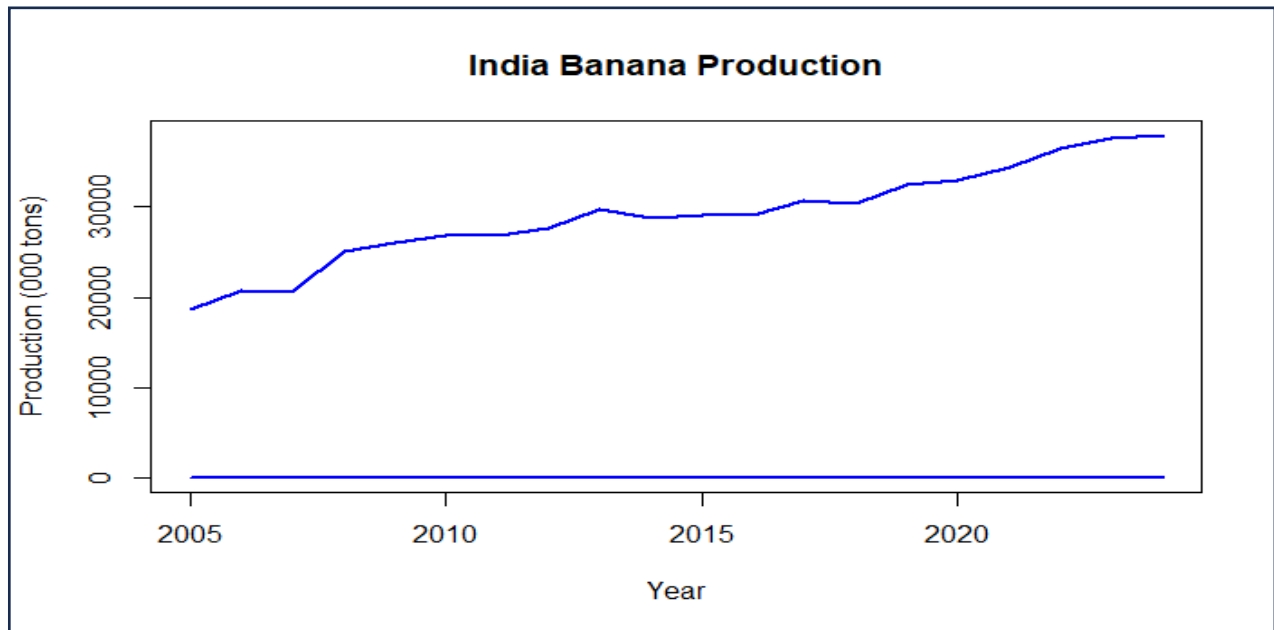


Figure 5.1: Time Series plot of year wise banana production of India

I. Model Fitting:

ARIMA(0,1,0) with drift

Coefficients:

drift

1019.2379

s.e. 275.9932

$\sigma^2 = 1527685$: log likelihood = -161.72

AIC=327.44 AICc=328.19 BIC=329.33

Training Set Measure:

ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
0.8807552	1172.568	888.5452	0.033995	3.208103	0.760097	-0.32529

The model demonstrates exceptional accuracy with a MAPE of only 3.2%, suggesting that historical patterns are highly Stable and well-captured. Consequently, the five-year forecast predicts production will reach 43,096 units by 2029, though widening confidence intervals reflect the inherent uncertainty of long-term agricultural projections. Overall, these findings highlight a strong, predicTable expansion in the sector that is statistically supported by both historical performance and future point estimates. This output shows the results of a time-series analysis for India's banana production using an ARIMA(0,1,0) model with drift. This specific model is essentially a Random Walk with Drift.

The forecast values for India's banana production indicate a steady, linear growth trend through 2029, driven by a consistent annual increase of 1,019 units. While the point forecast for 2029 stands at 43,096 units, the widening confidence intervals (reaching a

95% range of 37,679 to 48,513) reflect the inherent uncertainty of long-term agricultural forecasting. These results suggest a robust and positive outlook for the sector, assuming the historical growth factors continue to remain Stable.

II. Forecast values on India's Banana Production ('000 Tons)

Year	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
2025	39019	37435	40603	36597	41442
2026	40038	37798	42278	36612	43464
2027	41058	38314	43801	36862	45253
2028	42077	38909	45245	37232	46922
2029	43096	39554	46638	37679	48513

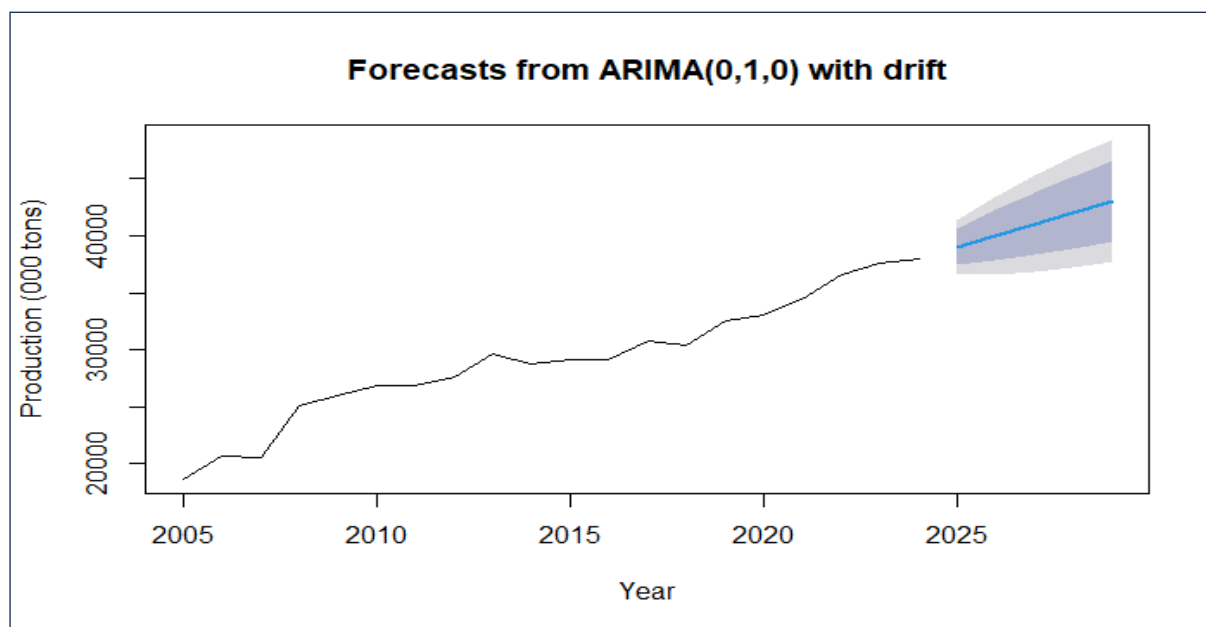


Figure 5.2: Forecast values of India's Banana Production

5.3 Data Discription of Statewise Banana Production

The dataset provides a comprehensive annual record of banana production in '000 tons across 24 Indian states and union territories from 2005 to 2024. It captures a significant national growth trend, with total production nearly doubling from 18,634 thousand tons in 2005 to nearly 38,000 thousand tons by 2024. The data highlights the emergence of major production hubs like Andhra Pradesh, Maharashtra, and Uttar Pradesh, while also documenting the high year-on-year volatility in smaller contributing regions

5.4 Time Series Analysis and Forecasting for Maharashtra

The banana production data for Maharashtra from 2005 to 2024 shows a general upward trend, meaning production has increased over time, especially in recent years. However, the data also has some ups and downs, indicating that production did not grow steadily and was affected by certain factors in between. There is a noticeable drop in production around 2013 to 2016, followed by a gradual recovery. After 2020, the increase becomes

much faster and more consistent, reaching the highest level in 2024. Since the data is yearly, there is no seasonal pattern, but overall, it shows growth with some fluctuations. Significant fluctuations occur between 2005 and 2015, with a notable dip around 2007 and 2015. Since 2016, there has been a sharp, consistent upward trend, with production reaching record highs (above 7,000 thousand tons) by the end of the observed period.

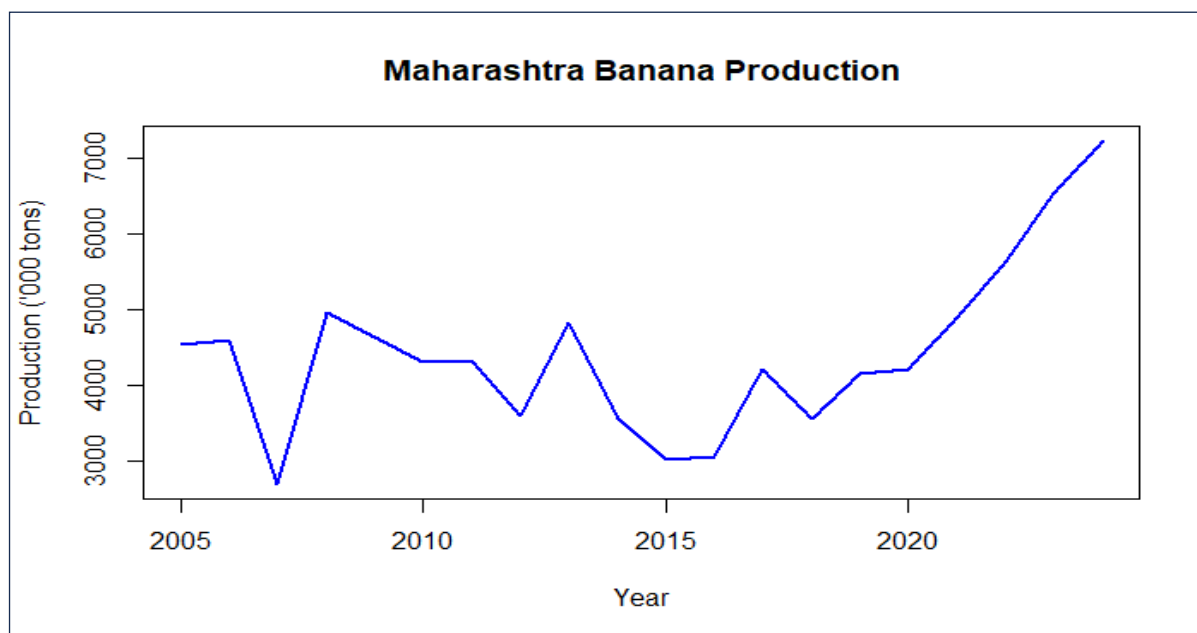


Figure 5.3: Time series Plot of Maharashtra's Banana Production

The analysis uses an ARIMA(1,0,0) model, also known as a First-order Autoregressive (AR1) model.

I. Model Fitting

Series: state_ts

ARIMA(1,0,0) with non-zero mean

Coefficients:

ar1 mean

0.6564 4662.4556

s.e. 0.2013 557.0875

sigma² = 874581: log likelihood = -164.42

AIC=334.84 AICc=336.34 BIC=337.83

Training Set Measures:

ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
5.748188	887.1993	714.2599	-4.28943	17.523	0.9583657	-0.08908

The ARIMA(1,0,0) model, or first-order autoregressive model, indicates that production in Maharashtra is strongly influenced by the previous year's value, as shown by the significant ar1 coefficient of 0.6564. With a non-zero mean of 4662.46, the series suggests Stable long-term average production level rather than the aggressive linear growth seen

in the national data. While the model is statistically sound, the MAPE of 17.5 indicates a higher degree of volatility and error compared to the India-wide model, likely due to more localized fluctuations in regional agricultural output.

II. Forecast for Maharashtra:

Year	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
2025	6347.574	5149.079	7546.068	4514.635	8180.513
2026	5768.484	4334.895	7202.074	3575.998	7960.971
2027	5388.399	3864.669	6912.129	3058.055	7718.743
2028	5138.929	3577.970	6699.888	2751.649	7526.209
2029	4975.189	3398.464	6551.915	2563.795	7386.583

The Table provides specific "Point Forecasts" for the next five years. Interestingly, the model predicts a gradual decline from the recent peak toward the long-term mean. Our ARIMA(1,0,0) model for Maharashtra predicts a decline from recent peaks back toward the long-term average of ~4,662 thousand tons. This "settling down" reflects the state's different market position:

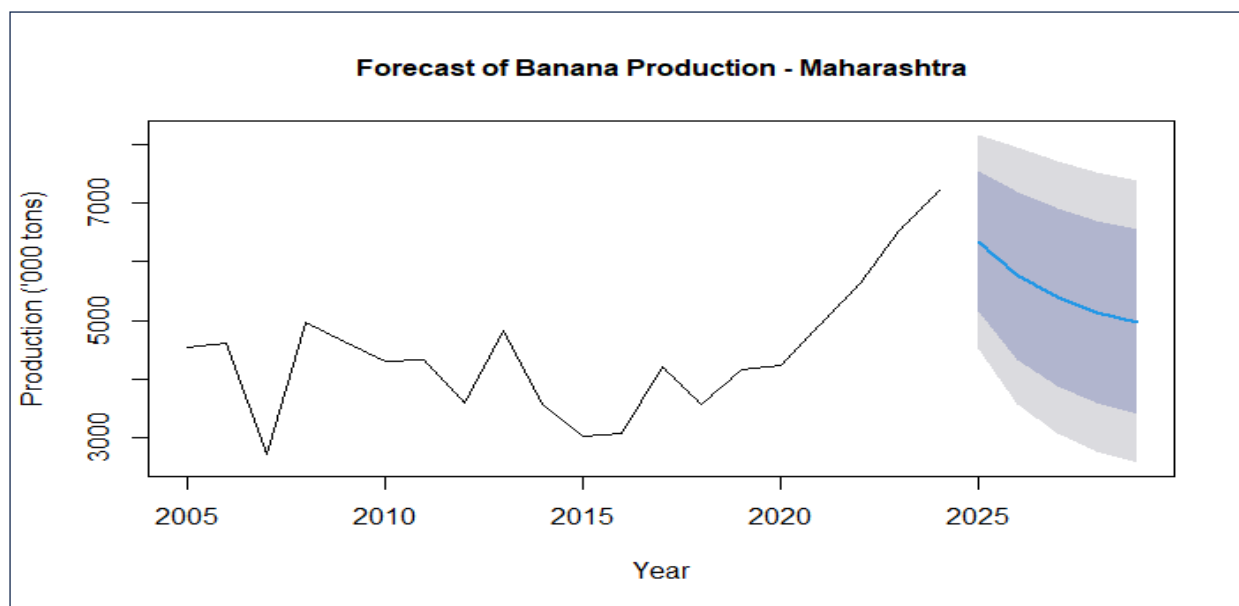


Figure 5.4: Forecast for Maharashtra Banana Production

- **Market Saturation & Volatility:** Maharashtra's production (hubs like Jalgaon and Solapur) is already very high and highly matured. The "dip" in our forecast often represents a natural market correction after record-breaking years, where oversupply can lead to lower prices and reduced planting in the following seasons.
- **Climatic Sensitivity:** Unlike Andhra Pradesh's coastal advantages, Maharashtra's banana belts are more susceptible to heatwaves and fluctuating water levels in dams. The model's tendency to return to the "mean" accounts for these cyclical environmental shocks.
- **Focus on Quality over Quantity:** In 2026, Maharashtra is shifting focus toward Premium Grade exports (18cm+ finger length). This often means farmers

prioritize the quality of the bunch rather than just increasing the total tonnage, which explains a stabilizing production volume.

For more better prediction we take a data from the year 2015 – 24.

I. Model Fitting:

Series: ts_data

ARIMA(0,1,0) with drift

Coefficients:

drift

467.1900

s.e. 171.9967

$\sigma^2 = 299524$: log likelihood = -68.99

AIC=141.97 AICc=143.97 BIC=142.36

Training set error measures:

ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
0.2557959	489.5089	386.5958	-1.656853	9.214556	0.6327896	-0.4334474

The fitted ARIMA(0,1,0) with drift model indicates that the banana production data is non-stationary and has been differenced once to achieve stationarity. Since there are no autoregressive or moving average components, the model essentially behaves like a random walk with a constant trend. The estimated drift coefficient is 467.19, which implies that banana production in Maharashtra is expected to increase by approximately 467 ('000 tons) each year on average. This reflects a consistent upward trend in production over time.

In terms of model performance, the error measures suggest that the model provides a reasonably good fit. The Mean Error (ME) is very close to zero, indicating that the model is unbiased. The RMSE (489.51) and MAE (386.60) show moderate prediction errors, while the MAPE of 9.21% suggests that the forecasts deviate by about 9% on average, which is acceptable for practical purposes. Additionally, the MASE value of 0.63, being less than 1, indicates that the model performs better than a naive forecasting approach. Although the residual autocorrelation (ACF1 = -0.43) shows a slight remaining pattern, it is not severe enough to invalidate the model.

II. Forecast of Maharashtra Banana Production ('000 Tons)

Year	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
2025	7697.05	6995.673	8398.427	6624.386	8769.714
2026	8164.24	7172.343	9156.137	6647.264	9681.216
2027	8631.43	7416.609	9846.251	6773.522	10489.338

The forecast for the next three years shows a steady increase in banana production in Maharashtra. Production is expected to rise from 7697 ('000 tons) in 2025 to 8631 ('000 tons) in 2027, indicating consistent growth driven by the positive drift in the model. The

prediction intervals suggest that while the central trend is upward, there is some uncertainty in the exact values, and this uncertainty increases slightly over time. Overall, the short-term forecast is Stable and reliable, making it useful for planning and decision-making.

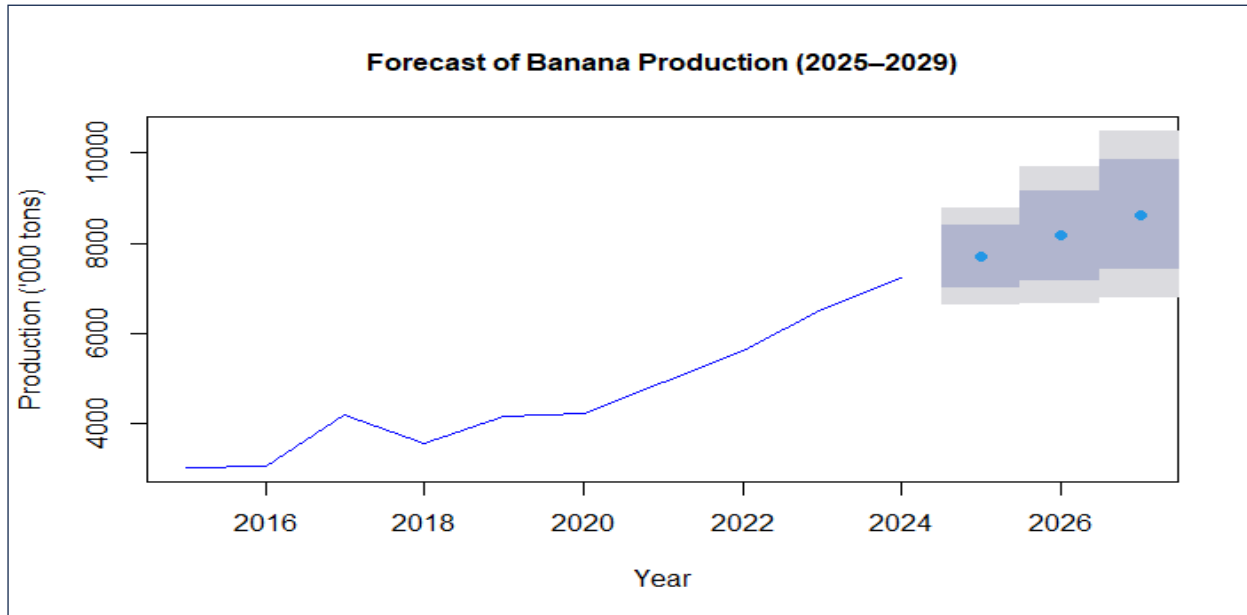


Figure 5.5: Forecast values of Maharashtra Banana Production ('000 Tons)

5.5 Time Series Analysis and Forecasting for Andhra Pradesh

In this section analysis focuses on banana production in Andhra Pradesh, using a more dynamic model compared to the Maharashtra analysis to account for a strong, consistent growth trend. In figure 5.3 The time series plot of banana production in Andhra Pradesh from 2005 to 2024 shows a clear upward trend over time, indicating continuous growth in production.

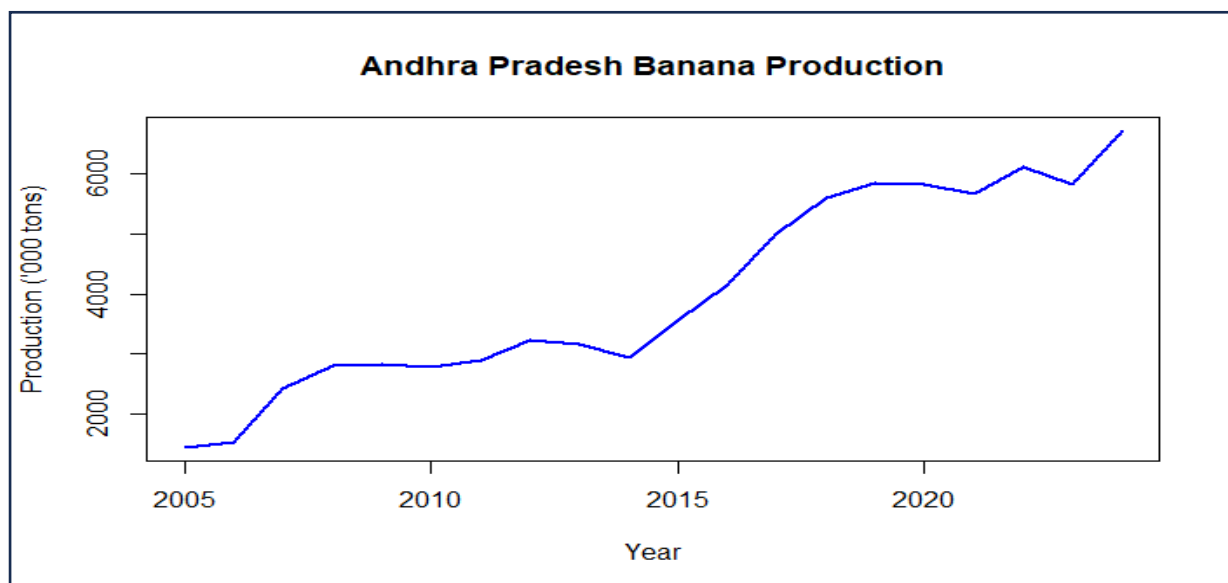


Figure 5.6: Time Series plot for Andhra Pradesh Banana Production

In the initial years (2005–2010), production increases steadily with slight fluctuations. During 2011–2014, there is a small decline and instability, but this is followed by a strong and consistent rise from 2015 onwards. The period after 2016 shows significant growth, with production reaching its highest levels in recent years (2022–2024).

Although there are minor fluctuations in between, the overall pattern is smooth and steadily increasing, unlike Maharashtra which had more volatility. Since the data is yearly, there is no seasonal pattern, and the series mainly reflects a strong positive growth trend with limited variability

I. Model Fitting:

Series: state_ts

ARIMA (0,1,0) with drift

Coefficients:

drift

278.7232

s.e. 85.8613

$\sigma^2 = 147852$: log likelihood = -139.53

AIC=283.07 AICC = 283.82 BIC=284.96

Training set error measures:

ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
0.05799881	364.7834	309.8832	-0.6290855	8.317486	0.8533806	0.0398376

The model chosen here is a Random Walk with Drift.

- The "1" in (0,1,0): This means the data was differenced. Instead of looking at the raw production numbers, the model looks at the change in production from year to year. This is necessary because the data has a strong trend and is not stationary.
- The Drift (278.72): This is the most important coefficient. It indicates that, on average, banana production in Andhra Pradesh increases by approximately 278.72 thousand tons every year.
- Model Accuracy: The MAPE (Mean Absolute Percentage Error) is 8.31%, which is quite good for agricultural forecasting, suggesting the model's predictions are historically reliable within a small margin of error.

II. Forecast for Andhra Pradesh Banana Production ('000 Tons):

Year	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
2025	7013.163	6520.387	7505.94	6259.527	7766.8
2026	7291.886	6594.995	7988.777	6226.083	8357.689
2027	7570.609	6717.096	8424.123	6265.273	8875.946
2028	7849.333	6863.78	8834.886	6342.06	9356.606
2029	8128.056	7026.174	9229.937	6442.873	9813.23

Our model predicts a steady climb toward 8,128 thousand tons by 2029. This aggressive growth is supported by several current factors:

- **Dominance of the G9 Variety:** Andhra Pradesh (specifically the Anantapur cluster) has transitioned heavily to the Grand Naine (G9) variety. These are tissue-cultured, disease-resistant plants that yield significantly more fruit per acre than traditional varieties.
- **Export Infrastructure:** As of early 2026, Andhra Pradesh is a key player in India's push to reach \$1 billion in banana exports. The state has an advantage in "cold-weather handling"—its fruit typically survives long-distance shipping (to Russia and Europe) better than fruit from other regions.
- **Government "Double Digit Growth" Plan:** The state has implemented the State Agriculture Plan (SAP), focusing on drip irrigation and price stabilization funds (ranging from ₹500 to ₹3,000 crore) to ensure farmers don't abandon the crop during price dips.

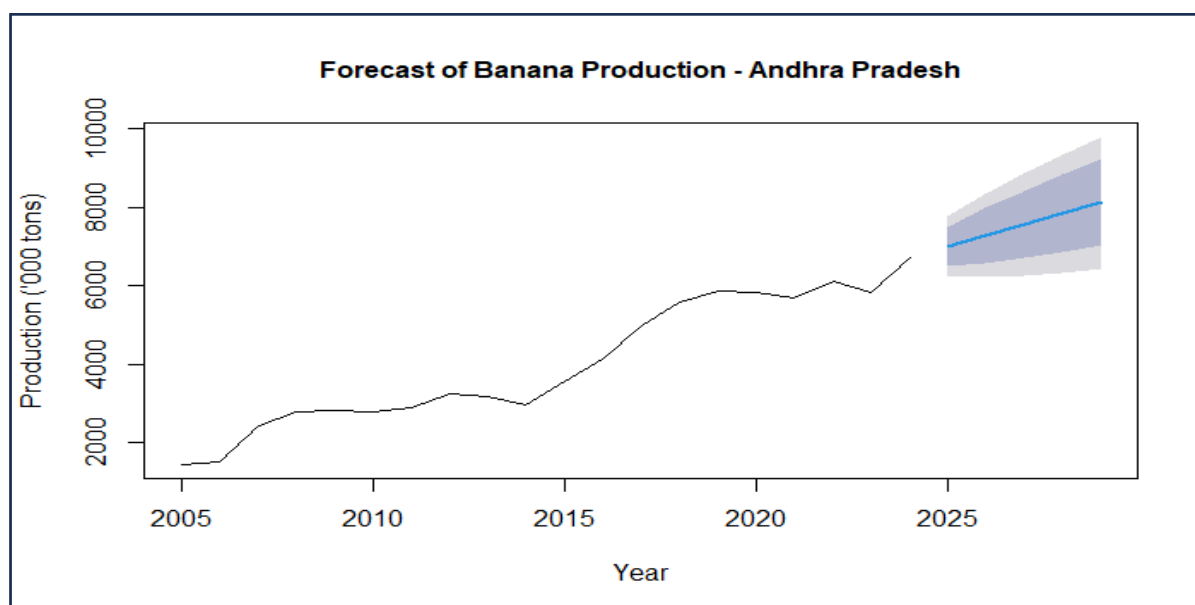


Figure 5.7: Forecast values of Andhra Pradesh Banana Production

5.6 Time Series Analysis and Forecasting for Tamil-Nadu

This analysis covers banana production in Tamil Nadu, using an ARIMA(1,0,0) model—similar to the one used for Maharashtra—to capture the state's cyclical production patterns. In figure 5.5 the time series plot for banana production in Tamil Nadu (2005–2024) illustrates a highly volatile and cyclical pattern, characterized by significant peaks followed by sharp contractions.

The data shows an initial period of robust growth, reaching an all-time peak of approximately 5,650 thousand tons around 2013. However, this was followed by a dramatic five-year decline that bottomed out in 2017–2018, likely due to severe regional drought conditions and climate-related stressors. Since 2019, the state has entered a recovery phase, with production numbers trending upward again toward the 4,900-thousand-ton mark.

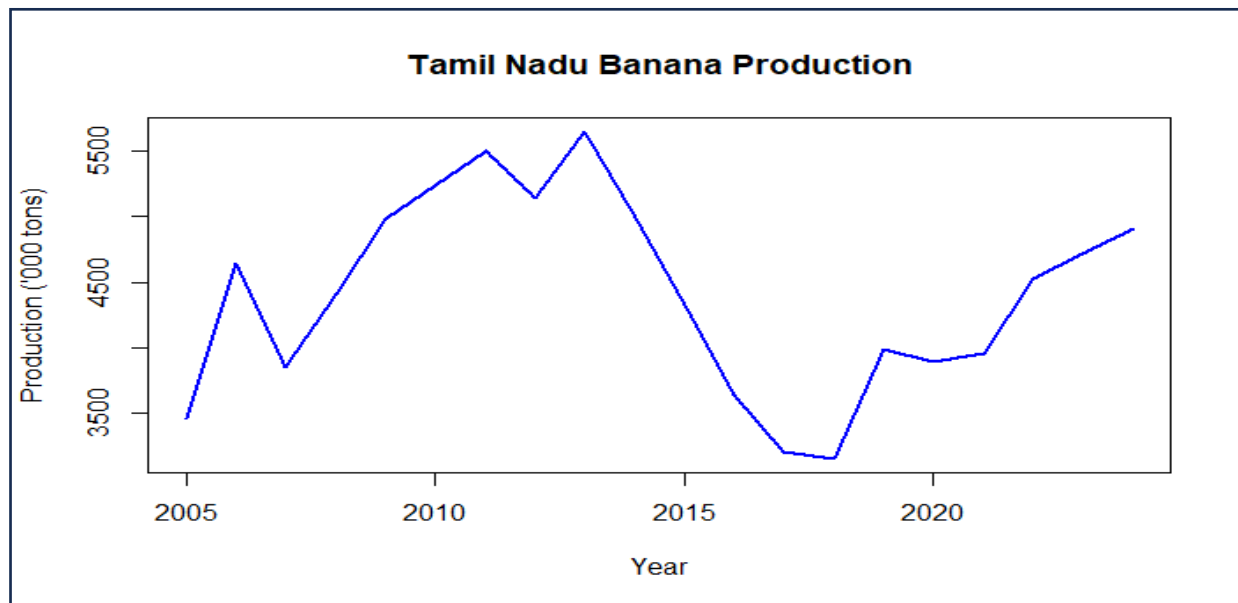


Figure 5.8: Time Series Plot for Banana Production in Tamil Nadu.

I. Model Fitting:

Series: state_ts

ARIMA (1,0,0) with non-zero mean

Coefficients:

ar1	mean
0.7090	4365.1704
s.e. 0.1552	354.5789

$\sigma^2 = 292499$: log likelihood = -153.54

AIC=313.07 AICc=314.57 BIC=316.06

Training set error measures:

ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
45.33712	513.0778	461.3715	-0.4259613	10.89079	0.9814447	-0.01569972

The "AR1" Term (0.7090): This is a strong coefficient. It indicates that about 71% of the previous year's performance influences the current year. This is higher than what you found for Maharashtra, suggesting that Tamil Nadu has a slightly stronger "momentum" or persistence in its trends.

- Mean (4365.17): The model identifies the long-term Stable average for the state at roughly 4,365 thousand tons.
- Error Metrics: The MAPE is 10.89%. This is slightly higher than Andhra Pradesh's model, likely due to the extreme fluctuations seen between 2013 and 2017, which are harder for a simple AR model to predict perfectly.

II. Forecasting for Tamil Nadu Banana Production ('000 Tons)

Year	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
2025	4748.268	4055.165	5441.372	3688.258	5808.279
2026	4636.800	3787.153	5486.447	3337.377	5936.222
2027	4557.765	3639.444	5476.085	3153.315	5962.215
2028	4501.726	3550.753	5452.700	3047.338	5956.115
2029	4461.993	3495.021	5428.966	2983.137	5940.850

Similar to the Maharashtra model, the Tamil Nadu forecast shows Mean Reversion. Because the production is currently above the long-term average, the model predicts it will slowly settle back down

- The 2016-2017 Drought: Tamil Nadu faced one of its worst droughts in decades during that period, which explains the sharp valley in your graph.
- Crop Diversification: Unlike Andhra Pradesh's heavy focus on G9, Tamil Nadu grows many traditional varieties (like Virupakshi and Sirumalai) which have lower yields but higher market value per ton. This often leads to lower total tonnage compared to "growth-focused" states.
- Climate Resilience: The current recovery reflects improved micro-irrigation and the adoption of more resilient suckers after the 2017 crash.

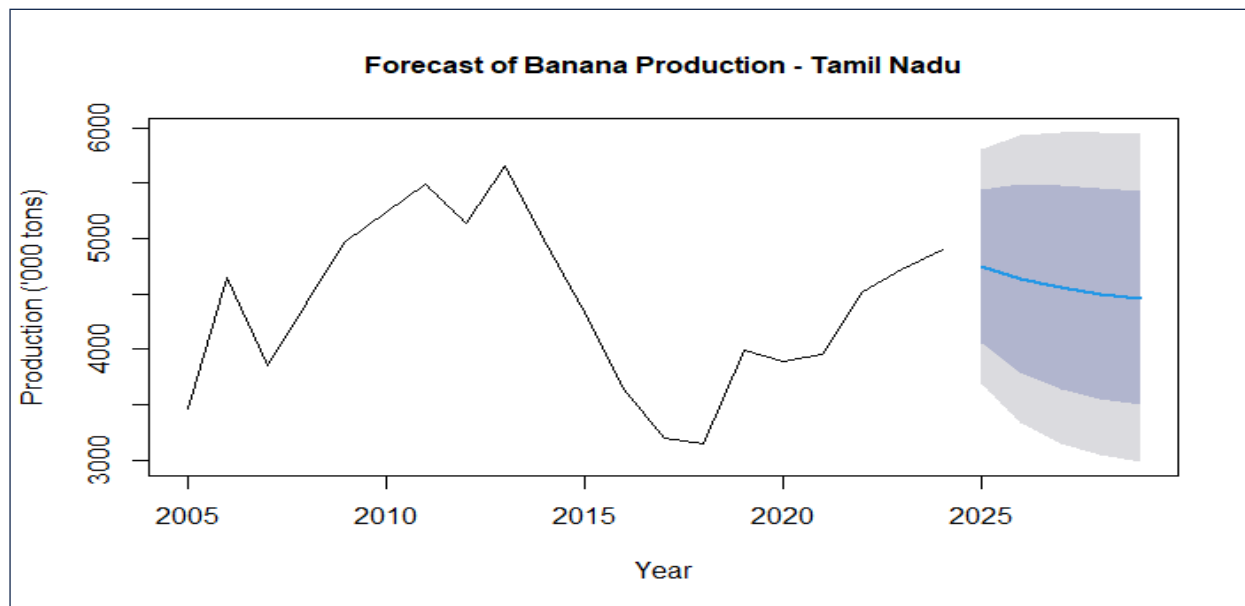


Figure 5.9: Forecast for Banana production In Tamil-Nadu

5.7 Time Series Analysis and Forecasting for Pradesh:

The time series plot of banana production in Uttar Pradesh shows a strong upward trend over the study period, indicating significant growth in production. In the initial years (around 2004–2011), production remains relatively low and fluctuates between

approximately 900 to 1300 ('000 tons), reflecting instability and slow growth. However, after 2012, a noticeable increase begins, with production rising steadily.

A sharp jump is observed around 2014–2015, where production increases rapidly to above 3000 ('000 tons), suggesting a structural change or improvement in cultivation practices. Following this period, production continues to grow at a more gradual and Stable rate, reaching its peak in the most recent years (around 2023–2024). Overall, the plot indicates a non-stationary series with a clear upward trend and changing growth patterns, implying that transformation or differencing would be required before applying time series models like ARIMA.

- Initial Stagnation (2005–2012): Production remained relatively low and flat, hovering around the 1,000-thousand-ton mark.
- The 2015 Surge: There was a massive jump in production around 2014–2015, where output nearly tripled.
- Continued Momentum: Since 2020, there has been another significant breakout, with production climbing toward nearly 5,000 thousand tons.

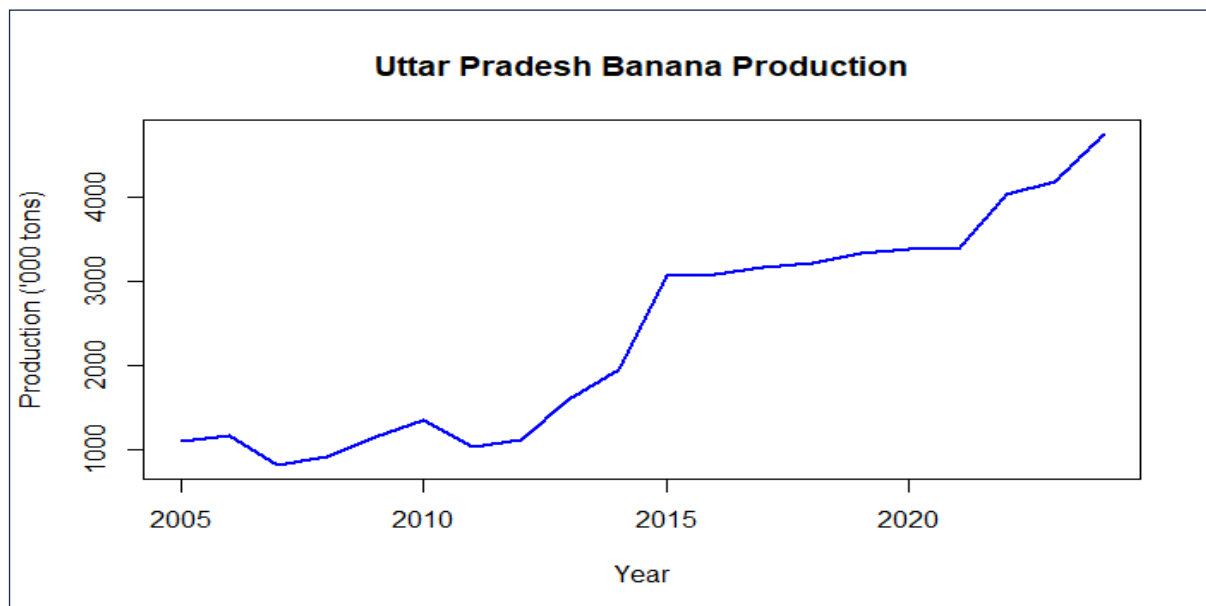


Figure 5.10: Uttar Pradesh: Time Series Plot of Banana Production

I. Model Fitting

Series: state_ts

ARIMA(0,1,0) with drift

Coefficients:

drift

192.5179

s.e. 74.9447

$\sigma^2 = 112646$: log likelihood = -136.95

AIC=277.9 AICc=278.65 BIC=279.79

Training set error measures:

ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
0.04537408	318.4041	224.5568	-4.729729	12.68318	0.8503873	0.06138703

- The Drift Coefficient (192.5179): This is the "growth engine" of your model. It suggests that, on average, banana production in Uttar Pradesh is increasing by approximately 192.52 thousand tons every year.
- Model Accuracy: The MAPE is 12.68%. This is slightly higher than Andhra Pradesh (8.31%), likely because the sudden "jumps" in the plot are a bit more difficult for the model to smooth out than a steady line.
- AIC/BIC: With an AIC of 277.9, this model is statistically efficient for the data provided.

II. Forecasting values for Uttar Pradesh Banana Production in ('000 Tons)

Year	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
2025	4950.358	4520.234	5380.482	4292.540	5608.176
2026	5142.876	4534.589	5751.163	4212.581	6073.170
2027	5335.394	4590.397	6080.390	4196.020	6474.767
2028	5527.912	4667.664	6388.159	4212.276	6843.547
2029	5720.429	4758.643	6682.216	4249.505	7191.354

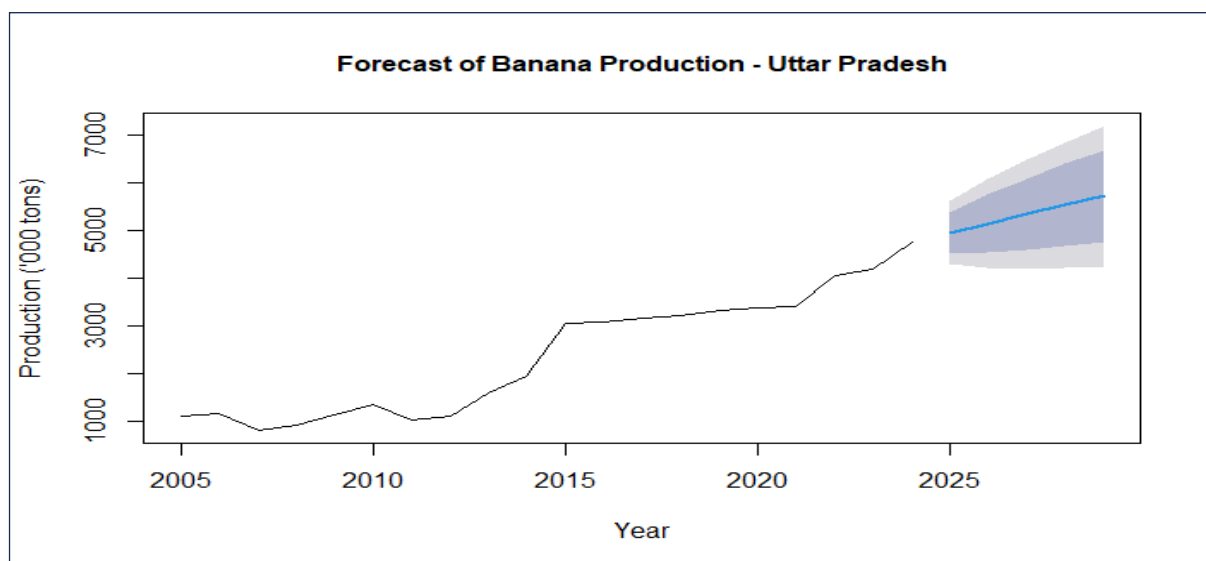


Figure 5.11: Forecast Plot of Banana Production in Uttar Pradesh

The reason behind the fast growth of Uttar Pradesh Banana Production is "One District One Product" (ODOP) initiative. Districts like Kushinagar have been designated specifically for banana development, which has led to better cold storage and the processing of banana fibre, encouraging farmers to increase their acreage significantly.

5.8 Time Series Analysis and Forecasting for Gujarat

In this section we analysis on Banana Production of Gujarat (2005 – 2024). This analysis focuses on banana production in Gujarat, using an ARIMA(0,2,1) model. The time series

plot for Gujarat's banana production reveals a strong upward trajectory from 2005 to 2012, followed by a period of volatility with a notable peak near 2018 and a sharp subsequent dip. Despite these fluctuations, the production levels stabilized after 2020, maintaining a consistent output of approximately 4,000 thousand tons in recent years.

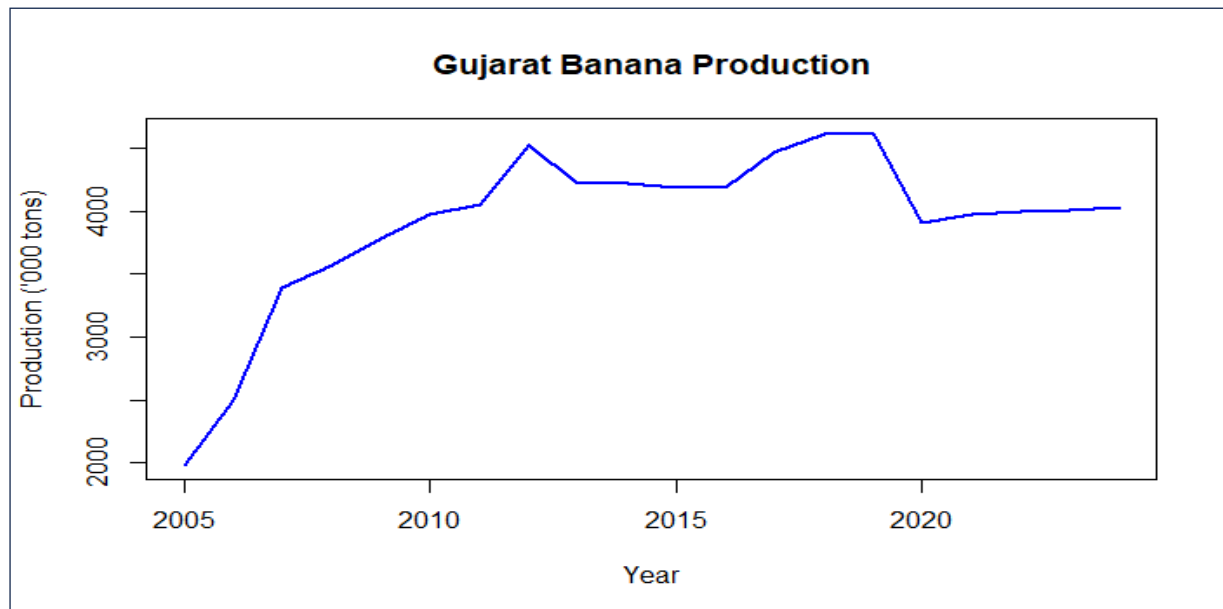


Figure 5.12: Gujarat: Time Series Plot of Banana Production

I. Model Fitting

Series: state_ts

ARIMA(0,2,1)

Coefficients: ma1 s.e.
 -0.6847 0.1774

$\sigma^2 = 106225$: log likelihood = -129.5

AIC=263.01 AICc=263.81 BIC=264.79

Training set error measures:

ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
-88.74733	300.4856	217.8926	-2.260733	5.490512	0.9945034	-0.1522747

Double Differencing (d=2): This is the most significant part of your model. By using a "2," the model indicates that the rate of change in production was changing over time. This was likely necessary to handle the sharp transition from rapid growth to the recent plateau.

Moving Average term (ma1 = -0.6847): This negative coefficient suggests that the model is "correcting" itself based on previous forecast errors. It helps smooth out the sudden shocks (like the 2020 drop) seen in the historical data.

High Accuracy: This model has the lowest MAPE (5.49%) of all the states you've shown so far, making it a very highly reliable fit for the Gujarat data series.

II. Forecast values of Gujarat Banana Production ('000 MT)

Year	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
2025	4006.530	3588.843	4424.216	3367.733	4645.326
2026	3987.780	3297.634	4677.925	2932.293	5043.266
2027	3969.029	2999.384	4938.674	2486.085	5451.974
2028	3950.279	2685.016	5215.542	2015.226	5885.332
2029	3931.529	2352.594	5510.464	1516.756	6346.301

The following are key factors behind these future values of Banana production in Gujarat.

- **Market Maturity:** The forecast of ~3,931 thousand tons by 2029 reflects a "mature market" phase where production has reached the capacity of current land allocations in key belts like Bharuch and Narmada, shifting the focus from quantity to quality.
- **Technological Ceiling:** The slight decline in point forecasts suggests that without a significant new technological breakthrough or a major increase in plantation area, the state will continue to consolidate its current high-yield levels rather than seeing aggressive new growth.
- **Climate Resilience:** While the point forecast is steady, the widening confidence intervals (Lo 95 to Hi 95) account for Gujarat's vulnerability to cyclonic activity and salinity ingress in coastal banana-growing regions.

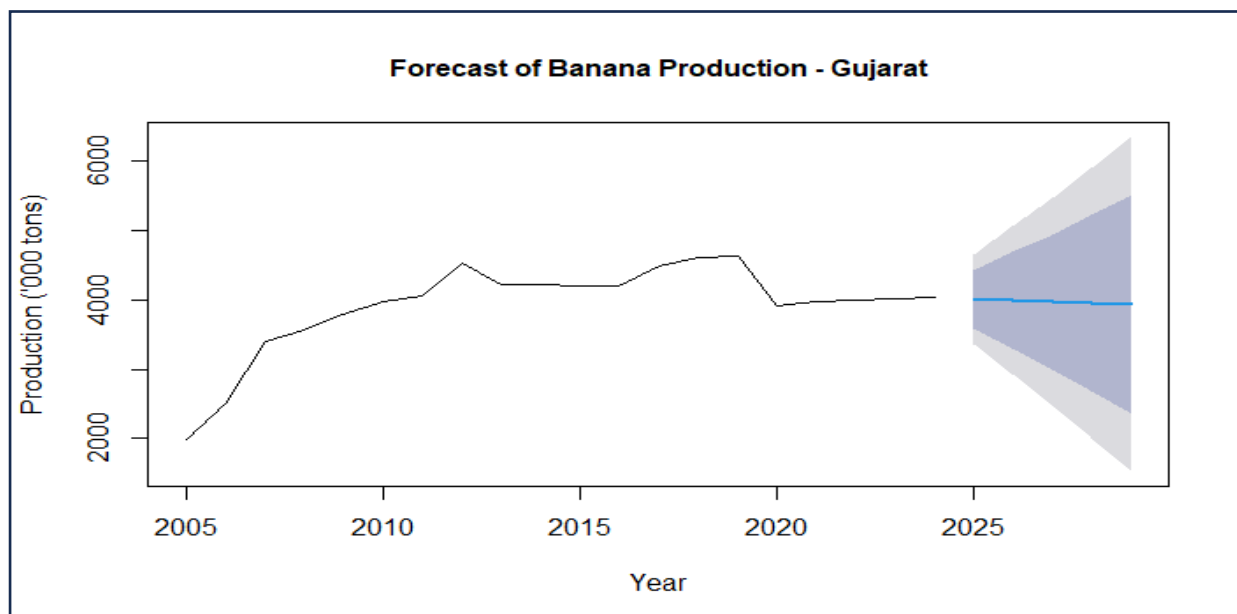


Figure 5.13: Forecast Plot of Banana Production in Gujarat

5.9 Time Series Analysis and Forecasting for Madhya Pradesh

In figure 5.11 the time series plot for banana production in Madhya Pradesh (2005–2024) illustrates a remarkably consistent and resilient upward trajectory. Starting from a relatively low base of under 1,000 thousand tons in 2005, the state has successfully transformed its agricultural landscape into a major production powerhouse, recently peaking at over 2,400 thousand tons. While the data shows minor cyclical fluctuations—most notably around 2011 and 2016—the overall trend is defined by a strong linear climb rather than extreme volatility.

This steady growth reflects a disciplined expansion in specialized hubs like the Nimar region, where the integration of modern irrigation and tissue culture technology has allowed the state to maintain momentum and reach record-breaking output levels by the end of the observed period.

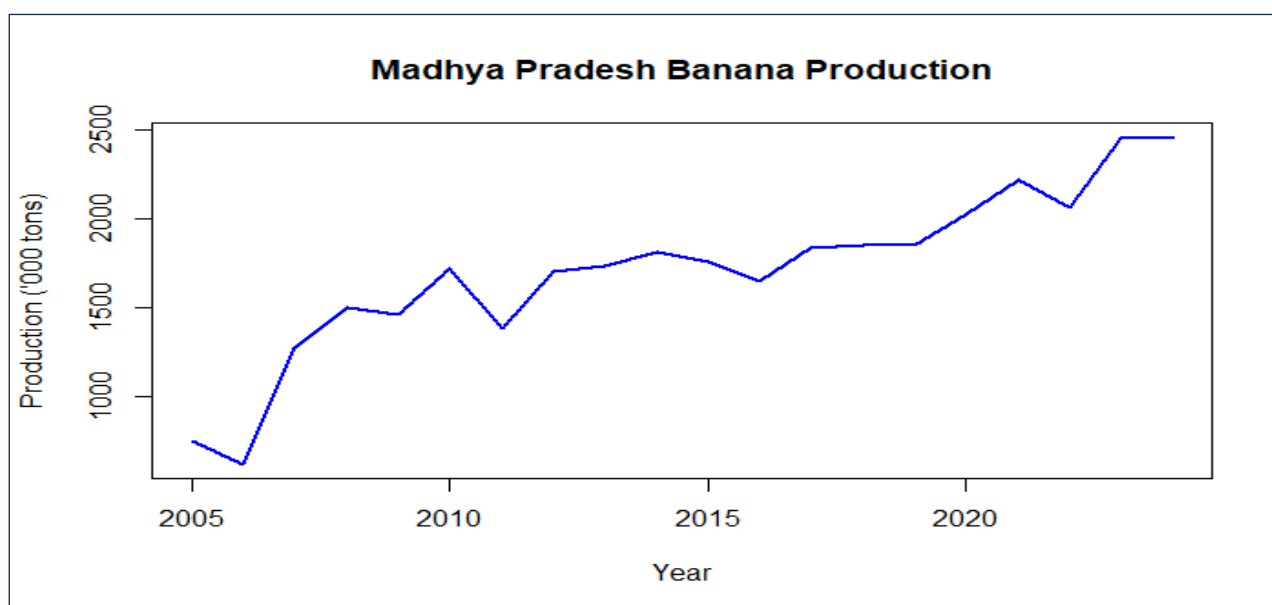


Figure 5.14: Time Series Plot of Banana Production in Madhya Pradesh

I. Model Fitting

Series: state_ts

ARIMA(1,1,0) with drift

Coefficients:

ar1 drift

-0.4190 95.0724

s.e. 0.2068 33.1665

$\sigma^2 = 45442$: log likelihood = -127.88

AIC=261.76 AICc=263.36 BIC=264.59

Training set error measures:

ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
-3.719145	196.5341	146.9177	-1.162704	10.1418	0.8208851	-0.04006582

This model is unique because it combines two different types of logic:

- The Drift (95.07): This indicates a baseline growth of approximately 95 thousand tons per year.
- The AR1 Term (-0.4190): Interestingly, this value is negative. In a differenced model, a negative AR term often acts as a "buffer." It suggests that if production jumped unusually high one year, it tends to pull back slightly the next to stay on its steady growth path, preventing the data from spiralling out of control.
- High Accuracy: The MAPE of 10.14% shows the model is highly reliable for forecasting in this region.

II. Forecasting values for Madhya Pradesh Banana Production ('000 Tons)

Year	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
2025	2594.271	2321.081	2867.460	2176.463	3012.078
2026	2674.329	2358.370	2990.287	2191.111	3157.546
2027	2775.691	2398.134	3153.249	2198.267	3353.115
2028	2868.128	2446.977	3289.280	2224.033	3512.223
2029	2964.305	2500.194	3428.416	2254.509	3674.101

The future forecast for banana production in Madhya Pradesh predicts a sustained and disciplined expansion, with output expected to climb from approximately 2,594 thousand tons in 2025 to nearly 2,964 thousand tons by 2029. This positive outlook is mathematically grounded in a "drift" model, suggesting a reliable annual growth increment of about 95 thousand tons that overrides minor year-to-year fluctuations. The widening confidence intervals toward 2029 reflect typical agricultural uncertainties, yet the central point forecasts indicate that Madhya Pradesh is on a firm path to nearly triple its original 2005 production levels by the end of the decade.

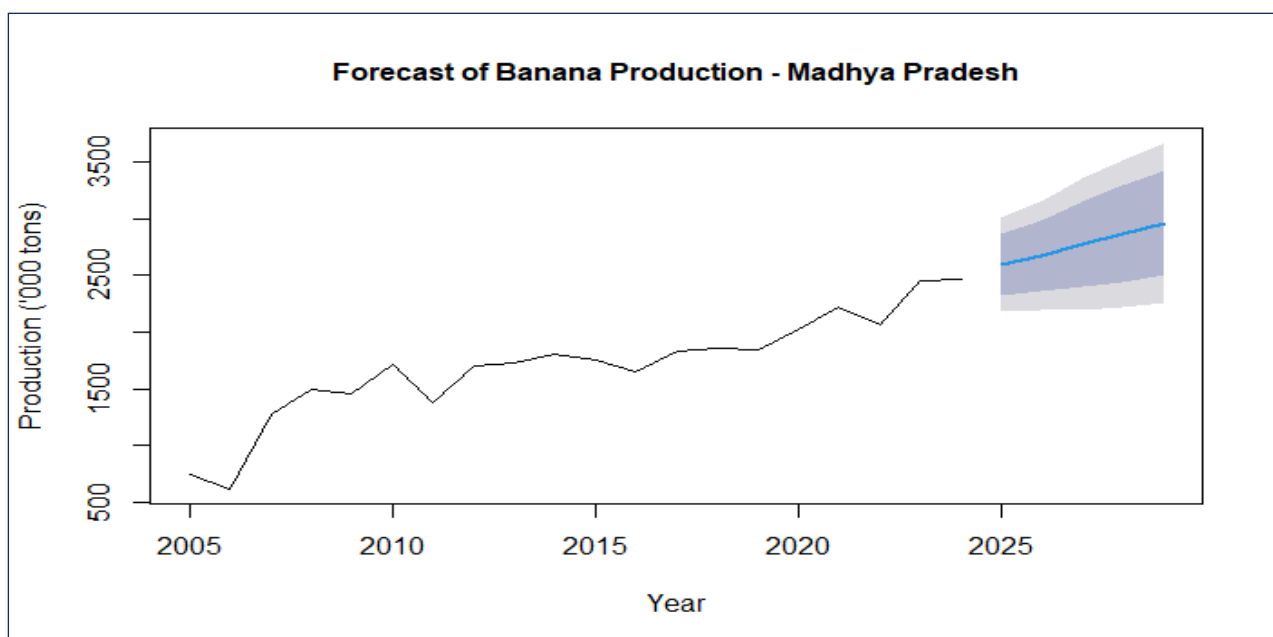


Figure 5.15: Madhya Pradesh: Forecast Plot for Banana Production

This projected growth is largely fuelled by the continued maturation of the Burhanpur banana cluster and increasing export-oriented farming practices, which are transitioning the state from a regional supplier to a national leader.

Chapter 6: District-wise Analysis of Banana Production in Maharashtra

6.1 Introduction

Maharashtra is one of the leading producers of bananas in India, playing a vital role in the agricultural economy of the country. Banana cultivation in the state is largely concentrated in specific districts due to favorable climatic conditions, availability of irrigation facilities, fertile soil, and adoption of improved farming practices. In this chapter, we undertake a comprehensive district-wise analysis of banana production in Maharashtra using data spanning multiple years. This analysis helps in understanding regional disparities, identifying high-performing and low-performing districts, and examining production trends over time. Furthermore, to highlight the contribution of major producing regions, a Pareto analysis has been carried out for the year 2024. This approach helps in identifying the top districts that contribute the most to total banana production, thereby providing valuable insights for policymakers, planners, and researchers for effective decision-making and strategic agricultural development.

6.2 Data Description

The district-wise data covering an 11-year period from 2014 to 2024. Each row corresponds to a district, while the columns represent yearly observations for the selected variable. The first column lists district names, followed by annual data from 2014 to 2024, capturing the temporal variation across districts. This dataset forms a panel structure (combining cross-sectional and time-series data), which is useful for analysing long-term trends, measuring growth rates, and studying year-to-year changes. It enables comprehensive analysis such as time series modelling, inter-district comparison, and identification of patterns in the data over time.

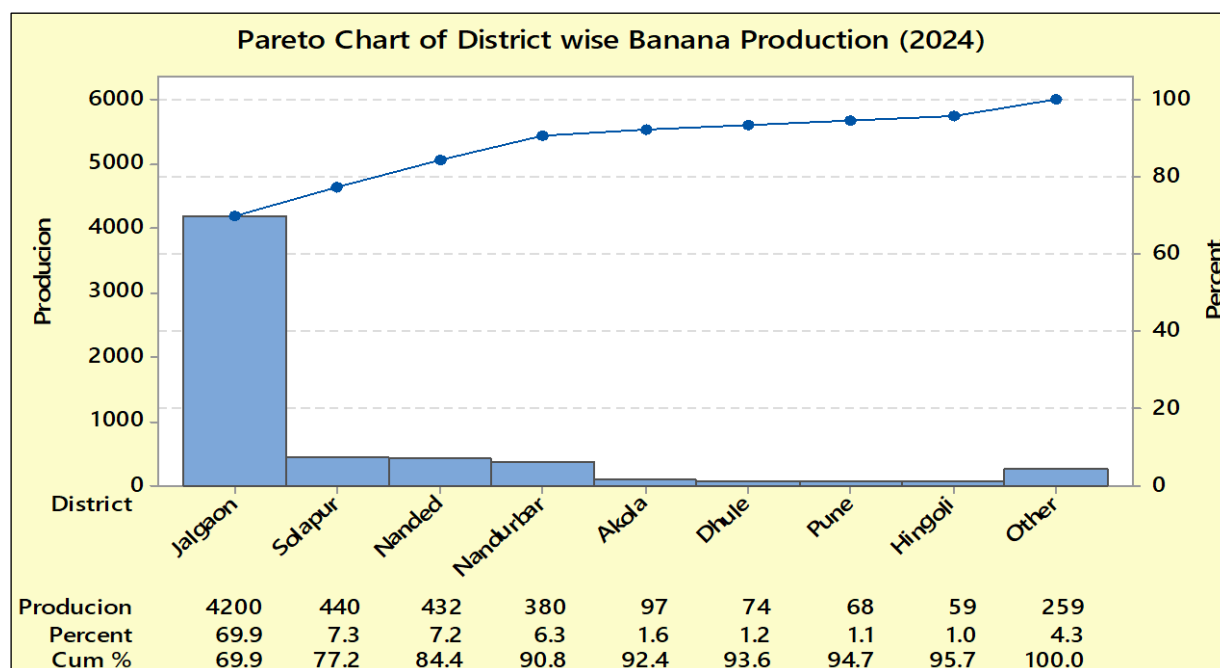


Figure 6.1: Pareto Chart of District wise Banana Production of Maharashtra

The Pareto chart of district-wise banana production for the year 2024 shows a highly unequal distribution of production across districts. Jalgaon district dominates banana production with around 4200 ('000 MT), contributing approximately 69.9% of the total production. This indicates that a single district accounts for the majority share, highlighting its leading role in banana cultivation. The next contributors—Solapur, Nanded, and Nandurbar—have much smaller shares, each contributing around 6–7%, and together raise the cumulative percentage to about 90.8%.

The remaining districts such as Akola, Dhule, Pune, Hingoli, and the “Other” category contribute only marginally to the total production, each accounting for less than 5%. The cumulative percentage curve rises steeply at the beginning and then flattens, clearly reflecting the Pareto principle (80/20 rule), where a small number of districts contribute to a large proportion of production. This analysis helps identify key production centers and suggests that policy focus on leading districts like Jalgaon can significantly impact overall banana production.

Now we are going to analysis the top 3 Banana producing districts namely Jalgaon, Solapur and Nanded which contributes more than 80% of the total production.

6.3 Time Series Analysis and Forecasting for Jalgaon District

According to Banana production data of the year 2024 and the corresponding pareto chart we observed that the Jalgaon district contributes approximately 70% of the total Maharashtra's Banana production.

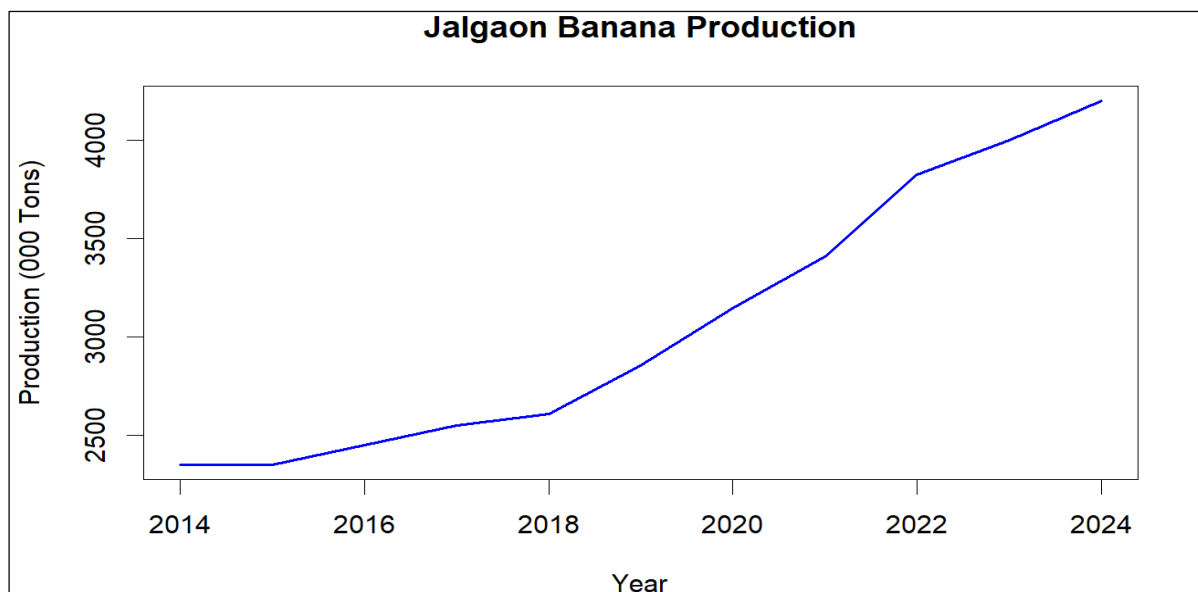


Figure 6.2: Time Series plot of Banana production in Jalgaon (2024-24)

The time series plot of banana production in Jalgaon district shows a consistent upward trend over the period 2014 to 2024. Production increased from around 2300 ('000 tons) in 2014 to nearly 4200 ('000 tons) in 2024. The growth is gradual in the initial years (2014–2018), followed by a sharper increase after 2019, indicating improved cultivation practices, expansion in area, or favourable climatic and market conditions. Overall, the

trend suggests strong and sustained growth, making Jalgaon the leading contributor to banana production

I. Model Fitting:

Series: jal_ts

ARIMA(0,2,0)

$\sigma^2 = 14536$: log likelihood = -55.9

AIC=113.8 AICc=114.37 BIC=114

Training set Measures:

ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
17.99074	109.0572	74.81635	0.740718	2.30447	0.4044127	-0.396001

The selected model ARIMA(0,2,0) indicates that the data required second-order differencing to achieve stationarity, which confirms a strong upward trend in the original series. The absence of AR and MA terms suggests that the model primarily captures the trend component without short-term dependencies. The model diagnostics show reasonably low error measures (MAPE $\approx$ 2.3%), indicating a good fit for forecasting purposes.

II. Forecast for Jalgaon

Year	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
2025	4400	4245.5	4554.5	4163.7	4636.3
2026	4600	4254.5	4945.5	4071.6	5128.4
2027	4800	4221.9	5378.1	3915.8	5684.2
2028	5000	4153.7	5846.3	3705.7	6294.3
2029	5200	4054.1	6345.9	3447.5	6952.5

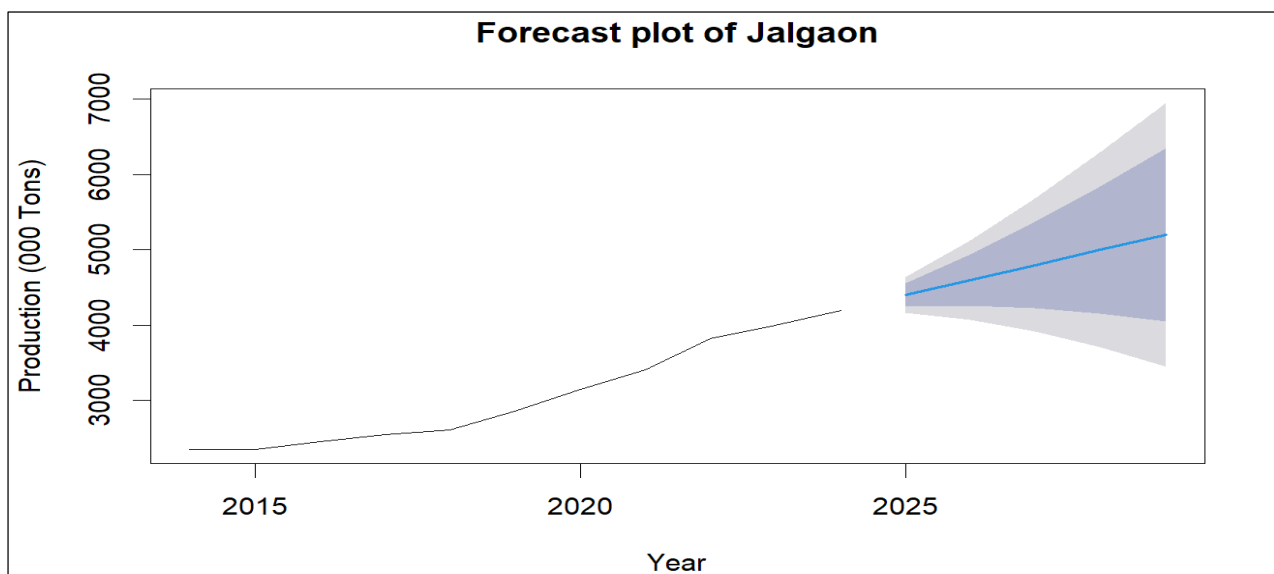


Figure 6.3: Forecast plot of Jalgaon Banana Production

The forecast results indicate a continued increasing trend in banana production for Jalgaon district over the period 2025 to 2029. Production is expected to rise from about 4400 ('000 tons) in 2025 to 5200 ('000 tons) by 2029, reflecting a steady annual increase of nearly 200 ('000 tons). However, the prediction intervals (80% and 95%) widen as the forecast horizon increases, indicating growing uncertainty in long-term projections, which is a common feature of time series models. For instance, in 2025, production is likely to lie between 4163 and 4636 ('000 tons) at the 95% confidence level, whereas by 2029, this range expands considerably from 3447 to 6952 ('000 tons), highlighting higher uncertainty in future estimates.

6.4 Time Series Analysis and Forecasting for Solapur District

In figure 6.1 we observed that after a Jalgaon district Solapur contributes 7.3% of the total Banana production and both districts contributes 77.2% of the total. In figure 6.4 the time series of Solapur district shows how the values have changed over the years. From the graph/data, we can observe the overall trend, which may indicate whether production (or the variable studied) is increasing, decreasing, or remaining Stable over time. There are some year-to-year fluctuations, which are normal and may be due to factors like climate conditions, market changes, or other local influences. However, despite these variations, the general pattern helps us understand the performance of Solapur district over time and gives an idea about its future direction based on past trends.

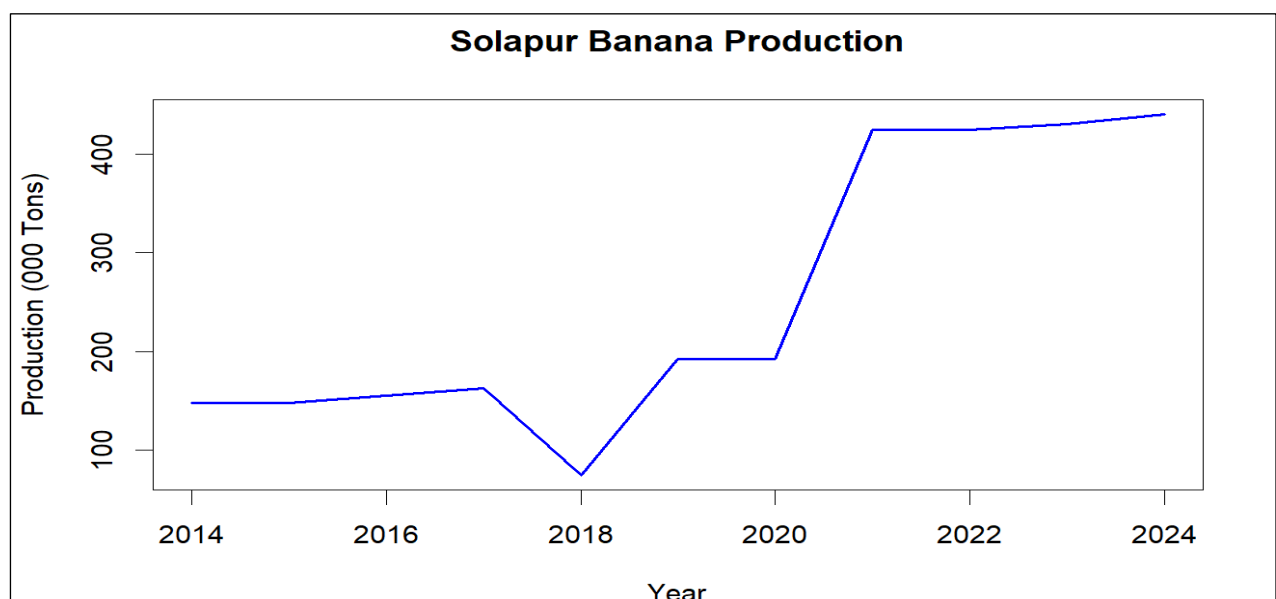


Figure 6.4: Time Series Plot of Banana Production in Solapur

I. Model Fitting

Series: sol_ts

ARIMA(0,1,0)

$\sigma^2 = 7539$: log likelihood = -58.83

AIC=119.66 AICc=120.16 BIC=119.96

Training set error measures:

ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
26.55891	82.7874	42.53764	1.010902	22.39428	0.9093785	-0.2826602

The time series model fitted for Solapur district is ARIMA(0,1,0), which means the data mainly shows changes over time without any strong pattern or relationship with past values. This model indicates that the series follows a simple random movement with a general trend captured through differencing. The model's error values (RMSE = 82.78 and MAE = 42.54) suggest that there is some variation between actual and predicted values, but the errors are within a reasonable range. The MAPE value of about 22.39% shows a moderate level of forecasting accuracy. The negative ACF1 value (-0.28) indicates a slight inverse relationship between consecutive errors, but it is not very strong. Overall, the model is simple and Suitable for short-term forecasting, showing that the data of Solapur district does not have a complex structure and mainly follows a gradual and somewhat irregular trend over time.

II. Forecast values for Solapur's Banana Production ('000 MT)

Year	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
2025	440	328.73	551.27	269.82	610.1801
2026	440	282.63	597.37	199.329	680.671
2027	440	247.27	632.73	145.239	734.7605
2028	440	217.45	662.55	99.6399	780.3601
2029	440	191.18	688.82	59.4658	820.5342

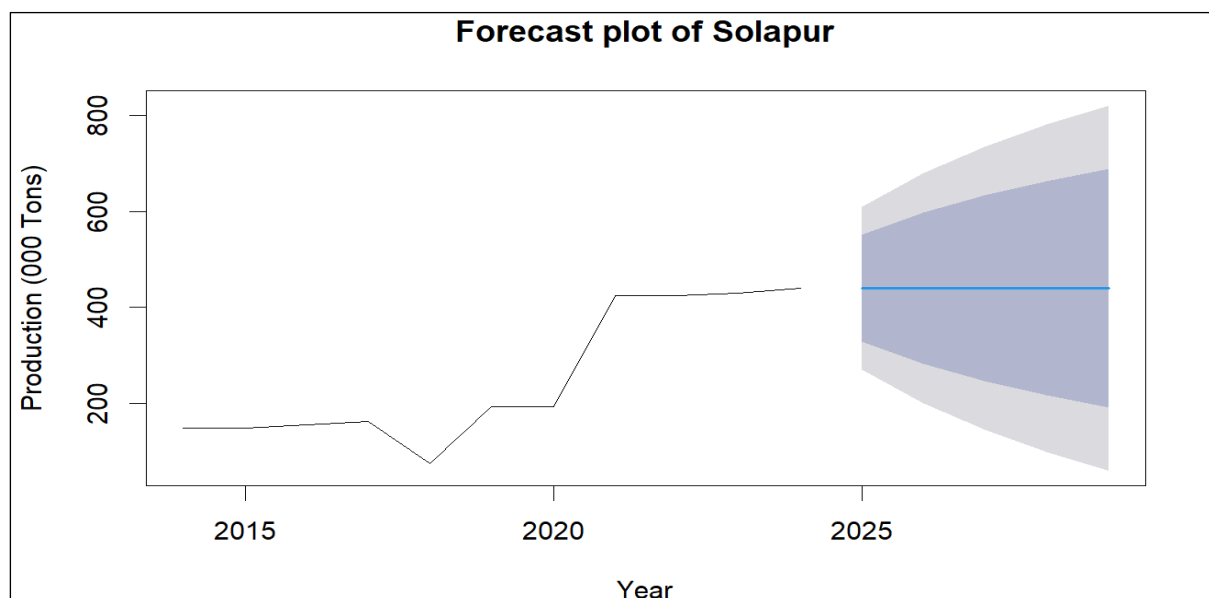


Figure 6.5: Forecast Plot for Banana Production of Solapur

The forecast suggests that Solapur's banana production is expected to remain nearly Stable at about 440 thousand tons from 2025 to 2029. The prediction line is flat, but the confidence intervals widen over time, showing that uncertainty increases further into the future. Overall, the historical trend shows a rise in production after 2020, and the model

projects this higher level to continue in the coming years. Now we are going to do analysis on Nanded district in the section

6.5 Time Series Analysis and Forecasting for Nanded District

In this section we study the time series analysis for Nanded district, which is third major Banana producing district in Maharashtra as see in figure 6.1. Nanded produces near about 7.2% of the total Banana production and if we considered these three districts namely Jalgaon, Solapur and Nanded are contributes approximately 84 – 85% of the total. In figure 6.6, the time series plot of Solapur district shows how the values have changed over time. From the plot, we can observe that the data does not follow a perfectly smooth path but shows some ups and downs from year to year. However, there is an overall pattern that may indicate a gradual increase (or Stable movement, depending on your graph) over the period. These fluctuations may be due to factors like weather conditions, market demand, or other external influences.

The absence of a regular repeating pattern suggests that the data is mostly non-seasonal. Overall, the plot helps us understand that while there are short-term variations, the long-term movement gives a clear idea about the trend and performance of Solapur district over time.

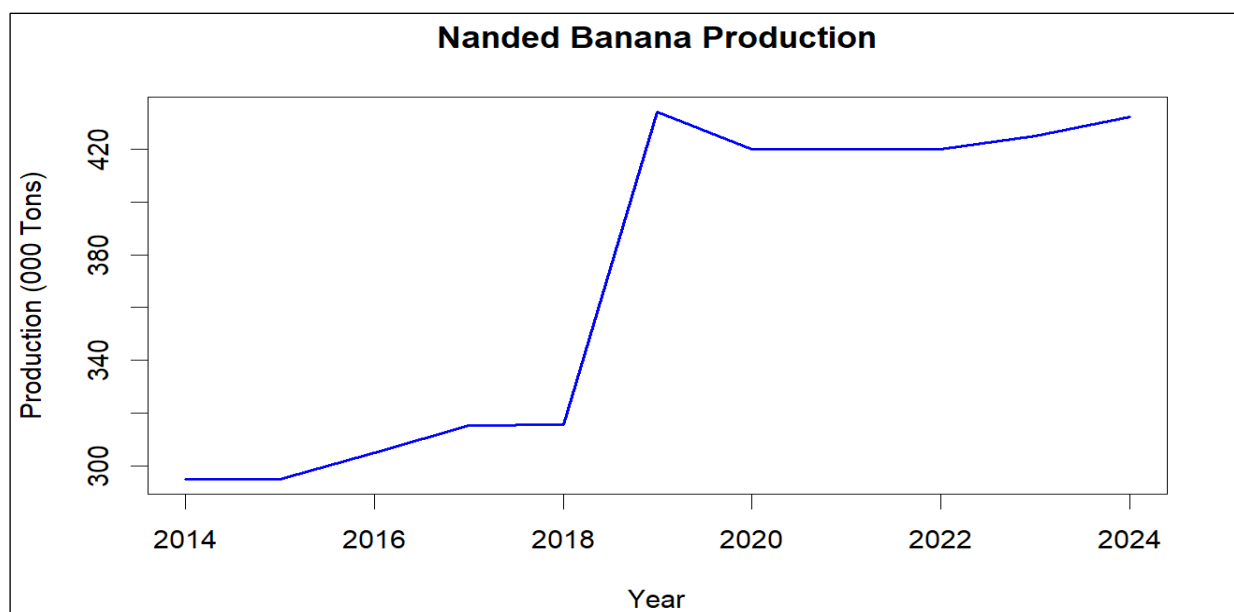


Figure 6.6: Time Series Plot for Banana Production of Nanded

I. Model Fitting:

Series: sol_ts

ARIMA(0,1,0)

$\sigma^2 = 7539$: log likelihood = -58.83

AIC=119.66 AICc=120.16 BIC=119.96

Training Set Measures:

ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
26.55891	82.7874	42.53764	1.010902	22.39428	0.9093785	-0.2826602

The model fitted for Nanded district is ARIMA(0,1,0), which indicates that the data mainly follows a simple random pattern with a gradual trend over time, without any strong dependence on past values. This means the changes from one year to the next are mostly unpredictable but follow a general direction. The error values (RMSE = 36.31 and MAE = 15.08) are relatively low, showing that the model fits the data reasonably well. The MAPE value of about 3.66% indicates very good forecasting accuracy, meaning the predictions are quite close to actual values.

The MASE value is less than 1, which suggests the model performs better than a simple average-based method. The negative ACF1 value (-0.26) shows a slight negative correlation between errors, but it is not very strong. Overall, this model is simple yet effective for short-term forecasting and suggests that the time series data of Nanded district is fairly Stable with minor variations over time.

II Forecast values for Nanded's Banana Production ('000 Tons)

Year	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
2025	432	383.2	480.8	357.36	506.637
2026	432	362.98	501.02	326.45	537.553
2027	432	347.47	516.53	302.72	561.276
2028	432	334.39	529.61	282.73	581.275
2029	432	322.87	541.13	265.11	598.894

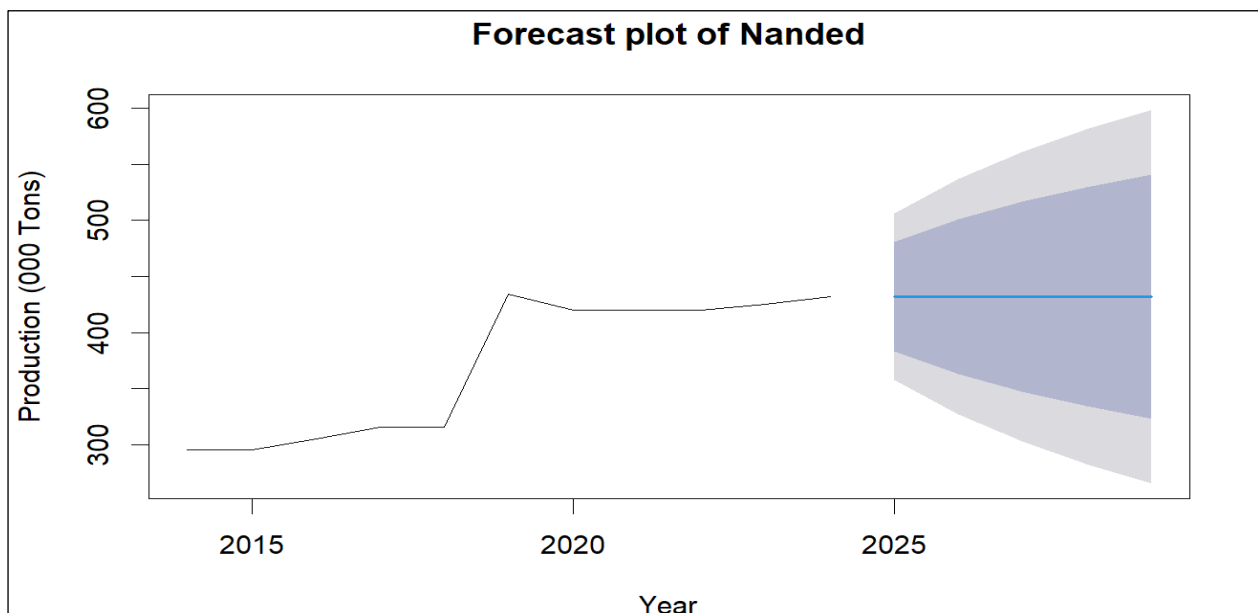


Figure 6.7: Forecast plot for Banana Production of Nanded

The banana production in Nanded is forecast to remain Stable at around 432 thousand tons from 2025 to 2029. The prediction intervals gradually widen over time, indicating

increasing uncertainty as the forecast horizon extends. Even under the lower 95% limit, production stays above roughly 265 thousand tons by 2029, while the upper 95% limit approaches about 599 thousand tons, suggesting that substantial deviations from the central forecast, though unlikely, are still possible in the long term.

7. Prices of Banana in India

7.1 Introduction to Banana Prices in India

The pricing of banana is highly important for farmers, traders, and consumers, as it directly affects income, market decisions, and profitability. Due to its perishable nature, banana prices tend to fluctuate frequently depending on supply and demand conditions in the market. The behaviour of banana prices is influenced by several factors such as seasonal production, weather conditions, transportation, storage facilities, and market structure. During peak harvesting seasons, an increase in supply often leads to a fall in prices, whereas during off-season periods, limited availability results in higher prices. These variations create uncertainty for farmers and make price analysis essential.

In the banana marketing system, the final price paid by consumers is distributed among different participants such as farmers, commission agents, wholesalers, and retailers. Commission agents generally charge a fixed percentage for facilitating sales in regulated markets, while retailers add their margin to cover handling costs and earn profit. As a result, the share received by farmers is often lower compared to the final retail price, which directly affects their profitability. Understanding this price spread and the role of intermediaries is important for evaluating market efficiency and farmer income.

Therefore, studying the trend, seasonal patterns, and fluctuations in banana prices, along with the structure of price distribution among market participants, is essential to understand overall market dynamics. This chapter focuses on analyzing banana price movements over a given period and provides insights that can support better decision-making and future planning for farmers and policymakers.

7.2 Data Sources and Methodology

The data used in this study has been collected from the publication “Agricultural Prices in India 2021 – 24” published by the Department of Agriculture and Farmers Welfare under the Government of India. The dataset represents wholesale prices of banana across selected major states in India.

The data is secondary in nature and follows a time series structure, as it records prices over multiple time periods. The dataset includes monthly observations for the years 2020 to 2024, allowing for trend and seasonal analysis.

The variables included in the dataset are:

- **Year:** Indicates the calendar year (2020–2024).
- **Month:** Represents the time period within each year (January to December).
- **State-wise Prices:** Wholesale prices of banana (in ₹ per 100 Nos.) for selected states according to the market prices of major Banana producing districts in those selected states.

State-wise wholesale prices of banana, expressed in ₹ per 100 numbers, were collected from the major banana-producing districts of the selected states. The study included Rajahmundry in Andhra Pradesh, Jalgaon market in Maharashtra, Vellore market in Tamil

Nadu, and Kanpur market in Uttar Pradesh. West Bengal data was excluded as per the study requirement. The data is quantitative and continuous, as it measures price values.

7.2 Time Series Analysis and Forecasting for Andhra Pradesh Banana Prices

This section presents the time series analysis and forecasting of banana prices (Rs. Per 100 Nos.) in Andhra Pradesh. We can see the figure 7.1 the time series plot for Andhra Pradesh displays a strong and consistent upward trend throughout the entire period. Prices increase steadily from 2020 to 2025 with only small fluctuations in between, indicating that the trend component dominates the series. There is very little randomness compared to other states, and the growth appears smooth and continuous. Due to this persistent rise in prices, the series is clearly non-stationary and would require differencing before applying any time series model.

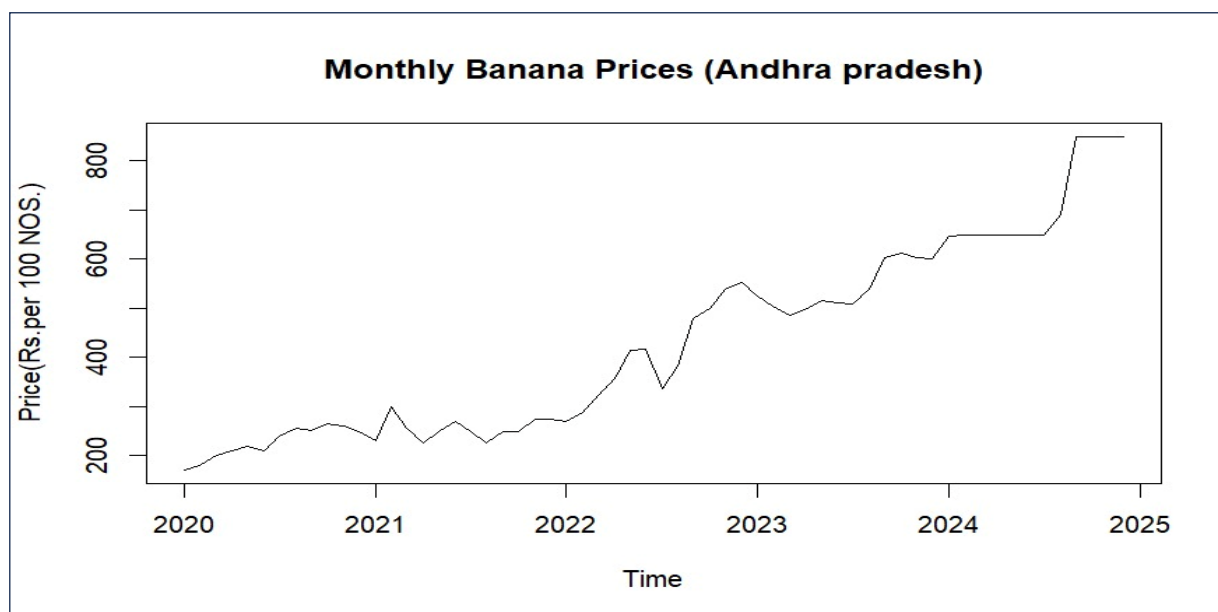


Figure 7.1: Time Series Plot for Monthly wholesale Prices of Banana in Andhra Pradesh

I. Model summary for Andhra :

Series: state_ts

ARIMA(0,1,0)(1,0,0)[12] with drift

Coefficients:

sar1 drift

0.2845 11.7076

s.e. 0.1376 5.5159

sigma² = 1105: log likelihood = -289.92

AIC=585.84 AICc=586.28 BIC=592.07

Training Set Measures:

ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
0.43978	32.3939	22.3989	-0.7124	6.04302	0.17795	0.09458

This output summarizes an ARIMA(0,1,0)(1,0,0)[12] with drift model fitted to the banana price time series for Andhra Pradesh. This is a seasonal model that accounts for both trends and annual seasonality (the [12] indicates monthly data).

Model Components

- **ARIMA(0,1,0):** This part of the model indicates that the data is non-stationary and has been differenced once to remove a trend. It suggests a "random walk" component in the underlying price levels.
- **Seasonal Component (1,0,0):** The model includes a seasonal autoregressive (SAR) term of order 1. This means the price in a given month is influenced by the price from the same month in the previous year (a 12-month lag).
- **Drift:** The model includes a drift term (coefficient 11.7076), indicating an underlying linear trend or consistent increase in prices over the observed period.

Coefficient Significance

- **sar1 (0.2845):** This is the seasonal autoregressive coefficient. With a standard error of 0.1376, it is statistically significant, suggesting that previous year's seasonal patterns have a moderate, positive influence on current prices.
- **Drift (11.7076):** This represents the average change in price per period after accounting for other factors.

Model Fit and Performance

- **Information Criteria:** The AIC (585.84) and BIC (592.07) are used for model selection; these values serve as a baseline to compare against other potential models (a lower value generally indicates a better model).
- **MAPE (6.04%):** The Mean Absolute Percentage Error of approximately 6.04% suggests that, on average, the model's forecasts deviate from actual values by about 6%.
- **ACF1 (0.09458):** This measures the autocorrelation of the residuals at lag 1. A value close to zero (like 0.09) is generally good, indicating that the model has successfully captured most of the predictable information, leaving little correlation in the remaining errors.

II. Forecast prices of Andhra Pradesh Banana (in Rs. Per 100 Nos.)

Month	Year	Point Forecast	Lo_80	Hi_80	Lo_95	Hi_95
Jan	2025	871.4651	828.8721	914.058	806.3248	936.6054
Feb	2025	880.4105	820.175	940.646	788.2882	972.5328
Mar	2025	888.7868	815.0137	962.56	775.9605	1001.613

The forecast suggests banana prices in Andhra Pradesh will increase slightly from January to March 2025, from about 871.5 to 888.8. The 80% and 95% intervals show the likely range of prices, and the wider 95% interval means there is more uncertainty in the forecast.

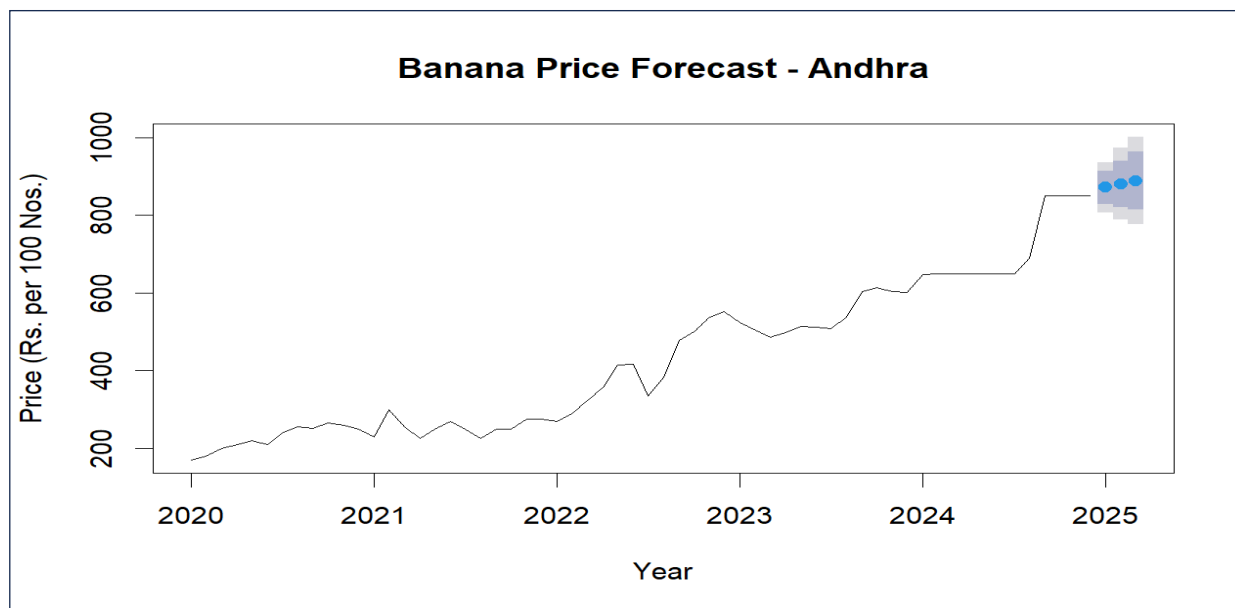


Figure 7.2: Forecast Plot of Monthly wholesale Prices of Banana in Andhra Pradesh

7.3 Time Series Analysis and Forecasting for Maharashtra Banana Prices:

In figure 7.3 the time series plot of banana prices in Maharashtra shows a mixed pattern of stability and fluctuation over the study period. Initially, from 2020 to around 2022, prices remain relatively low and stable with minor variations, indicating a nearly stationary phase. However, from 2023 onwards, a clear upward trend is observed, where prices increase significantly and reach peak levels during 2024. This is followed by some decline towards the end, suggesting short-term volatility. Overall, the series exhibits a non-stationary behavior due to the presence of trend and changing variability over time.

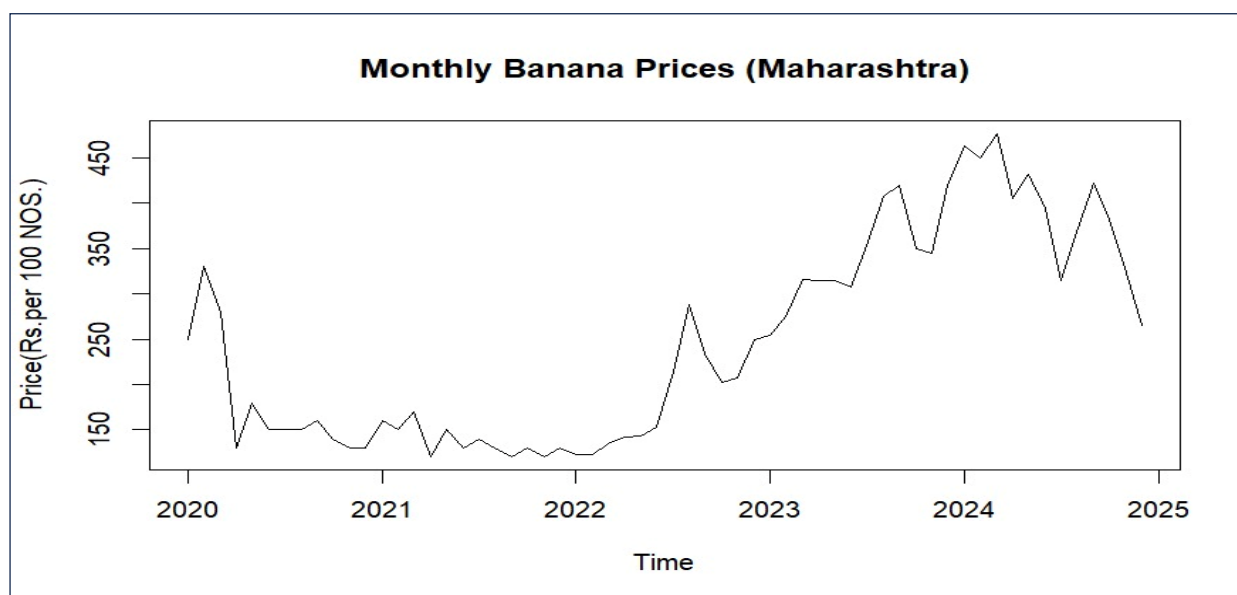


Figure 7.3: Time Series plot for wholesale Banana Prices in Maharashtra

I. Model summary for Maharashtra:

Series: state_ts

ARIMA(0,1,0)(0,0,1)[12]

Coefficients:

sma1

0.5306

s.e. 0.1915

$\sigma^2 = 1518$: log likelihood = -301.28

AIC=606.56 AICc=606.77 BIC=610.71

Training set error measures:

ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
-1.034027	38.30183	27.61201	-1.42077	12.1486	0.3085859	-0.01755291

1. Model Quality

- **Model Specification:** You used an ARIMA(0,1,0)(0,0,1)[12] model, which is well-suited for data with seasonal patterns (indicated by the [12] period).
- **Coefficient Significance:** The sma1 coefficient is 0.5306 with a standard error of 0.1915. Since the coefficient is more than twice its standard error, it indicates that the seasonal moving average component is a statistically significant part of your model.
- **Fit Metrics:** An AIC (Akaike Information Criterion) of 606.56 serves as a baseline. You can use this value to compare against other models—the lower the AIC, the better the balance between model accuracy and simplicity.

2. Accuracy (Training Set Error)

The MAPE (Mean Absolute Percentage Error) is 12.1486.

This means that, on average, your model's predictions deviate from the actual historical prices by about 12.15%. This is generally considered a reasonable accuracy level for agricultural commodity forecasting, where prices are often volatile.

II. Forecast prices of Banana for Maharashtra (in Rs. Per 100 Nos.)

Month	Year	Point Forecast	Lo_80	Hi_80	Lo_95	Hi_95
Jan	2025	283.6588	233.702	333.615	207.257	360.061
Feb	2025	274.3199	203.687	344.953	166.296	382.344
Mar	2025	275.9323	189.431	362.433	143.64	408.224

The forecast shows that banana prices in Maharashtra are expected to remain fairly stable in early 2025 with slight fluctuations. The point forecasts indicate a small decline from

January (₹283.66) to February (₹274.32), followed by a marginal increase in March (₹275.93).

The prediction intervals (both 80% and 95%) are quite wide, suggesting a high level of uncertainty in price movements. This means actual prices could vary significantly, but they are most likely to stay within the given ranges, with no strong upward or downward trend in the short term.

7.4 Time Series Analysis and Forecasting for Tamil Nadu Banana Prices

In figure the banana price series for Tamil Nadu shows a highly irregular and volatile pattern over time. There are frequent sharp increases and decreases in prices, particularly between 2021 and 2023, indicating high variability. Unlike other states, there is no clear long-term upward or downward trend visible in the data. However, towards the later period, prices appear to stabilize slightly at a moderate level. Overall, the series is dominated by random fluctuations, making it less predictable and more challenging for modelling.

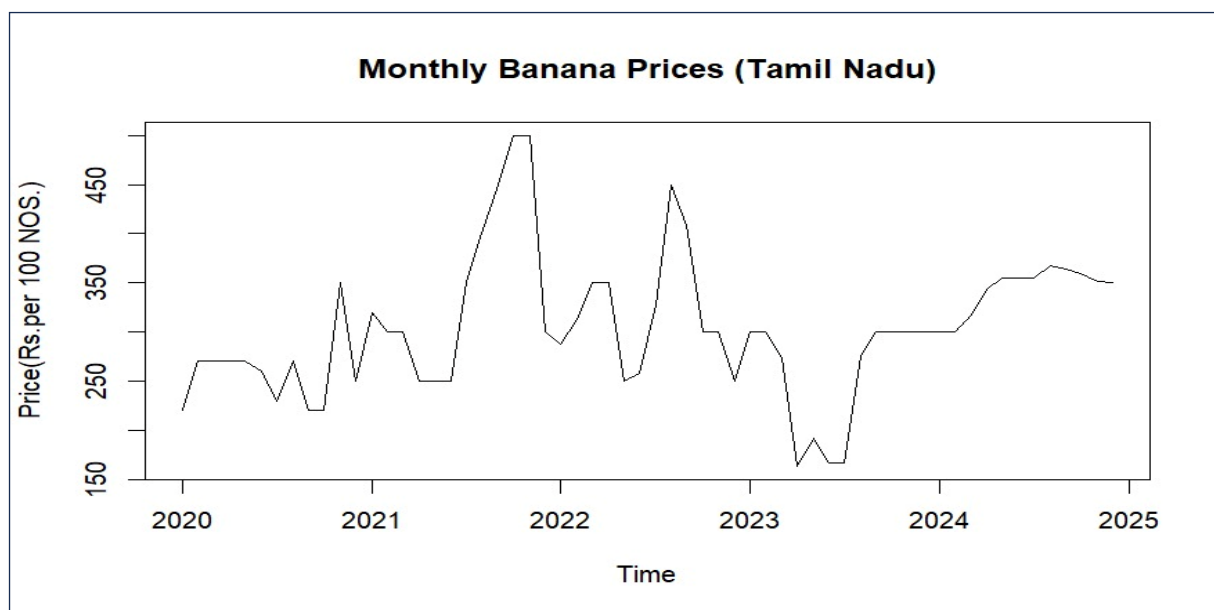


Figure 7.4: Time Series Plot for Wholesale Prices of Banana in Tamil Nadu

I. Model summary for Tamil Nadu :

Series: state_ts

ARIMA(1,0,0) with non-zero mean

Coefficients:

ar1 mean

0.6970 303.1192

s.e. 0.0916 20.7891

sigma^2 = 2650: log likelihood = -320.93

AIC=647.85 AICc=648.28 BIC=654.14

Training set error measures:

ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
1.359942	50.61704	36.82376	-2.509717	12.92949	0.4477053	0.09358298

The fitted ARIMA (1,0,0) model with non-zero mean suggests that banana prices in Tamil Nadu are mainly influenced by their immediate past values. The AR(1) coefficient (0.697) is positive and relatively high, indicating strong persistence—meaning if prices increase or decrease, the effect tends to continue in the short run.

The estimated mean price is around ₹303.12 per 100 Nos., representing the average level of the series. The standard errors are reasonably small, showing that the parameter estimates are reliable.

The model performance measures indicate a moderate fit:

- MAPE $\approx$ 12.93%, meaning the average forecasting error is about 13%, which is acceptable for agricultural price data.
- RMSE and MAE values show moderate prediction errors.
- ACF1 $\approx$ 0.094 is close to zero, indicating that residuals are nearly uncorrelated, so the model has captured most of the pattern in the data.

II. Forecast prices of Banana for Tamil Nadu (in Rs. Per 100 Nos.):

Month	Year	Point forecast	Lo_80	Hi_80	Lo_95	Hi_95
Jan	2025	335.7931	269.8158	401.7703	234.8895	436.6966
Feb	2025	325.8914	245.4709	406.312	202.8988	448.884
Mar	2025	318.9904	232.4193	405.5616	186.5914	451.3895

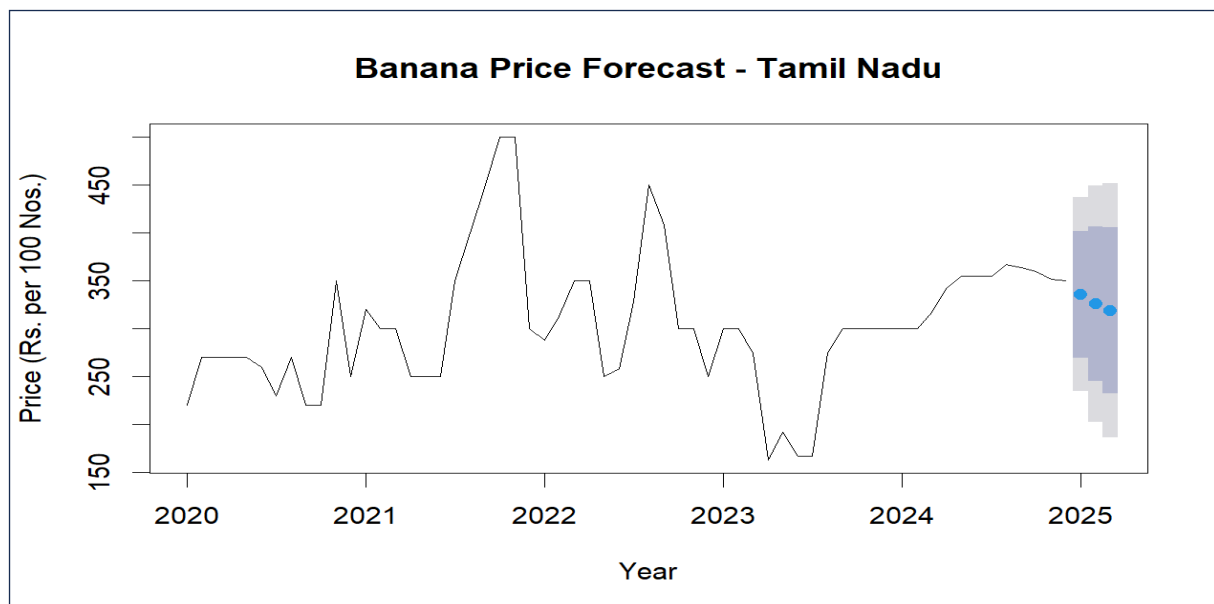


Figure 7.5: Time Series Plot for Wholesale Prices of Banana in Tamil Nadu

The forecast of banana prices for Tamil Nadu indicates a gradual declining trend during the early months of 2025. The point estimates show that prices are expected to decrease

from about ₹335.79 per 100 Nos. in January to ₹325.89 in February and further to ₹318.99 in March. This suggests a slight softening of prices over time. However, the forecast intervals (both 80% and 95%) are quite wide, indicating a high degree of uncertainty in future price movements. This implies that actual market prices may vary considerably from the predicted values due to factors such as supply fluctuations, seasonal effects, and market demand. Overall, while the model predicts a mild downward trend, the variability in prices remains significant.

7.6 Time Series Analysis and Forecasting for Uttar Pradesh Banana Prices

The time series of banana prices in Uttar Pradesh exhibits a cyclical pattern with repeated rises and falls over time. These fluctuations suggest the presence of seasonal or periodic behavior in the data. In addition to this, there is a slight upward trend observed in the later years, indicating gradual price growth. The variability is moderate, and the pattern is more structured compared to Tamil Nadu. Overall, the series is non-stationary due to both trend and possible seasonality, and it may be effectively modeled using seasonal time series techniques.

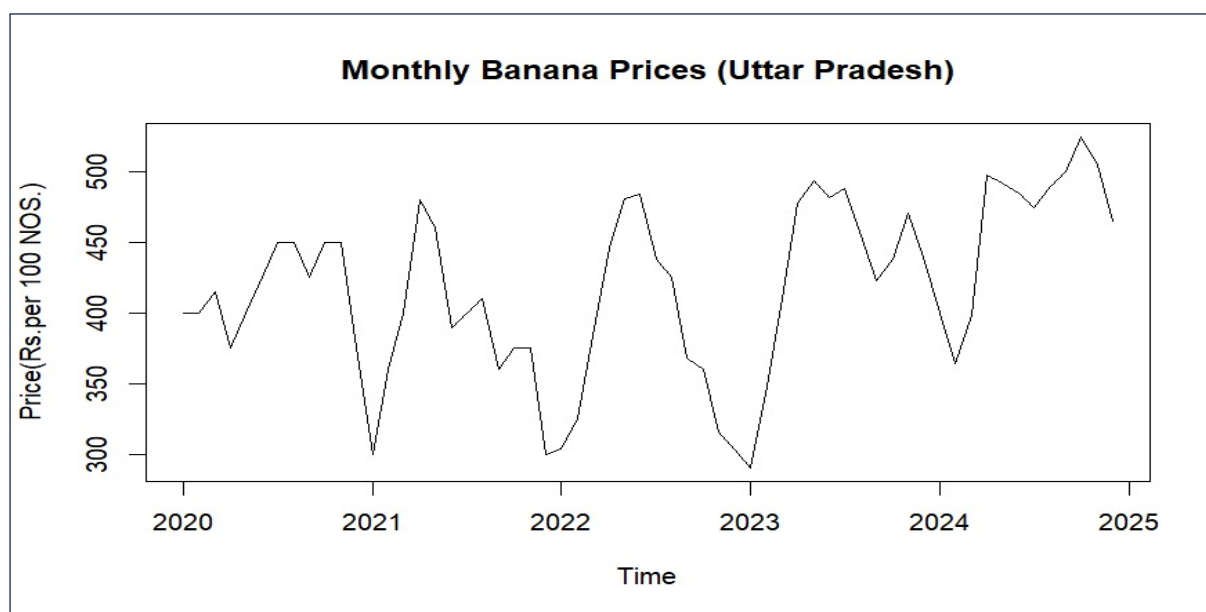


Figure 7.6: Time Series plot for wholesale Banana Prices in Uttar Pradesh

I. Model summary for UP :

Series: state_ts

ARIMA(0,1,0)(1,0,0)[12]

Coefficients:

sar1

0.4046

s.e. 0.1247

sigma² = 1322: log likelihood = -296.3

AIC=596.61 AICc=596.82 BIC=600.76

Training set error measures:

ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
0.9627769	35.7513	28.52924	-0.1850744	7.083161	0.6441221	0.1447196

1. Model Fitting Interpretation

- **Model Specification:** You used an ARIMA(0,1,0)(1,0,0)[12] model. Unlike the Maharashtra model (which used a seasonal moving average component), this model uses a seasonal autoregressive (SAR) term ($\text{sar1} = 0.4046$). This implies that the price in a given month is influenced by the price from the same month in the previous year.
- **Significance:** The sar1 coefficient is 0.4046 with a standard error of 0.1247. Since the coefficient is more than three times its standard error, this seasonal autoregressive component is statistically highly significant.
- **Goodness of Fit:** With an AIC of 596.61, this model suggests a solid fit. When comparing models, a lower AIC generally indicates a better model; you may want to compare this AIC against other potential configurations to confirm it is the optimal choice for the UP dataset.

2. Accuracy (Training Set Error)

MAPE (Mean Absolute Percentage Error): The MAPE is 7.083161.

This is an excellent accuracy metric. An error of $\sim 7.08\%$ suggests that your model is quite reliable at predicting historical price movements for bananas in UP. This is a lower (more accurate) error rate compared to the 12.15% MAPE observed in your Maharashtra dataset.

II. Forecast prices of Banana for Uttar Pradesh (in Rs. Per 100 Nos.):

Month	Year	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
Jan	2025	448.8144	402.214	495.4148	377.5452	520.0835
Feb	2025	434.2473	368.3444	500.1502	333.4575	535.0371
Mar	2025	448.0051	367.2909	528.7193	324.5633	571.4469

The forecasted banana prices for early 2025 show fluctuating behavior rather than a clear upward or downward trend. Prices are expected to decrease slightly from ₹448.81 in January to ₹434.25 in February, and then increase again to ₹448.01 in March.

The prediction intervals (80% and 95%) are moderately wide, indicating some uncertainty, but they are narrower compared to other states, suggesting relatively more stable forecasts. For example, in January, the 95% interval ranges from about ₹377.55 to ₹520.08.

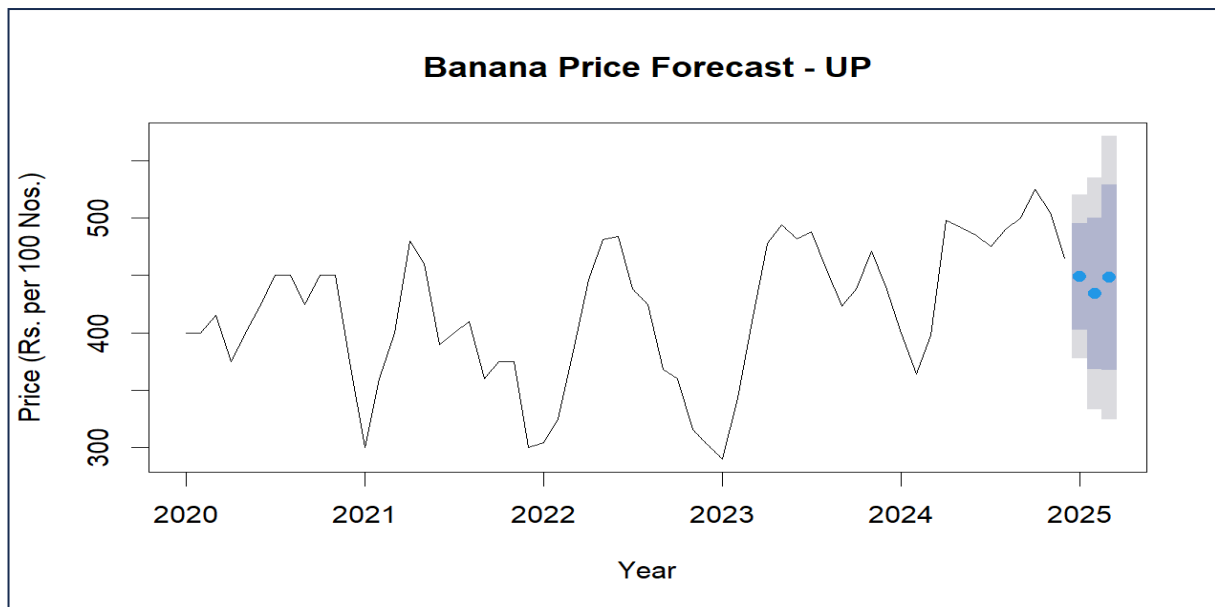


Figure 7.7: Time Series Plot for Wholesale Prices of Banana in Uttar Pradesh

8. Products From Banana

8.1 Introduction

Banana is one of the most widely cultivated and consumed fruit crops in India. India ranks first in banana production globally, contributing a significant share to the world's total output. Bananas are not only consumed as fresh fruit but are also processed into a wide variety of products that add economic value and reduce post-harvest losses. The banana plant is unique as almost every part of it fruit, stem, leaf, flower, and fibre can be utilized. This chapter presents a detailed study of various products derived from bananas in India, highlighting their processing methods, economic importance, nutritional value, and future prospects.

8.2 Importance of Banana in Indian Agriculture

Banana holds a vital position in Indian horticulture due to its adaptability to diverse agro climatic conditions. It is cultivated in tropical and subtropical regions across the country. Banana farming provides year-round employment to millions of farmers and agricultural labourers. The crop has a short gestation period and high productivity per unit area, making it economically attractive. Bananas also contribute significantly to food security and nutritional intake among low- and middle-income groups.

Classification of Banana Products

Banana products in India can be classified into four major categories:

- (i) Fresh consumption products
- (ii) Processed food products
- (iii) Industrial and fibre-based products
- (iv) By-products and eco-friendly products

This classification helps in understanding the wide scope of banana utilization beyond fresh consumption.

1. Fresh Banana Products:

Fresh bananas are the most commonly consumed form in India. They are marketed through local vendors, wholesale markets, supermarkets, and export channels. Popular varieties include Cavendish, Robusta, Poovan, Rasthali, Nendran, and Red Banana. Fresh bananas are consumed directly or used in traditional dishes such as banana fritters, halwa, and sweets.

2. Processed Banana Products:

Processing bananas into value-added products increases shelf life, reduces wastage, and improves marketability. Processed products are gaining popularity due to urbanization and changing food habits.

➤ **Banana Chips**

Banana chips are one of the most popular processed banana products in India. They are prepared from raw bananas, sliced thinly and fried or baked. Kerala is famous for Nendran banana chips, while Maharashtra and Tamil Nadu are also major producers.



➤ **Banana Flour and Powder**

Banana flour and powder are produced by drying unripe bananas and grinding them into fine powder. These products are gluten-free and rich in dietary fibre. They are used in baby food, bakery products, health supplements, and traditional foods.



➤ **Banana Pulp and Puree**

Banana pulp is produced from ripe bananas and is widely used in baby food, beverages, ice creams, bakery items, and confectionery industries. It helps in utilizing overripe bananas that may otherwise go to waste.



➤ **Banana Jam, Jelly, and Candy**

Banana jam, jelly, and candies are sweet products made from banana pulp, sugar, and preservatives. These products are popular among children and have a longer shelf life compared to fresh bananas.



➤ **Dehydrated Banana Products**

Dehydrated banana slices and flakes are prepared by removing moisture through drying techniques. These products are used in breakfast cereals, snack mixes, and health foods.

➤ **Banana Beverages**

Banana-based beverages include banana juice, shakes, wine, beer, and vinegar. Banana wine and vinegar are produced through fermentation and provide opportunities for small-scale rural enterprises.



3. Banana Fiber Products:

Banana fiber is extracted from the pseudstem of the banana plant. It is strong, biodegradable, and eco-friendly. Banana fiber is used in making textiles, ropes, mats, carpets, handicrafts, paper, and packaging materials. In recent years, banana fiber products have gained importance as sustainable alternatives to synthetic materials.

Table 8.1: Classification of Banana Products

Category	Sub-products	Uses
Fresh	Raw & ripe banana	Direct consumption
Processed	Chips, flour, pulp	Food industry
Fiber	Rope, textiles	Industrial use
By-products	Leaf plates, stem juice	Eco-friendly uses

- **Banana Leaf Products**

Banana leaves are traditionally used in India as plates, food wrappers, and for serving meals during festivals and religious ceremonies. Banana leaf plates and packaging materials are now commercially produced as eco-friendly substitutes for plastic.

- **Banana Stem and Flower Products**

Banana stem and flower are used as vegetables in Indian cuisine. They are rich in fiber, minerals, and antioxidants. Banana stem juice is known for its medicinal properties, particularly in digestive health.

- **Nutritional Significance of Banana Products**

Banana and its products are rich in carbohydrates, potassium, vitamins (B6 and C), dietary fiber, and natural sugars. Regular consumption helps in maintaining energy levels, digestive health, and heart health.

Table 8.2: Nutritional Value of Banana (per 100 grams)

Nutrient	Value	Nutrient	Value
Energy	89 kcal	Dietary Fiber	2.6 g
Carbohydrates	22.8 g	Potassium	358 mg
Protein	1.1 g	Vitamin C	8.7 mg
Fat	0.3 g	Vitamin B6	0.4 mg

- **Regional Distribution of Banana-Based Industries**

Different regions of India specialize in various banana products. Kerala is known for chips, Tamil Nadu for fiber products, Maharashtra for chips and powder, and north-eastern states for banana wine.

- **Challenges in Banana Processing Industry**

Despite its potential, the banana processing industry faces challenges such as perishability, lack of cold storage facilities, high transportation costs, limited access to technology, and inadequate market linkages.

- **Government Initiatives and Support**

The Government of India supports banana processing through schemes like the National Horticulture Mission (NHM), PMFME scheme, and food processing subsidies. Training and financial assistance are provided to encourage value addition

- **Environmental Benefits of Banana-Based Products**

Banana by-products contribute to environmental sustainability by reducing agricultural waste. Banana fiber and leaf products help in reducing plastic usage

- **Future Prospects of Banana Products in India**

With rising demand for healthy foods and eco-friendly products, banana-based industries have immense growth potential. Export opportunities for processed banana products are also increasing.

Table 8.3: Major Processed Banana Products and Their Uses

Product	Raw Material	Processing Method	Uses
Banana Chips	Raw banana	Slicing, frying/baking	Snacks
Banana Flour	Unripe banana	Drying, grinding	Bakery, baby food
Banana Pulp	Ripe banana	Crushing, preservation	Beverages, ice cream
Banana Jam/Jelly	Banana pulp	Cooking with sugar	Spreads
Dehydrated Banana	Banana slices	Drying	Cereals, snacks
Banana Wine/Vinegar	Ripe banana	Fermentation	Beverage, food processing

Banana products play a crucial role in strengthening India's agricultural and industrial economy. Efficient processing, marketing, and policy support can significantly enhance income generation, reduce post-harvest losses, and promote sustainable development.

9. Review of Research Paper

Paper 1: Unlocking India's Banana Export Potential: Trends, Challenges and Industrial Prospects (Chaudhari et al., 2025)

1. Introduction

Banana is one of the most important horticultural crops in India. India produces about 37.8 million tons of bananas annually, making it the largest producer in the world. However, despite this large production, India contributes only a small share to the global banana export market. The study conducted by Chaudhari et al. (2025) analyzed production trends, export performance, industrial utilization, and challenges in India's banana sector. The study highlights that post-harvest losses (20–30%), poor cold chain infrastructure, and limited value addition restrict India's export competitiveness.

2. Banana Production Trends in India

Banana cultivation in India has increased steadily over the past decade. The cultivated area increased from 796 thousand hectares in 2011–12 to 1,000 thousand hectares in 2023–24. Production increased from 28.5 million tons to 37.8 million tons, showing strong growth in banana farming.

Table 9.1: Growth Percentage in Area, Production and Productivity

Indicator	2011–12	2023–24	Growth (%)
Area ('000 Ha)	796	1000	25.6
Production (MMT)	28.5	37.8	32.6
Productivity (MT/Ha)	35.8	37.8	5.6

This shows that production growth (32.6%) is higher than productivity growth (5.6%), meaning the increase in production is mainly due to expansion of cultivation area rather than major productivity improvements. The major banana producing states includes, Maharashtra, Tamil Nadu, Gujarat, Andhra Pradesh These states contribute a significant share of India's total banana production.

3. Export Growth of Bananas from India

The study reports a remarkable increase in banana exports during the last decade. Export quantity increased from 34.9 thousand tonnes in 2013–14 to 596.4 thousand tonnes in 2023–24.

Table 9.2: Export Growth Percentage

Indicator	2013–14	2023–24	Growth (%)
Export Quantity	34.9 thousand tonnes	596.4 thousand tonnes	≈1609
Export Value	₹153.8 crore	₹2410.9 crore	≈1467

This large growth shows increasing demand for Indian bananas in international markets.

4. Country-wise Export Share

The study also analysed the percentage distribution of banana exports across countries.

Table 9.3: Country-wise Share of Banana Export

Country	Export Share (%)
Iraq	30%
Iran	23%
UAE	18%
Uzbekistan	10%
Oman	7%
Saudi Arabia	5%
Nepal	3%
Kuwait	2%
Qatar	1%
Bahrain	1%

These results show that more than 80% of India's banana exports go to Middle Eastern and South Asian countries.

5. Industrial Utilization of Banana Varieties

The study also analysed the percentage of bananas used for industrial processing.

Table 9.4: Industrial Utilization of Banana Varieties

Variety	Direct Consumption (%)	Industrial Use (%)
Grand Naine	95%	5%
Robusta	70%	30%
Nendran	60%	40%
Dwarf Cavendish	85%	15%
Poovan	90%	10%
Red Banana	98%	2%
Yelakki	95%	5%

This indicates that industrial processing varies between 2% and 40% depending on the variety. Varieties such as Nendran (40%) and Robusta (30%) have higher processing potential for producing banana chips, flour, powder, and snacks.

6. Post-Harvest Losses and Processing Level

Another major finding of the study is that post-harvest losses in bananas range from 20% to 30% due to improper storage and handling. At the same time, only about 5–10% of total banana production in India is processed into industrial products, such as Banana

chips, Banana puree, Banana flour and Banana beverages. This indicates a large opportunity for value-added processing industries.

The literature suggests that India has a strong production base but still underperforms in the global banana export market. While banana production increased by 32.6%, export growth has been much higher due to increasing international demand. However, challenges such as 20–30% post-harvest losses, limited cold chain infrastructure, and low processing levels (5–10%) restrict export competitiveness. Improving cold storage facilities, processing industries, and export quality standards can significantly enhance India's banana export potential and increase farmers' income.

Paper 2: A Figuration of Banana Production in India (Kumar et al., 2013)

1. Introduction

Banana is one of the most important fruit crops in the world and contributes significantly to global fruit production. Globally, banana accounts for approximately 16% of total fruit production. India is the largest producer of bananas, contributing about 27% of the world's banana production. Bananas are cultivated widely due to their year-round availability, high nutritional value, and strong market demand. The study by Kumar et al. (2013) analysed the growth rate of banana area, production, and productivity across major Indian states using statistical growth analysis.

2. Global Banana Production

Globally, banana cultivation covers about 4.8 million hectares with total production of approximately 99 million tonnes and an average productivity of 20.8 MT per hectare.

Table 9.5: Major Exporting Countries (Share by Quantity)

Country	Export Share (%)
Ecuador	29%
Costa Rica	13%
Philippines	12%
Colombia	10%
Guatemala	7%
Others	29%

These figures show that Ecuador dominates global banana exports with nearly one-third (29%) of total exports.

3. Banana Production in India

In India, banana cultivation covers about 830 thousand hectares, producing approximately 29.8 million tonnes, with an average productivity of 35.9 MT per hectare.

Table 9.6: State-wise Banana Production in India (2010-11)

State	Area ('000 ha)	Production ('000 MT)	Productivity (MT/ha)
Tamil Nadu	125.4	8253	65.8
Maharashtra	82	4303	52.5
Gujarat	64.7	3978	61.5
Andhra Pradesh	79.3	2774	35
Karnataka	111.8	2281	20.4
Bihar	31.9	1517	47.6
Uttar Pradesh	32.4	1346	41.5
West Bengal	42	1010	24
Assam	47.6	723.6	15.2

4. Percentage Share of Production by State

Based on production data, the study estimated the percentage share of major banana producing states.

Table 9.7: Major Banana Producing States

State	Production Share (%)
Tamil Nadu	27.7%
Maharashtra	14.4%
Gujarat	13.4%
Andhra Pradesh	9.3%
Karnataka	7.6%
Madhya Pradesh	5.8%
Bihar	5.1%
Uttar Pradesh	4.5%
West Bengal	3.4%
Assam	2.4%

These results show that Tamil Nadu alone contributes about 27.7% of total banana production in India, making it the leading producing state.

5. Compound Growth Rate of Banana Production

The study used Compound Growth Rate (CGR) to measure the growth in area, production, and productivity of banana cultivation during the period 2000-01 to 2010-11.

Table 9.8: Compound Growth Rate (%)

State	Area Growth (%)	Production Growth (%)	Yield Growth (%)
Tamil Nadu	2.47%	4.47%	1.95%
Maharashtra	1.16%	-0.06%	-1.21%
Gujarat	7.57%	13.37%	5.40%
Andhra Pradesh	8.42%	11.72%	3.07%
Karnataka	6.02%	5.71%	-0.31%
Madhya Pradesh	1.54%	2.56%	1.00%
Bihar	1.53%	9.84%	8.20%
West Bengal	7.53%	10.55%	2.81%

India (Overall)	5.05%	5.71%	0.63%
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The study shows that Andhra Pradesh recorded the highest growth in area at 8.42%, while Gujarat achieved the highest growth in production at 13.37%. Bihar, on the other hand, registered the highest productivity growth at 8.20%. These findings suggest that although cultivation area expanded across states, gains in productivity were not uniform. Overall, the results indicate mixed progress in agricultural performance across the region

7. Structure of Banana Farming in India

Unlike many countries where banana is produced on large commercial plantations, the Indian banana sector is dominated by small and marginal farmers, usually cultivating less than 3 acres of land. This structure influences production efficiency, technology adoption, and export potential.

8. Conclusion

The literature indicates that banana production in India has shown steady growth, with an overall compound growth rate of 5.71% in production and 5.05% in area. However, productivity growth is relatively low (0.63%), indicating the need for improved agricultural practices and technology adoption. States such as Gujarat, Andhra Pradesh, and Bihar have demonstrated strong production growth, suggesting that these regions have significant potential for expanding banana cultivation in India.

Paper 3: Environmental impact of Banana Cultivation and Production in India: A Geographical Perspective_(Dr. Arvind Vaman Dalavi)

1. Introduction and Overview

The paper examines the growth and environmental impact of banana cultivation in India using a geographical and statistical perspective. Bananas (*Musa paradisiaca*) are one of the most important fruit crops globally and serve as a major source of nutrition and income, especially in developing countries. India holds the largest share in banana production and cultivation area globally, making this study highly relevant. The study highlights that banana cultivation has expanded due to:

- Favorable tropical climate
- Rising global demand (especially Gulf countries)
- Adoption of tissue culture technology
- Better irrigation and farm management

However, environmental concerns such as deforestation, water overuse, and soil degradation are also emphasized.

2. Objectives and Methodology

Objectives: The study focuses on two main objectives:

1. To analyse the increase in banana cultivation area in India
2. To examine the increase in banana production and productivity

Methodology

- Based on secondary data (FAOSTAT database)
- Comparative analysis between 2017–18 and 2023–24
- Use of state-wise Tables for area, production, and productivity

This approach is simple but effective for identifying trends.

3. Growth in Banana Cultivation: Key Findings

Area expanded from 884 thousand hectares in 2017–18 to 987 thousand hectares in 2023–24, an increase of 103 thousand hectares. During the same period, production rose from 30,808 MT to 38,145 MT, while productivity improved from 34.85 MT/ha to 38.64 MT/ha. This indicates a clear expansion in cultivation along with better efficiency in output per hectare. Overall, the figures reflect strong agricultural growth and improved performance

Table 9.9: Area, Production and Productivity (India)

Year	Area ('000 ha)	Production (000 MT)	Productivity (MT/ha)
2017–18	884	30,808	34.85
2023–24	987	38,145	38.64
Change	+103	+7,337	+3.79

4. State-wise Analysis of Cultivation

Maharashtra recorded the highest increase in area at 18.9 thousand hectares, followed by Andhra Pradesh with 18.45 thousand hectares and Kerala with 15.27 thousand hectares. On the other hand, Chhattisgarh showed the largest decline at 5.41 thousand hectares, while Odisha declined by 3.93 thousand hectares. This reflects notable regional differences in area expansion and contraction across states

Table 9.10: Area under Banana Cultivation

State	2017–18	2023–24	Change
Maharashtra	80.88	99.78	+18.90
Andhra Pradesh	88.96	107.41	+18.45
Kerala	109.26	124.53	+15.27
Chhattisgarh	26.57	21.16	–5.41
Odisha	24.2	20.27	–3.93

5. Production Trends

Total production increased by 7,337 MT, showing a substantial overall rise. Among the states, Andhra Pradesh recorded the highest growth at 1,621.12 MT, followed by Maharashtra with 1,331.98 MT and Karnataka with 648.55 MT. In contrast, Chhattisgarh experienced a decline of 219.89 MT. This suggests that production growth was concentrated in a few major states, despite a setback in one region

Table 9.11: Production of Banana

State	2017-18	2023-24	Change
Andhra Pradesh	5003.07	6624.19	+1621.12
Maharashtra	4209.27	5541.25	+1331.98
Karnataka	2328.9	2977.45	+648.55
Chhattisgarh	745.78	525.89	-219.89

Production growth is strongly linked to market demand and export opportunities.

6. Productivity Analysis

Productivity increased by 3.79 MT/ha, which is about 10%, indicating a meaningful improvement over the period. The highest gains were seen in Andhra Pradesh at 5.43 MT/ha, followed by Karnataka at 5.09 MT/ha and Odisha at 4.62 MT/ha. This shows that productivity improvements were strongest in a few key states, contributing to the overall rise.

Table 9.12: Productivity of Banana

State	2017-18	2023-24	Change
Andhra Pradesh	56.24	61.67	+5.43
Karnataka	21.07	26.16	+5.09
Maharashtra	52.05	55.53	+3.48
Chhattisgarh	28.07	24.85	-3.22

This shows technological improvements and better farming practices.

7. Environmental Impact Analysis

The paper provides a detailed discussion of environmental consequences:

Negative Impacts:

1. Deforestation - Loss of biodiversity
2. Excessive water use - Depletion of water resources
3. Chemical fertilizers & pesticides - Soil and water pollution
4. Monoculture farming - Ecological imbalance
5. Soil fertility decline

Positive Impacts:

- Banana plants contribute to carbon sequestration
- Potential for sustainable farming using organic methods

The study emphasizes the need for eco-friendly cultivation practices.

8. Key Drivers of Growth

The increase in banana production is driven by:

- Rising international demand (Gulf countries)

- Better market prices
- Use of modern technology (tissue culture plants)
- Improved irrigation systems
- Government and institutional support

9. Conclusion

This paper shows that banana cultivation in India has grown significantly, supported by a production increase of 7,337 MT and a productivity improvement of around 10%. The growth appears to be driven by better technology and rising market demand. However, the paper also highlights that unsustainable practices could damage the environment. Therefore, sustainable agriculture is essential for ensuring long-term growth.

Paper 4: Growth & Instability Performance of Banana Production in India

1. Introduction

The horticulture sector plays a significant role in Indian agriculture by generating employment, enhancing farmers' income, and ensuring nutritional security. Among horticultural crops, banana holds a prominent position as one of the most widely consumed fruits globally. India is the largest producer of bananas, contributing approximately 26% of global production. Banana cultivation is widespread across several Indian states such as Gujarat, Tamil Nadu, Andhra Pradesh, Maharashtra, Karnataka, Madhya Pradesh, and Bihar, which together account for more than 75% of total production.

Globally, banana production increased from 78.66 million tonnes (2000–04) to 116.24 million tonnes (2015–19), showing rapid growth. India alone produced 30.02 million tonnes during 2015–19, reinforcing its dominant position.

Banana is not only economically important but also nutritionally valuable, offering health benefits such as reducing risks of heart disease, hypertension, and digestive disorders.

2. Objectives and Methodology

The main objective of the study is to analyse the growth and instability in banana production in India using time-series data from 1990 to 2017.

Data Sources

- Secondary data collected from:
 - Ministry of Agriculture, Government of India
 - FAO (Food and Agriculture Organization)
 - Ministry of Commerce

Analytical Techniques

1. Compound Annual Growth Rate (CAGR)
 - Used to measure growth trends in area, production, and productivity.

2. Cuddy-Della Valle Instability Index

- Used to measure variability by adjusting for trend effects.

3. Global and National Production Trends

Table 9.13: Country-wise Banana Production (Million Tonnes)

Country	2000–04	2005–09	2010–14	2015–19
India	14.23	22.50	28.73	30.02
China	5.75	7.77	11.37	11.44
Philippines	5.25	7.66	8.37	5.98
Brazil	6.38	6.91	7.01	6.72
Ecuador	5.95	6.52	7.02	6.62
World Total	78.66	101.38	114.07	116.24

India's share increased from 18.14% to 26.01%, showing strong growth.

Table 9.14: India's Banana Production and Share

Period	Production (MMT)	Share (%)
1990–94	10.10	17.80
2000–04	14.23	18.14
2005–09	22.50	22.40
2010–14	28.73	22.12
2015–19	30.02	26.01

This indicates a steady increase in both production and global share.

4. Growth Performance (CAGR Analysis)

Table 9.15: CAGR in Area, Production and Yield (1990–2017)

Period	Area (%)	Production (%)	Yield (%)
1990–1999	1.89	7.74	5.75
2000–2009	5.48	8.36	2.73
2010–2017	3.31	0.99	-2.24
Overall	3.53	5.22	1.64

Production growth of 5.22% was largely driven by area expansion (3.53%) rather than yield. The strongest growth occurred during 2000–2009, while yield growth weakened and even turned negative in 2010–17. Overall, expansion in cultivated area sustained production despite declining yield trends.

5. Instability Analysis

Table 9.16: Instability Index (1990–2017)

Category	CV (%)	Instability Index
Area	31.45	15.85
Production	41.39	35.35
Yield	16.69	14.26

Production of bananas shows the highest instability at 35.35%, indicating significant fluctuations over time. This instability is driven more by variability in the cultivated area than by yield performance, highlighting the influence of land allocation decisions on overall production trends. As a result, such instability directly increases the income risk faced by farmers, making their earnings more uncertain and vulnerable to market and environmental shocks.

6. State-wise Production Analysis

Table 9.17: State-wise Production Share (%) (2015–18)

State	Share (%)
Gujarat	14.50
Tamil Nadu	14.28
Andhra Pradesh	13.39
Maharashtra	13.07
Karnataka	8.25
Madhya Pradesh	6.06
Bihar	5.09

We observed that Gujarat leads banana production, with southern and western states dominating overall output, and total production has grown nearly threefold over time.

7. State-wise Growth and Instability

Growth Insights:

- Highest CAGR in production:
 - Andhra Pradesh → 7.95%
 - Bihar → 7.76%
 - Gujarat → 7.53%

Instability Insights:

- Karnataka shows highest instability (36.05%)
- Gujarat shows lowest instability (less than 1%)
- Bihar is relatively Stable.

8. Major Findings

India dominates global banana production with 26% share

- Growth is driven mainly by area expansion, not productivity
- High instability exists, especially in production
- Significant variation exists across states
- Weather conditions, irrigation, and technology influence production

9. Conclusion and Policy Implications

The study concludes that banana production in India has experienced significant growth, but also faces high instability, particularly in production levels.

Policy Suggestions:

- Promote sustainable agricultural practices
- Improve irrigation and infrastructure
- Introduce risk management strategies
- Encourage technological adoption to improve yield
- Focus on Stable production systems

Reducing instability is crucial to ensure income stability for farmers and balanced agricultural growth.

10. Overall Conclusion

- The study highlights that banana is one of the most important fruit crops globally, both in terms of nutritional value and economic significance. Its widespread cultivation across tropical and subtropical regions makes it a key contributor to global food security and agricultural income.
- From the historical analysis, it is evident that banana originated in Southeast Asia and gradually spread across the world through trade and migration. Over time, it has evolved into a globally cultivated crop with more than 1000 varieties, showing its adaptability and importance.
- Global banana production has increased significantly from 22.4 MMT in 1960 to around 142–145 MMT by 2025. This reflects more than a six-fold increase over time. The growth shows the expanding importance of banana in the global agricultural sector.
- The annual growth rate of banana production has remained between 1% to 3%, indicating a stable and consistent expansion. This steady growth highlights the reliability of banana as a crop. It also reflects continuous demand in global markets.
- Time series data from 2000 to 2024 shows production increasing from 70 MMT to over 135 MMT. This confirms a strong upward trend in production. The pattern indicates non-stationarity in the data.
- The fitted ARIMA (0,1,0) model with drift 3.0870 shows an average increase of 3.09 MMT per year. This indicates a consistent growth trend. The model is statistically suitable for forecasting purposes.
- Forecast results show that production may reach 155.03 MMT by 2029. However, the confidence intervals widen over time. This indicates increasing uncertainty in long-term predictions.
- Global banana production is highly concentrated, with the top 5 countries contributing 51.95%. This shows that a few countries dominate global supply. It reflects unequal distribution across regions.
- The top 15 countries contribute 76.45% of total production. This further confirms strong geographical concentration. Many smaller countries contribute only a minor share.
- India is the largest producer, with 36.61 MMT production. It contributes about 26–27% of global output. This highlights India's dominant role in banana production.
- China (12.06 MMT) and Indonesia (9.34 MMT) follow India. However, their production is much lower compared to India. This shows a large gap between the first and second rank.
- India's banana production increased from 28,455 ('000 MT) in 2011–12 to 37,807 ('000 MT) in 2023–24. This indicates continuous growth over time. The trend is stable and upward.

- The CAGR of production in India is 2.51%, while area increased from 796 to 1000 thousand hectares. This shows expansion in cultivation. It reflects increasing commercial importance.
- Productivity growth in India is relatively low with CAGR of 0.37%. This indicates that yield improvements are slow. More focus is needed on efficiency.
- The ARIMA model for India (drift 1019.2379) shows an annual increase of about 1,019 thousand tonnes. This indicates strong growth. The model is highly accurate.
- Forecast suggests that India's production may reach 43,096 ('000 MT) by 2029. This confirms a positive future outlook. Growth is expected to continue steadily.
- Maharashtra is the leading state with 7,229.86 ('000 MT) production. It plays a major role in India's output. The state has favourable conditions for banana cultivation.
- Andhra Pradesh (6,734.44) and Tamil Nadu (4,905.48) are also major contributors. These states show strong production levels. Together they form key production regions.
- The top 3 states contribute nearly 50% of total production. This indicates regional concentration. Production is not evenly distributed across India.
- Maharashtra shows the highest growth rate of 71.20% (2020–2024). This indicates rapid expansion. It reflects increased investment and cultivation.
- Uttar Pradesh shows strong growth of 40.45%. This indicates emerging importance. The state is becoming a major producer.
- Gujarat shows stable production with only 3.02% growth. This indicates maturity in production levels. Growth is slow but consistent.
- Highest productivity is observed in Madhya Pradesh (70.64 MT/ha) and Gujarat (67.20 MT/ha). These states are highly efficient. They achieve better yield per hectare.
- Andhra Pradesh shows the highest improvement in productivity. This reflects adoption of modern techniques. Technological advancement plays a key role.
- Maharashtra shows higher variability with MAPE of 17.5%. This indicates fluctuations in production. External factors affect its output.
- Andhra Pradesh shows stable growth with drift 278.72. This indicates consistent yearly increase. The state has less variability.
- Banana cultivation area increased significantly in Maharashtra (46.38%) and Uttar Pradesh (26.44%). This shows expansion of farming. It reflects rising demand.
- Overall, banana production shows strong growth, concentration, and improving efficiency. The sector is expanding steadily. It plays a key role in agriculture.
- However, risks such as climate change, water scarcity, and diseases create uncertainty. These factors affect production stability. They must be managed properly.
- Statistical models like ARIMA provide reliable forecasts. These models help in planning and decision-making. They are useful for policy formulation.
- Future growth depends on technology adoption, risk management, and balanced development. Sustainable practices are necessary. Proper planning can ensure long-term growth.

- Overall, this research provides a comprehensive understanding of global and Indian banana production trends, highlighting key patterns, growth drivers, and future prospects. It serves as a useful reference for policymakers, researchers, and farmers for planning and decision-making.

References

1. FAO. 2026. Banana Market Review - Preliminary Results 2025. Rome.
<https://openknowledge.fao.org/handle/20.500.14283/cd8079en>
2. De Langhe, E., Vrydaghs, L., de Maret, P., Perrier, X., & Denham, T. (2009). Why Bananas Matter: An introduction to the history of banana domestication. *Ethnobotany Research and Applications*, 7, 165–177. Retrieved from <https://ethnobotanyjournal.org/index.php/era/article/view/356>
3. Aquino, César Fernandes, et al. "Physical, chemical and morphological characteristics of banana cultivars depending on maturation stages." *Revista Caatinga* 30.1 (2017): 87-96.
4. Musa, L. "The Biology of Musa L.(banana)."
[https://scholar.google.com/scholar?q=The+Biology+of+Musa+L.+\(banana\)&hl=en&as_sdt=0,5&as_rr=1](https://scholar.google.com/scholar?q=The+Biology+of+Musa+L.+(banana)&hl=en&as_sdt=0,5&as_rr=1)
5. Venkataramani, K. S. "Studies on Indian Bananas: I. A descriptive study of twenty-four varieties." *Proceedings/Indian Academy of Sciences*. Vol. 23. No. 3. New Delhi: Springer India, 1946.
6. Chaudhari VP, Mahera AB, Vashi HG, Chaudhari BD, Rajput YV. Unlocking India's banana export potential: Trends, challenges and industrial prospects. *Int J Agric Extension Social Dev* 2025;8(12):43-46
7. Dalavi, A. V. (2025). Environmental Impact of Banana Cultivation and Production in India: A Geographical Perspective. *IIP: International Multidisciplinary Research Journal (IIPIMRJ)*, 2(IV), 518–526.
8. Report: Agricultural Prices in India 2021 – 24, Department of Agriculture And Farmers Welfare Department of Agriculture and Farmers Welfare, Government of India.

❖ Data Sources:

1. <https://ourworldindata.org/grapher/banana-production-by-region?tab=Table&time=earliest..2018&country=POL~PHL>
2. <https://desagri.gov.in/HI/document-report/agricultural-statistics-at-a-glance-2024/>
3. <https://agriwelfare.gov.in>
4. <https://oldror.lbp.world>

❖ Tools and Technology

- R software
- Minitab
- Excel